

EEBus UC Technical Specification

Monitoring of Inverter

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1 Scope of the document

This document describes the Use Case "Monitoring of Inverter" (short-name: MOI). Chapter 2 specifies the High-Level Use Case. Chapter 3 details the technical solution for SPINE for this Use Case. Within this document, a top-down approach is used to derive the requirements for the technical solution from the High-Level description.

1.1 References

1.1.1 EEBUS documents

[ProtocolSpecification] EEBus_SPINE_TS_ProtocolSpecification.pdf

[ResourceSpecification] EEBus_SPINE_TS_ResourceSpecification.pdf

[SHIP] EEBus_SHIP_TS_Specification_v1.0.1.pdf

1.1.2 Normative references

[RFC2119] IETF RFC 2119: 1997, Key words for use in RFCs to indicate requirement levels
Please see section 1.3.1 for details.

1.2 Terms and definitions

Actor

An Actor models a role within a Use Case definition (e.g. an energy manager or an electric vehicle).

AC

Abbreviation for alternating current.

Active sign convention

An electrical current is positive if the current is flowing out of device or component. In this case, the device or component produces electrical power and the active power is greater than zero. An electrical current is negative if the current is flowing into a device or component. In this case, the device or component consumes electrical power and the active power is smaller than zero.

Note: By convention, apparent power is always expressed as non-negative value.

CEM

Abbreviation for Customer Energy Manager. The CEM is an energy manager located at the home or premises of the user or in a cloud application. The energy manager enables energy-optimized operation of the connected devices by harmonising energy demand and availability.

DC

Abbreviation for direct current.

MGCP

Short name of the Use Case "Monitoring of Grid Connection Point"

207 MOB

208 Short name of the Use Case "Monitoring of Battery"

209 MOI

210 Monitoring of Inverter (short-name of this Use Case)

211 Monitoring Appliance

212 The Actor Monitoring Appliance evaluates particular data of another Actor.

213 MPC

214 Short name of the Use Case "Monitoring of Power Consumption"

215 MPS

216 Short name of the Use Case "Monitoring of PV String"

217 Passive sign convention

218 An electrical current is positive if the current is flowing into a device or component. In this case, the
219 device or component consumes electrical power and the active power is greater than zero. An

220 electrical current is negative if the current is flowing out of a device or component. In this case, the
221 device or component produces electrical power and the active power is smaller than zero.

222 Note: By convention, apparent power is always expressed as non-negative value.

223 PV

224 Abbreviation for photovoltaic.

225 PV String

226 Larger quantities of PV modules are connected in groups parallel to the inverter. A set of modules
227 connected in series is called a PV string.

228 Scenario

229 Part of a Use Case. Splitting a Use Case into Scenarios helps readers understand the Use Case more
230 quickly. Some Scenarios are mandatory for a Use Case, whereas others may be recommended or
231 optional.

232 Specialization

233 Reusable data collection for a specific functionality.

234 SPINE

235 Smart Premises Interoperable Neutral-message Exchange: Technical Specification of EEBus Initiative
236 e.V.

237 UI

238 Abbreviation for user interface.

239 UTC

240 Abbreviation for Coordinated Universal Time.

241

1.3 Requirements

1.3.1 Requirements wording

The following keywords are used:

- SHALL
- SHALL NOT
- SHOULD
- SHOULD NOT
- MAY

Note: They apply only if written in capital letters.

For the meaning of the keywords, please refer to [RFC2119].

1.3.2 Mapping of High-Level requirements

Within the High-Level Use Case description, the following abbreviation is used:

[MOI-xyz]

e.g.: [MOI-007]

The abbreviation is used to mark High-Level requirements or rules of this Use Case with a unique number xyz. These requirements are referenced throughout the technical solution to show how each High-Level requirement is realized in the technical part.

2 High-Level description

2.1 Introduction

Regenerative energy generators produce and batteries store direct current (DC). To use this in the normal alternating current (AC) grid, an inverter ensures that alternating current is generated from the direct current. An inverter is also required to integrate a battery (which is charged and discharged with direct current) into the regular power grid. Hybrid devices are often used to which photovoltaic (PV) strings and one or more batteries can be connected. This simplifies the use of renewable energies.

Within this Use Case, a client (e.g. a Customer Energy Manager (CEM) or a user interface (UI)) monitors data provided by any type of inverter, such as PV, battery, hybrid, etc. The inverter is responsible for the AC-connection-related data (e.g. power production or voltage) and the inverter itself (e.g., state or temperature). Other Use Cases provide additional data as well as the DC-connection-related data of the PV string(s) or a battery via the inverter as data-providing instance (see section 2.5.2 and 2.5.3 for details).

The inverter provides information needed for identification (like vendor name or serial number), its current state (e.g. normal operation or failure), additional details on its electrical (AC) connection*¹ (further power values, energy details as well as the power factor), corresponding nominal min/max values (like the inverter's nominal max power production) and further data points needed for diagnosis or efficiency calculation.

*¹: Basic AC measurement values (power, current, voltage, etc.) of the Inverter are provided by the Use Case "Monitoring of Power Consumption" (see section 2.5.1).

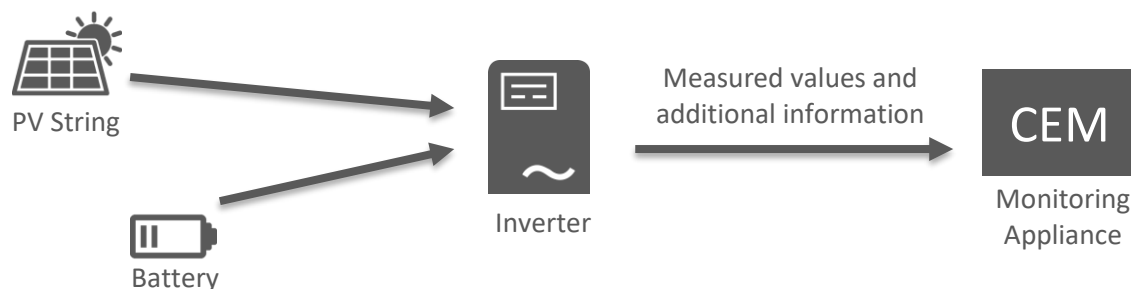


Figure 1: High-Level Use Case functionality overview

The retrieved data may be used for energy management purposes or to just visualize the data to the customer. Some examples are given in the following list:

- Optimization of self-consumption
- Increase of autarky rate
- Error detection
- Inverter identification
- Efficiency calculation
- Remote monitoring of inverter data

2.2 Detailed background information

This Use Case describes an Inverter. Further Use Cases consider its sub-devices like PV strings or batteries (see section 2.5).

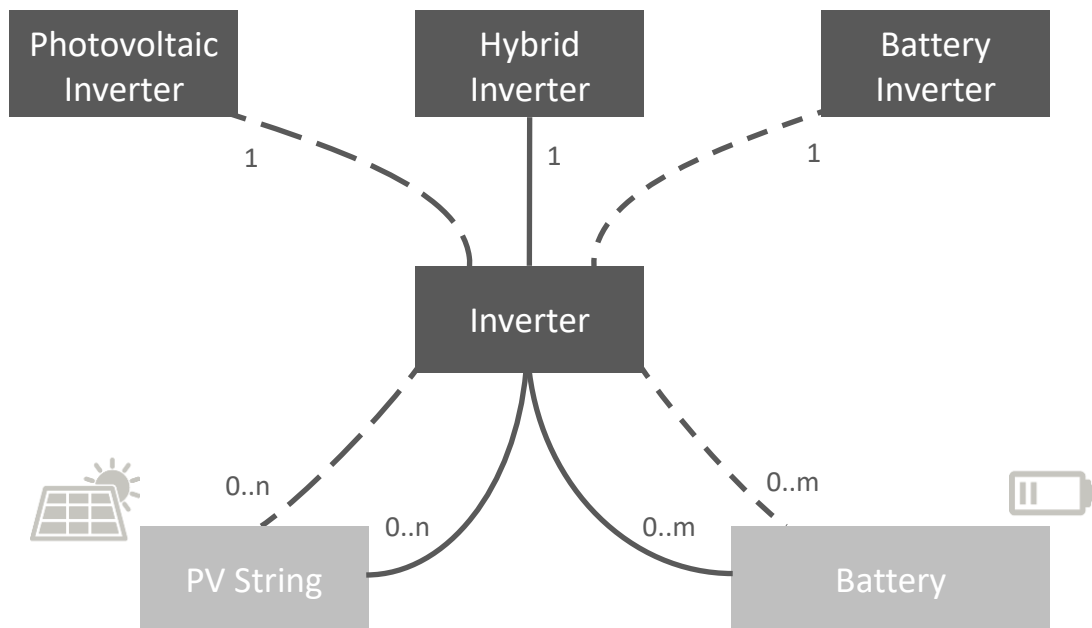


Figure 2: Example for Inverter device-hierarchy

The context of the Inverter is given by its so-called "parent" in the device-hierarchy ([MOI-004]). This can be:

- "Photovoltaic Inverter" ([MOI-004/1]), meaning that the Actor Inverter has only PV strings (up to "n" in Figure 2) as so-called "children" in its device hierarchy; a photovoltaic Inverter is only capable of producing energy
- "Hybrid Inverter" ([MOI-004/2]), meaning that the Actor Inverter is a hybrid Inverter that has both PV strings (up to "n" in Figure 2) and batteries (up to "m" in Figure 2) as "children" in its device hierarchy; a hybrid Inverter is capable of consuming and producing energy
- "Battery Inverter" ([MOI-004/3]), meaning that the Actor Inverter has only batteries (up to "m" in Figure 2) as "children" in its device hierarchy; a battery Inverter is capable of consuming and producing energy.

Note: The "0" in "0..n" and "0..m" of Figure 2 just expresses that this Use Case does neither require nor describe data of any PV string or battery itself, as such information is covered by related Use Cases discussed in section 2.5. Instead, the Use Case covers electrical constraints that are dedicated to the circuit for attached PV strings or to the circuit for attached batteries.

2.3 Actors

2.3.1 Monitoring Appliance

The Actor Monitoring Appliance (e.g. a CEM) may manage the home's appliances in such a way that lets the customer benefit from the energy optimized operation of the appliances or visualize the

information. In this Use Case, the Monitoring Appliance retrieves data provided by an Inverter and its sub-devices, such as PV strings or batteries.

2.3.2 Inverter

The Inverter transforms DC power to AC power and vice versa (only with connected battery). It may receive the DC power via PV strings or batteries and may pass on the DC power to a battery.

2.4 Scenarios

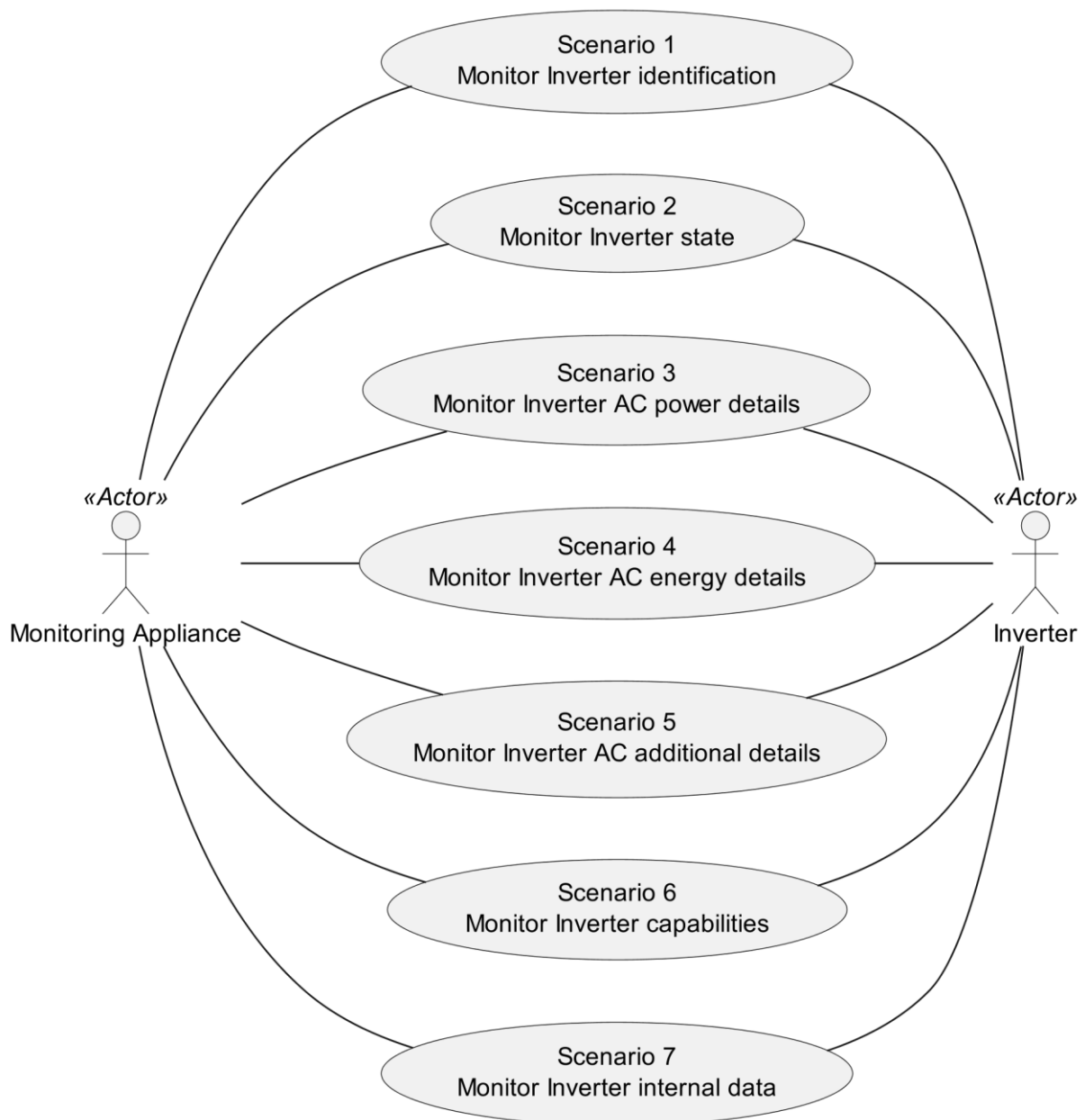


Figure 3: Scenario overview

Scenario number	Scenario name	Monitoring Appliance	Inverter
1	Monitor Inverter identification	M	M
2	Monitor Inverter state	M	M
3	Monitor Inverter AC power details	R	M
4	Monitor Inverter AC energy details	M	M
5	Monitor Inverter AC additional details	R	M
6	Monitor Inverter capabilities	R	M
7	Monitor Inverter internal data	O	O

Table 1: Scenario implementation requirements for Actors

2.4.1 Scenario 1 - Monitor Inverter identification

2.4.1.1 Description

This Scenario is used to identify the Actor Inverter and to provide descriptive information.

Data point name	Data point description	Support indication	High-Level requirement
Device Name	Name of the Inverter as indicated on the packaging.	M	[MOI-011]
Device Code	Code of the Inverter to uniquely identify the exact model.	M	[MOI-012]
Serial Number	Unique serial number of the Inverter.	M	[MOI-013]
Software Revision	Version of the software on the Inverter.	M	[MOI-014]
Hardware Revision	Hardware version of the Inverter.	M	[MOI-015]
Manufacturer Name	Name of the manufacturer of the Inverter.	M	[MOI-016a]
Brand Name	Name of the brand of the Inverter.	M	[MOI-016b]
Inverter Label	Label for the Inverter (e.g. for display purposes).	O	[MOI-017]
Inverter Description	Description of the Inverter (e.g. for display purposes).	O	[MOI-018]
Inverter Grid Code	Grid code of the Inverter ^{*1} . In some cases, it is possible that the parameters of the Inverter are changed to fit the needs of the local grid operator. In these cases, the parameters described by the grid code are (partly) overwritten.	M	[MOI-019]

Table 2: Scenario 1 - Monitor Inverter identification - Data point list

^{*1}: Information elements of an "Inverter grid code":

- country (optional): contains the country (e.g. AT, DE, SE, UK, etc.)

- technical connection rule (mandatory): contains the name of the connection rule itself (e.g. TORD4, VDE-AR-N4105, EN50438, G98/1-3, etc.)
- year (optional): the year in which the connection rule was published
- ancillary (optional): further information about the (sub part of the) rule (e.g. LV, etc.).

The type of Inverter (PV-Inverter ([MOI-004/1]), Hybrid-Inverter ([MOI-004/2]), Battery-Inverter ([MOI-004/3])) can be determined from the device hierarchy (see section 2.2).

2.4.1.2 Conditions

Triggering Event:

The Actor Monitoring Appliance connects to the Actor Inverter.

Pre-condition:

The Actor Monitoring Appliance does not know the Inverter's identification information.

Post-condition:

The Actor Monitoring Appliance knows the identification information provided by the Actor Inverter.

2.4.2 Scenario 2 - Monitor Inverter state

2.4.2.1 Description

The state information of the Actor Inverter can be retrieved by this Scenario. The following data points can be provided by the Inverter:

Data point name	Data point description [High-Level requirement]	Support indication	High-Level requirement
Inverter State	State of the Inverter. Possible states: - Normal operation [MOI-021/1] - Standby [MOI-021/2] - Temporarily not ready [MOI-021/3] - Off [MOI-021/4] - Failure [MOI-021/5]	M	[MOI-021]
Vendor State Code	Vendor specific state code of the Inverter.	R	[MOI-022]
PV String Availability Status	Indicates the availability of the PV String(s). Possible values: - Unknown - o [MOI-023/1] - o The status of the PV String(s) is unknown. - Disconnected - o [MOI-023/2] - o The PV String(s) is/are disconnected. - Connected - o [MOI-023/3]	O	[MOI-023]

	○ The PV String(s) is/are connected, but not ready yet. - Available ○ [MOI-023/4] ○ The PV String(s) is/are connected and ready for operation (but currently not producing energy). - In operation ○ [MOI-023/5] ○ The PV String(s) is/are producing energy.		
Battery Availability Status	Indicates the availability of the Battery (or Batteries). Possible values: - Unknown ○ [MOI-024/1] ○ The status of the Battery or Batteries is unknown. - Disconnected ○ [MOI-024/2] ○ The Battery is (or the Batteries are) disconnected. - Connected ○ [MOI-024/3] ○ The Battery is (or the Batteries are) connected, but not ready yet. - Available ○ [MOI-024/4] ○ The Battery is (or the Batteries are) connected and ready for operation (but currently not consuming or producing energy). - In operation ○ [MOI-024/5] ○ The Battery is (or the Batteries are) consuming or producing energy.	O	[MOI-024]
Grid Connection	Indicates whether the Inverter is connected to the public grid or not. Possible values: - Unknown ○ [MOI-025/1] ○ The status of the Inverter's public grid connection is unknown. - Disconnected ○ [MOI-025/2] ○ The Inverter is disconnected from the public grid.	R	[MOI-025]

	- Connected ○ [MOI-025/3] ○ The Inverter is connected to the public grid.		
Detailed Inverter State Information	Detailed Inverter state information of the different categories (some of these states may be active in parallel): <i>Category: Functionality</i> [MOI-026/1] - External override from grid*¹ ○ [MOI-026/1/1] - Autonomous grid support*¹ ○ [MOI-026/1/2] - Islanding mode*³ ○ [MOI-026/1/3] - Balancing*³ ○ [MOI-026/1/4] - Trickle charging*³ ○ [MOI-026/1/5] - Calibration*⁴ ○ [MOI-026/1/6] - Commissioning missing*⁴ ○ [MOI-026/1/7] - Sleeping*¹ ○ [MOI-026/1/8] - Starting*¹ ○ [MOI-026/1/9] - MPPT*² ○ [MOI-026/1/10] - Throttled*² ○ [MOI-026/1/11] - Shutting down*¹ ○ [MOI-026/1/12] - Manual shutdown*¹ ○ [MOI-026/1/13] <i>Category: Failure</i> [MOI-026/2] - Inverter defective*¹ ○ [MOI-026/2/1] - Battery overcurrent protection*³ ○ [MOI-026/2/2] - PV String overcurrent protection*² ○ [MOI-026/2/3] - Grid fault*¹ ○ [MOI-026/2/4] - Ground fault*¹ ○ [MOI-026/2/5] - AC disconnected*¹ ○ [MOI-026/2/6] - DC disconnected*⁴ ○ [MOI-026/2/7] - Cabinet open*¹ ○ [MOI-026/2/8] - Over temperature*^{1, *3} ○ [MOI-026/2/9]	R	[MOI-026]

	- Under temperature^{*1, *3} ○ [MOI-026/2/10] - Frequency above limit^{*1} ○ [MOI-026/2/11] - Frequency below limit^{*1} ○ [MOI-026/2/12] - AC voltage above limit^{*1} ○ [MOI-026/2/13] - AC voltage below limit^{*1} ○ [MOI-026/2/14] - DC voltage above limit^{*4} ○ [MOI-026/2/15] - DC voltage below limit^{*4} ○ [MOI-026/2/16] - Hardware test failure^{*1, *4} ○ [MOI-026/2/17] - Generic internal error^{*1, *4} ○ [MOI-026/2/18]		
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Table 3: Scenario 2 - Monitor Inverter state - Data point list

^{*1}: Inverter specific state information

^{*2}: PV-related Inverter state information

^{*3}: Battery-related Inverter state information

^{*4}: PV- or Battery-related Inverter state information

2.4.2.2 Conditions

Triggering Event:

The Actor Monitoring Appliance connects to the Actor Inverter. Subsequently, the Inverter state information should be gathered frequently.

Pre-condition:

The Actor Monitoring Appliance does not know the state information of the Actor Inverter.

Post-condition:

The Actor Monitoring Appliance knows the state information of the Actor Inverter.

2.4.3 Scenario 3 - Monitor Inverter AC power details

2.4.3.1 Description

The following AC power values of the Actor Inverter are provided by this Scenario:

Data point name	Data point description [High-Level requirement]	Support indication	High-Level requirement
Inverter Total Apparent Power	The total apparent AC power of the Inverter.	R	[MOI-031]

Inverter Phase-Specific Apparent Power	The phase-specific apparent power of the Inverter: - Apparent power on phase A ○ [MOI-032/1] - Apparent power on phase B ○ [MOI-032/2] - Apparent power on phase C ○ [MOI-032/3]	O* ¹	[MOI-032]
Inverter Total Reactive Power	The total reactive AC power of the Inverter.	R	[MOI-033]
Inverter Phase-Specific Reactive Power	The phase-specific reactive power of the Inverter: - Reactive power on phase A ○ [MOI-034/1] - Reactive power on phase B ○ [MOI-034/2] - Reactive power on phase C ○ [MOI-034/3]	O* ¹	[MOI-034]

Table 4: Scenario 3 - Monitor Inverter AC power details - Data point list

*¹: If this data point is supported, at least one phase SHALL be provided.

Note: If this Scenario is supported, at least one of the values stated above SHALL be supported.

2.4.3.2 Conditions

Triggering Event:

The Actor Monitoring Appliance is interested in the AC power values of the Actor Inverter.

Pre-condition:

The Actor Monitoring Appliance does not know the AC power values of the Actor Inverter.

Post-condition:

The Actor Monitoring Appliance knows the AC power values of the Actor Inverter.

2.4.4 Scenario 4 - Monitor Inverter AC energy details

2.4.4.1 Description

The AC energy detail values of the Actor Inverter are provided by this Scenario. The following data points can be provided by the Inverter:

Data point name	Data point description	Support indication	High-Level requirement
Inverter Cumulated Total AC Yield	Yield (real/active energy) of all PV strings of the Inverter. Lifetime accumulated or since last reset.	M	[MOI-041]
Inverter Total Yield Day	Yield (real/active energy) of all PV strings of the Inverter of this day, starting not earlier than 00:00 (local time* ¹).	R	[MOI-042a]

Inverter Total Yield Day Historical	Yield (real/active energy) of all PV strings of the Inverter. List of previous days (e.g. of the last 30 days; the current day SHALL NOT be included), each starting not earlier than 00:00 (local time ^{*1}) and ending not later than 24:00 (local time ^{*1}).	O	[MOI-042b]
Inverter Total Yield Month	Yield (real/active energy) of all PV strings of the Inverter of this month, beginning not earlier than the first day of the current month at 00:00 (local time ^{*1}).	R	[MOI-043a]
Inverter Total Yield Month Historical	Yield (real/active energy) of all PV strings of the Inverter. List of previous months (e.g. of the last 12 months; the current month SHALL NOT be included), each beginning not earlier than the first day of the month at 00:00 (local time ^{*1}) and ending not later than 24:00 (local time ^{*1}) of the month's last day.	O	[MOI-043b]
Inverter Total Yield Year	Yield (real/active energy) of all PV strings of the Inverter of this year, beginning not earlier than the first day of the current year at 00:00 (local time ^{*1}).	R	[MOI-044a]
Inverter Total Yield Year Historical	Yield (real/active energy) of all PV strings of the Inverter. List of previous years (e.g. of the last 10 years; the current year SHALL NOT be included), each beginning not earlier than the first day of the year at 00:00 (local time ^{*1}) and ending not later than 24:00 (local time ^{*1}) of the year's last day.	O	[MOI-044b]

Table 5: Scenario 4 - Monitor Inverter AC energy details - Data point list

^{*1}: Since counting the yields only makes sense for the local time, Actors may need to convert the timestamps internally accordingly (e.g., if timestamps are only provided as UTC times).

The given start and end times describe the regular maximum period covered by the respective data point or list entry. If the Inverter was not installed during the complete period, the period reduces accordingly.

Note: Please also consider section 2.5.4 for further data points that may be of interest in combination with the ones provided by this Scenario.

2.4.4.2 Conditions

Triggering Event:

The Actor Monitoring Appliance is interested in the AC energy detail values of the Actor Inverter.

Pre-condition:

The Actor Monitoring Appliance does not know the AC energy detail values of the Actor Inverter.

Post-condition:

The Actor Monitoring Appliance knows the AC energy detail values of the Actor Inverter.

2.4.5 Scenario 5 - Monitor Inverter AC additional details

2.4.5.1 Description

Some additional values of the Actor Inverter are provided by this Scenario. The following data points are specified:

Data point name	Data point description	Support indication	High-Level requirement
Inverter Power Factor Cos φ	Power factor $\cos \varphi$ of the Inverter (not the grid-power-factor- $\cos \varphi$). This value holds valid for all connected phases (not phase-specific).	R	[MOI-051]

Table 6: Scenario 5 - Monitor Inverter AC additional details - Data point list

Note: If this Scenario is supported, at least one of the values stated above SHALL be supported.

2.4.5.2 Conditions

Triggering Event:

The Actor Monitoring Appliance is interested in the AC additional detail values of the Actor Inverter.

Pre-condition:

The Actor Monitoring Appliance does not know about the AC additional detail values of the Actor Inverter.

Post-condition:

The Actor Monitoring Appliance knows the AC additional detail values of the Actor Inverter.

2.4.6 Scenario 6 - Monitor Inverter capabilities

2.4.6.1 Description

The capabilities of the Actor Inverter are provided by this Scenario. The following data points can be provided by the Inverter:

Data point name	Data point description	Support indication	High-Level requirement
Inverter Active Power Production Nominal Max	The nominal maximum active (real) power the Inverter can produce, as specified in the manufacturer's data sheet* ¹ . For the measured value see [MPC-011].	M	[MOI-061a]
Inverter Active Power Consumption Nominal Max	The nominal maximum active (real) power the Inverter can consume, as specified in the manufacturer's data sheet* ¹ . For the measured value see [MPC-011].	O	[MOI-061b]
Inverter Apparent Power Production Nominal Max	The nominal maximum apparent power the Inverter can produce, as specified in the manufacturer's data sheet* ¹ . For the measured value see [MOI-031].	R* ⁴	[MOI-062a]

Inverter Apparent Power Consumption Nominal Max	The nominal maximum apparent power the Inverter can consume, as specified in the manufacturer's data sheet* ¹ . For the measured value see [MOI-031].	O* ⁴	[MOI-062b]
Inverter's Battery AC Charge Power Nominal Min	The nominal minimum active power of the Inverter's battery or batteries to charge attached batteries, as specified in the manufacturer's data sheet* ¹ . Sum of all phases. For the measured value see [MPC-011]* ³ .	O	[MOI-063a]
Inverter's Battery AC Charge Power Nominal Max	The nominal maximum active power the Inverter's battery or batteries can be charged with, as specified in the manufacturer's data sheet* ¹ . Sum of all phases. For the measured value see [MPC-011]* ³ .	M* ²	[MOI-063b]
Inverter's Battery AC Charge Power Max	The maximum active power the Inverter's battery or batteries can be charged with* ¹ . Sum of all phases. For the measured value see [MPC-011]* ³ .	M* ²	[MOI-064]
Inverter's Battery AC Discharge Power Nominal Min	The nominal minimum active power the Inverter's battery or batteries must be discharged with, as specified in the manufacturer's data sheet* ¹ . Sum of all phases. For the measured value see [MPC-011]* ³ .	O* ²	[MOI-065a]
Inverter's Battery AC Discharge Power Nominal Max	The nominal maximum active power the Inverter's battery or batteries can be discharged with, as specified in the manufacturer's data sheet* ¹ . Sum of all phases. For the measured value see [MPC-011]* ³ .	M* ²	[MOI-065b]
Inverter's Battery AC Discharge Power Max	The maximum active power the Inverter's battery or batteries can be discharged with* ¹ . Sum of all phases. For the measured value see [MPC-011]* ³ .	M* ²	[MOI-066]
Inverter's Battery Time To AC Charge Power Max	The time (in seconds) currently required by the Inverter and the battery or batteries to adjust the active power until the currently maximum possible charge power ("Inverter Battery AC charge power max") is reached. This value is variable, e.g. depending on the current operating state of the Inverter and the battery or batteries* ¹ .	O* ²	[MOI-067a]
Inverter's Battery Time To AC Discharge Power Max	The time (in seconds) currently required by the Inverter and the battery or batteries to adjust the active power until the currently maximum possible discharge power ("Inverter Battery AC discharge power max") is reached. This value is variable, e.g. depending on the current operating state of the Inverter and the battery or batteries* ¹ .	O* ²	[MOI-067b]

*¹: This is a non-negative (absolute) value ([MOI-003]).

*²: Only mandatory / recommended / optional if a Battery is connected to the Inverter.

*³: The measured value is not a battery-specific one, but an Inverter-global one.

*⁴: This value SHALL only be provided, if [MOI-031] is provided, too.

2.4.6.2 Conditions

Triggering Event:

The Actor Monitoring Appliance is interested in the capabilities of the Actor Inverter.

Pre-condition:

The Actor Monitoring Appliance does not know the capabilities of the Actor Inverter.

Post-condition:

The Actor Monitoring Appliance knows the capabilities of the Actor Inverter.

2.4.7 Scenario 7 - Monitor Inverter internal data

2.4.7.1 Description

Some internal data of the Actor Inverter is provided by this Scenario. The following data points are specified:

Data point name	Data point description	Support indication	High-Level requirement
Internal Temperature	Internal component temperature. It is not further specified where the temperature sensor is located within the Actor.	O	[MOI-071]

Table 8: Scenario 7 - Monitor Inverter internal data - Data point list

Note: If this Scenario is supported, at least one of the values stated above SHALL be supported.

2.4.7.2 Conditions

Triggering Event:

The Actor Monitoring Appliance is interested in the internal data of the Actor Inverter.

Pre-condition:

The Actor Monitoring Appliance does not know the internal data of the Actor Inverter.

Post-condition:

The Actor Monitoring Appliance knows the internal data of the Actor Inverter.

2.5 Dependencies to other Use Cases

2.5.1 "Monitoring of Power Consumption"

The Use Case "Monitoring of Power Consumption" (MPC) delivers basic measurement values like the current power, phase specific current, etc. of the Inverter.

The Actor Monitoring Appliance SHOULD support the Use Case "Monitoring of Power Consumption".

The Actor Inverter SHALL support the Use Case "Monitoring of Power Consumption".

The Actor Monitoring Appliance of this Use Case acts as Actor Monitoring Appliance within the Use Case "Monitoring of Power Consumption".

The Actor Inverter of this Use Case acts as Actor Monitored Unit within the Use Case "Monitoring of Power Consumption".

2.5.2 "Monitoring of PV String"

In case of photovoltaic or hybrid inverters, the Use Case "Monitoring of PV String" (MPS) enables to monitor additional parameters of the PV string(s).

The Actor Monitoring Appliance MAY support the Use Case "Monitoring of PV String".

If the Actor Inverter is a photovoltaic inverter or hybrid inverter, the Actor Inverter SHOULD support the Use Case "Monitoring of PV String", providing the corresponding Actor PV String as so-called "child". This means that the Actor PV String is located behind the Actor Inverter in a hierarchical "device tree" (see section 2.2).

The Actor Monitoring Appliance of this Use Case acts as Actor Monitoring Appliance within the Use Case "Monitoring of PV String".

The Actor Inverter of this Use Case acts as parent Actor of the Actor PV String within the Use Case "Monitoring of PV String".

2.5.3 "Monitoring of Battery"

In case of battery inverters or hybrid inverters, the Use Case "Monitoring of Battery" (MOB) enables to monitor additional parameters of the battery (or batteries).

The Actor Monitoring Appliance MAY support the Use Case "Monitoring of Battery".

If the Actor Inverter is a battery inverter or hybrid inverter, the Actor Inverter SHOULD support the Use Case "Monitoring of Battery", providing the corresponding Actor Battery as so-called "child". This means that the Actor Battery is located behind the Actor Inverter in a hierarchical "device tree" (see section 2.2).

The Actor Monitoring Appliance of this Use Case acts as Actor Monitoring Appliance within the Use Case "Monitoring of Battery".

492 The Actor Inverter of this Use Case acts as parent Actor of the Actor Battery within the Use Case
493 "Monitoring of Battery".

494

495 **2.5.4 "Monitoring of Grid Connection Point"**

496 The Use Case "Monitoring of Grid Connection Point" (MGCP) can provide the data points "Total Grid
497 Feed-In Energy" ([MGCP-031]) and "Total Grid Consumed Energy" ([MGCP-041]) that may be of
498 interest in context with the Use Case "Monitoring of Inverter".

499

500 **2.6 Further information and rules**

501 **2.6.1 Sign conventions**

502 Throughout the whole Use Case the "load convention" (i.e. "passive sign convention") is applied for
503 measurement values ([MOI-001]). This means measured electrical current, active power and
504 exchanged energy are expressed with positive values in case of energy consumption whereas
505 negative values are used in case of energy production. This holds valid for the following values of this
506 Use Case:

- 507 - Inverter Cumulated Total AC Yield ([MOI-041])
- 508 - Inverter Total Yield Day ([MOI-042a])
- 509 - Inverter Total Yield Day Historical ([MOI-042b])
- 510 - Inverter Total Yield Month ([MOI-043a])
- 511 - Inverter Total Yield Month Historical ([MOI-043b])
- 512 - Inverter Total Yield Year ([MOI-044a])
- 513 - Inverter Total Yield Year Historical ([MOI-044b])

514 Apparent power ([MOI-031] and [MOI-032]) is always expressed as non-negative value ([MOI-
515 002/1]).

516 For the reactive power ([MOI-033] and [MOI-034]), the following relations apply ([MOI-002/2]):

- 517 1. If inductive power is consumed / capacitive power is produced:
518 a. $[MOI-033] \text{ and } [MOI-034] \geq 0$
- 519 2. If inductive power is produced / capacitive power is consumed:
520 a. $[MOI-033] \text{ and } [MOI-034] \leq 0$

521 For the Inverter Power Factor $\cos \varphi$ ([MOI-051]), the following relations apply ([MOI-002/3]):

- 522 1. If active power is consumed:
523 a. Power Factor $\cos \varphi \geq 0$
- 524 2. If active power is produced:
525 a. Power Factor $\cos \varphi \leq 0$

526 The sign convention described above does NOT apply to nominal values. For nominal values that are
527 dedicated to a particular energy direction it is common practice to express them as positive value.
528 Their name indicates whether they are used for consumption or production direction ([MOI-003]):

- 529 - Inverter Active Power Production Nominal Max ([MOI-061a]): production direction
- 530 - Inverter Active Power Consumption Nominal Max ([MOI-061b]): consumption direction
- 531 - Inverter Apparent Power Production Nominal Max ([MOI-062a]): production direction
- 532 - Inverter Apparent Power Consumption Nominal Max ([MOI-062b]): consumption direction
- 533 - Inverter's Battery AC Charge Power Nominal Min ([MOI-063a]): consumption direction
- 534 - Inverter's Battery AC Charge Power Nominal Max ([MOI-063b]): consumption direction
- 535 - Inverter's Battery AC Charge Power Max ([MOI-064]): consumption direction
- 536 - Inverter's Battery AC Discharge Power Nominal Min ([MOI-065a]): production direction
- 537 - Inverter's Battery AC Discharge Power Nominal Max ([MOI-065b]): production direction
- 538 - Inverter's Battery AC Discharge Power Max ([MOI-066]): production direction

539

540 **2.6.2 Value state rules**

541 Each Inverter's measurement value has a state that defines whether the value is correct or is to be
542 ignored by the Monitoring Appliance ([MOI-005]):

- 543 - normal: value correct
- 544 - out of range: value is out of range and SHALL be ignored by the Monitoring Appliance
- 545 - error: value is erroneous and SHALL be ignored by the Monitoring Appliance.

546

547 **2.7 Assumptions and Prerequisites**

548 None.

549

3 Technical SPINE solution

3.1 General rules and information

3.1.1 Underlying technology documents

This technical solution relies on the SPINE Resources Specification version 1.3.0 [ResourceSpecification].

For interoperable connectivity this technical solution relies on:

- SPINE Protocol Specification version 1.3.0 [ProtocolSpecification] as application protocol.
- SHIP Specification version 1.0.1 [SHIP] as transport protocol.

3.1.2 Use Case discovery rules

Use Case discovery SHALL be supported by each Actor and the following rules SHALL apply:

- The string content for the Element "nodeManagementUseCaseData. useCaseInformation. useCaseSupport. useCaseName" within the Use Case discovery (please refer to [ProtocolSpecification]) SHALL be "monitoringOfInverter". The string content SHALL only be defined by this Use Case (regardless of the Use Case version).
- The string content of the Element "nodeManagementUseCaseData. useCaseInformation. actor" within the Use Case discovery (please refer to [ProtocolSpecification]) SHALL be set to the according value stated within the corresponding Actor's section.
- An Actor A that is implemented to support this Use Case specification SHALL set the Element "nodeManagementUseCaseData. useCaseInformation. useCaseSupport. useCaseVersion" within the Use Case discovery (please refer to [ProtocolSpecification]) to "1.0.0".
- If an Actor A supports multiple versions of this Use Case with the same major version number, only the highest one SHOULD be set within the Use Case discovery.
- If an Actor A finds a proper counterpart Actor B for this Use Case that supports multiple versions of this Use Case with the same major version number as supported by Actor A, the Actor A SHOULD evaluate from these versions of Actor B only the highest version number.
- If an Actor A supports multiple versions of this Use Case with different major version numbers, for each major version number only the highest version number SHOULD be set within the Use Case discovery.
- If an Actor A finds a proper counterpart Actor B for this Use Case that supports only versions with a major version number not implemented by Actor A, it still might be possible to run the Use Case or parts of the Use Case. Therefore, the Actor A should try to evaluate the Actor B as a valid partner for this Use Case.

3.1.3 Rules for "Content of Specialization..." tables and "Content of Function..." tables

3.1.3.1 General presence indication definitions

Abbreviations for the presence indication of Elements listed in the tables are defined as follows:

Abbreviation	Meaning	Link to requirement keywords
--------------	---------	------------------------------

M	Mandatory	SHALL
R	Recommended	SHOULD
O	Optional	MAY

Table 9: Presence indication description

An Actor MAY support Elements that are not listed in the tables. However, another Actor MAY ignore these Elements.

The presence indications "M", "R" and "O" are always meant relative to the respective parent Element. I.e. if a parent Element is optional ("O") and a child is mandatory ("M") the child Element can only be present if the parent Element is present as well.

Note: The indications and the aforementioned rules apply for "complete messages" (so-called "full function exchange", please refer to [ProtocolSpecification]). In contrast, the so-called "restricted function exchange" is designed to permit exchange of specific excerpts of data, i.e. fewer Elements than potentially available from the data owner (partially even not all "mandatory" Elements).

3.1.3.2 Presence indications for "Content of Specialization..." tables

This section only defines rules for the client side.

Elements that are marked with "M" SHALL be supported by the client in case of readable as well as writeable data. This Element may be optional on the server side.

The following applies for readable data that is exchanged in a "read/reply" or "notify" operation:

- "R" means that the data SHOULD be supported by the client. In other words: If the server responds with the according Element, the client SHOULD be able to interpret the according Elements.
- "O" means that the data MAY be supported by the client. In other words: If the server responds with the according Element, the client MAY be able to interpret the according Elements.

The following applies for writeable data that is exchanged in a "write" operation:

- "R" means that the data SHOULD be written by the client.
- "O" means that the data MAY be written by the client.
- "F" means that the data SHALL NOT be written by the client.

The following applies for Elements that are not listed in the Actor section:

- In case of a received "reply" message: The client MAY ignore the Element.
- In case of a "write" operation to be created: The client MAY set the Element but SHALL consider that the server may ignore the Element.
- In case of a received "notify" message: The client MAY ignore the Element.

M, R or O may be combined with the suffix "(event)" to express that a supported Element or value only has to be supported during a certain event and hence does not need to be present at all times. If the event is not active the Element may be omitted or another value may be set. In most cases a High-Level requirement reference for the event is given in the rules column.

622 In some cases, an Element may have a double presence indication, separated by a colon. Examples:

- 623 - M, O
- 624 - M \W, O \W

625 The description (rules) of the Element details in such a case the presence requirement. A typical
626 example is a list-based Element where the "first" (or "last") Element has a different presence
627 requirement than the other list Elements.

628

629 **3.1.3.3 Presence indications for "Content of Function..." tables**

630 This section only defines rules for the server side.

631 Elements that are marked with "M" SHALL be supported by the server in case of readable as well as
632 writeable data. In case of writeable data (marked with "M \W") the server does not need to set the
633 Element, because the Element is set only by the client.

634 The following applies for readable data that is exchanged in a "read/reply" or "notify" operation:

- 635 - "R" means that the data SHOULD be provided by the server.
- 636 - "O" means that the data MAY be provided by the server.
- 637 - "F" means that the data SHALL NOT be provided by the server.

638 The following applies for writeable data that is exchanged in a "write" operation:

- 639 - "R" means that the data SHOULD be supported. In other words: If the client writes the
640 Element, the server SHOULD accept those messages and the contained Elements.
- 641 - "O" means that the data MAY be supported. In other words: If the client writes the Element,
642 the server MAY accept those messages and the contained Elements.

643 The following applies for Elements that are not listed in the Actor section:

- 644 - In case of a received "read" request: The according Element MAY be set in the reply.
- 645 - In case of a received "write" operation: The server MAY ignore the Element.
- 646 - In case of a "notify" operation to be created: The server MAY set the Element.

647 Note: The server will only accept write operations if the result fulfils the server Function
648 requirements (permitted values, e.g.). Write operations on Elements that are not writeable MAY
649 result in an error message.

650 M, R or O may be combined with the suffix "(event)" to express that a supported Element or value
651 only has to be supported during a certain event and hence does not need to be present at all times. If
652 the event is not active the Element may be omitted or another value may be set. In most cases a
653 High-Level requirement reference for the event is given in the rules column.

654 In some cases, an Element may have a double presence indication, separated by a colon. Examples:

- 655 - M, O
- 656 - M \W, O \W

The description (rules) of the Element details in such a case the presence requirement. A typical example is a list-based Element where the "first" (or "last") Element has a different presence requirement than the other list Elements.

3.1.3.4 Cardinality indications on Elements and list items

A cardinality indication on an Element or list item expresses constraints on the number of occurrences of a given Element or data set. In this section we use "X" as representation for such an Element or data set. Furthermore, "a" and "b" represent constraints. The following rules apply for the occurrence of "X" and its content related to a specific Scenario (see note underneath the list):

1. X
No cardinality indication.
2. X (a..b)
This means "X" SHALL occur at least "a" times and at maximum "b" times.
3. X (a..
This means "X" SHALL occur at least "a" times and MAY occur more than "a" times.
4. X (..b)
This means "X" SHALL occur at maximum "b" times and MAY occur less than "b" times (even zero occurrences are permissive).

Note: These rules apply only under consideration of presence indications and with regards to the given Scenario or Function definition for this Use Case.

The following table is an example to explain this for two different placements.

Scenario [{...}]: M/R/O [W][C]	Element	Value	[High-Level Mapping] Element and value rules
...	...	...	...
2: M \W	xFeatureType. xListData. xData. (1..3)		
2: M \W	xId	<*(1..)>	PRIMARY IDENTIFIER
2: M \W	timePeriod		...
2: M \W	timePeriod. startTime	<xs:duration>	
2: M \W	xSlot. (1..)		
2: M \W	xSlot. xSlotId		...
2: M \W	xSlot. duration	<xs:duration>	...
...	...	...	...

Table 10: Example table for cardinality indications on Elements and list items

The field

xFeatureType. xListData. xData. (1..3)

introduces a data pattern (required Elements and values) for "xData" instances used for Scenario 2. The field itself specifies that such an "xData" instance SHALL occur at least 1 time and at maximum 3

times within "xListData" of Feature Type "xFeatureType". However, this constraint holds only for Scenario 2 and only if such "xData" are required. In this case, they are required, as the left field

2: M \W

denotes that this data set is mandatory for Scenario 2.

The field

xSlot. (1..)

expresses that the Element "xSlot" SHALL occur at least one time within its "xData", but MAY occur more than one time.

For the expression "<*(1..)>" of Element "xId" please see section 3.1.3.6.

The remaining fields do not have an explicit cardinality indication.

Note: Cardinality expressions are also used within placeholder expressions as defined in section 3.1.3.6. In many cases such placeholder expressions define the number of required or permitted Elements or list items as they explicitly define how many different values for a given Identifier are required or permitted for a given Scenario.

3.1.3.5 Writability and changeability indication

In the same column where the presence indications are denoted, a mark is used to distinguish between writeable, changeable or readable Elements:

- Elements that are marked with "\W" are written by a client and SHALL be writeable at the server according to their presence indications. The client is not obliged to read the according data. Received notifications do not need to be evaluated.
- Elements that are marked with "\C" are changed by a client and SHALL be changeable at the server according to their presence indications. The client is not obliged to read the according data. Received notifications do not need to be evaluated.
- Elements that are marked with "\RW" are read and written by a client and SHALL be writeable and provided by the server according to their presence indications. Received notifications SHALL be evaluated according to their presence indications.
- Elements that are marked with "\RC" are read and changed by a client and SHALL be changeable and provided by the server according to their presence indications. Received notifications SHALL be evaluated according to their presence indications.
- Elements that are not marked are only read by a client and SHALL be provided by the server according to their presence indications. Received notifications SHALL be evaluated according to their presence indications.

"Writeable" means that the Element and its value may be written by a client. This includes the possibility to modify (if the Element is already present), create (if the Element is not present yet), and delete the Element. The server SHALL adjust its Function according to the received "write" operation (unless the server cannot accept the "write" operation according to section 3.1.3.3).

"Changeable" means that the Element's value may be changed by a client. If the Element is not present at the resource before, it probably **cannot** be created by the client via the "write" operation. In this case the server MAY decline such a message.

Note: "\W" includes "\C" already.

Note: Depending on the resource a client might need to request a proper binding before the server accepts a "write" operation.

3.1.3.6 "Value" placeholders

3.1.3.6.1 Introduction

Specializations may use placeholders to model relations between different Elements or even list items of different Functions. The main purpose is to declare which Identifier values relate to each other. As a Use Case does not prescribe specific values to be used for a given Identifier, a placeholder like "<x1>" can be used in "Value" columns to express the intended relations.

There are two styles to identify a placeholder that can be referenced:

1. <xM>
2. <xM#S>

where

1. "x" is any alphabetical prefix like "m", "t", "ec", "cc", etc.
2. "M" is a (major) number like "1", "2", "15", etc.
3. "S" is a sub-number like "1", "7", "10", etc.

Examples for the first style are "<ec1>", "<z12>". Examples for the second style are "<p4#2>", "<m22#3>". For a given placeholder, only one of the styles can be used.

In addition, there are also styles for placeholders that do not need to be referenced:

1. <*>
2. <*#S>

The second style is only used with so-called cardinality expressions.

3.1.3.6.2 Uniqueness of placeholders

A given placeholder <xM> or <xM#S> represents the same value throughout a given Use Case specification for a given set of its parent Identifier values. This shall be explained in a brief example:

We assume a list item with PRIMARY IDENTIFIER "pId". It also has a SUB IDENTIFIER "sId" with placeholder "<s1>". Then, each occurrence of "<s1>" represents the same value for a given value of pId. This means that "<s1>" of a list item with pId=1 can differ from "<s1>" of a list item with pId=2. But it also means that "<s1>" represents the same value in all list items with pId=1.

754 Note: Typically, parent Identifiers like "pId" will also be expressed with a placeholder like "<p5>", e.g.
 755 In this case, the uniqueness rule applies for "<p5>" likewise.

756 Note: The uniqueness also applies for placeholders used as FOREIGN IDENTIFIER.

757

758 3.1.3.6.3 Placeholder expressions with cardinalities

759 For some Identifiers, more than one placeholder is needed. Several notations are used for this
 760 purpose, which make use of cardinality expressions. The general notation is as follows:

761 1. <xM#(a..b)>

762 This is equivalent to this explicit definition:

763 At least a and at maximum b placeholders of this list: <xM#1> <xM#2> ... <xM#b>

764 This means that the implementation of a given Use Case (or Scenario) requires a minimum of "a"
 765 distinct values of the respective Identifier. In total, there can be up to "b" distinct values of the
 766 respective Identifier.

767 Additionally, the following notations may occur:

768 2. <xM#(a..)>

769 This is equivalent to "<xM#(a..b)>" with "b" equal to infinity.

770 3. <xM#(..b)>

771 This is equivalent to "<xM#(a..b)>" with "a" equal to zero.

772 "<xM#(a..)>" has only a lower bound of "a" distinct values, but no upper bound. "<xM#(..b)>", on the
 773 other hand, expresses that the Identifier may not be required at all, but it is permitted to have up to
 774 "b" distinct values.

775 Similarly, the cardinality can be used for placeholders that are not referenced, i.e. <*(a..b)> etc.

776 Note: The cardinality does NOT express which of the sub-numbers have to be used! I.e., it does NOT
 777 mean that the Identifiers <xM#1> ... <xM#a> are always used and just those with larger sub-numbers
 778 (<xM#a+1> ... <xM#b>) are optional. If, for instance, a placeholder expression "<xM#(3..5)>" is given,
 779 it may well happen that an implementation makes use of <xM#2>, <xM#4>, and <xM#5> (i.e., it does
 780 NOT use <xM#1>, <xM#3>). Which sub-numbers are used usually depends on other parts of a
 781 Specialization and their references to required placeholders, which is explained in section 3.1.3.6.4.

782

783 3.1.3.6.4 References to placeholders and relations

784 According to the styles for placeholders that can be referenced, an enumeration value "e" can refer
 785 to a particular placeholder:

786 1. e(-><xM>)

787 2. e(-><xM#S>)

788 This denotes that "e" is to be used with "<xM>" or "<xM#S>", resp.

Example: A Specialization contains the Elements "mld" and "phase". "mld" is an Identifier with placeholder definition <m2#(1..3)>. "phase" is a string that permits the values "a", "b", and "c" using this expression:

"a"(-><m2#1>)

"b"(-><m2#2>)

"c"(-><m2#3>)

This expresses that the enumeration value "a" is to be used with the placeholder <m2#1>, "b" is to be used with <m2#2> and "c" with <m2#3>.

Similarly, a placeholder "yN" can refer to a particular placeholder:

3. <yN->xM>

4. <yN->xM#S>

5. <yN#T->xM>

6. <yN#T->xM#S>

where "T" is a sub-number of "yN".

It is also feasible to associate placeholders with cardinalities:

7. <yN#(a..b)->xM#(a..b)>

denotes that <yN#1> is to be used with <xM#1>, <yN#2> is to be used with <xM#2>, etc.

Note: In this case, the placeholder expressions of yN and xM must have the same cardinality.

In some cases, there is a need to express that multiple list items with similar values are feasible or required, but only particular combinations of these different data are permitted. The following example shows that several "fData" list items with different "phase" value are required, but that these list items may only express either the "phase" value combination { "a", "b", "c" } or the "phase" value combination { "a", "abc", "neutral" }. The permitted combinations are defined in a note below a table:

Scenario [...]: M/R/O [W][C]	Element	Value	[High-Level Mapping] Element and value rules
2: M	F. fListData. fData.		
2: M	zld	<z3#(3..5)>	
2: M	phase	"a"(-><z3#1>)	
		"b"(-><z3#2>)	
		"c"(-><z3#3>)	
		"abc"(-><z3#4>)	
		"neutral"(-><z3#5>)	

Table 11: Content of an example Specialization

Note: One of the following combinations SHALL be used at least: {<z3#1>, <z3#2>, <z3#3>} or {<z3#1>, <z3#4>, <z3#5>}.

3.1.3.7 Rules for content of "Value" column

For a given Scenario, the "Value" column may restrict the permitted content of a Function's Element to one or more particular values. This means that Elements with values deviating from the restriction (i.e. from the permitted values) do not belong to the respective Scenario and need to be considered as if the Element is not set. If more than one particular value is permitted for an Element, the values are in a single line each.

If a presence indication is set for the value (in an additional column before the value), the following rules SHALL be applied:

- "M" means that the value SHALL be supported. This means the value needs to be set at a certain point in time (depending on the value rules) or for a certain Element within a list entry.
- "R" means that the value SHOULD be supported.
- "O" means that the value MAY be supported.

If all possible values of a given mandatory Element are optional or recommended and this Element is used for the purpose of the respective Scenario, one of the values SHALL be set. If all possible values of a given optional or recommended Element are optional or recommended, this Element MAY contain also other values, but then this Element SHALL NOT be considered as part of the respective Scenario.

M, R or O may be combined with the suffix "(event)" to express that a supported value only has to be supported during a certain event and hence does not need to be present at all times. If the event is not active another value may be set. In most cases a High-Level requirement reference for the event is given in the rules column.

If no presence indication is set for the value, the following rules SHALL be applied:

- In case of Elements where the server may set or change an Element on its own (see section 3.1.3.5):
 - within the tables in the "Server data - Resources" sections:
 - the server SHALL support at least one of the listed values.
 - within the tables in the "Client data - Specializations" sections:
 - the client SHALL support all listed values.
- In case of Elements that are writeable or changeable (see section 3.1.3.5):
 - within the tables in the "Server data - Resources" sections:
 - the server SHALL support all listed values.
 - within the tables in the "Client data - Specializations" sections:
 - the client SHALL support at least one of the listed values.

Depending on the Element, different values may be used during runtime. If this is the case, those rules are described within the value rules.

853 If a value is placed in parenthesis, the corresponding value is a recommendation. The actual value
 854 MAY deviate from this, e.g. "(1024)".

855

856 **3.1.3.8 General information on how to interpret the "Content of Function..." and "Content of** 857 **Specialization..." tables**

858 Within the "Client data - Specializations" sections each Specialization is described in an own sub-
 859 section with the name "Specialization "<name of the Specialization>" (e.g. "Specialization
 860 "Measurement_GridFeedInEnergy"). It contains only one table that includes all Elements needed for
 861 this Specialization. The different Functions are mentioned in a continuous row, highlighted with grey
 862 background colour. This row contains the following parts:

863 <Feature Type>. <Function>.[<list entry instance name>.]

864 The <list entry instance name> is only included if the <Function> is a list-based Function. An example
 865 could be:

866 DeviceConfiguration. deviceConfigurationKeyValueDescriptionListData.
 867 deviceConfigurationKeyValueDescriptionData.

868 In the following rows, only the names of the Elements are stated, without the prefix described above.

869

870 Within the "Server data - Resources" sections each Feature Type is described in an own sub-section
 871 with the name "Feature Type "<name of the Feature Type>" (e.g. "Feature Type "Measurement").
 872 It contains sub-sections for each Function named "Function "<name of the Function>" (e.g.
 873 "Function "measurementListData"). These sections contain one table with all Elements needed for
 874 this resource. The list entries are mentioned in a continuous row, highlighted with grey background
 875 colour. This row contains the following parts:

876 <Feature Type>. <Function>.[<list entry instance name>.]

877 The <list entry instance name> is only included if the <Function> is a list-based Function. An example
 878 could be:

879 Measurement. measurementDescriptionListData. measurementDescriptionData.

880 In the following rows, only the names of the Elements are stated, without the prefix described above.

881

882 For both kinds of tables, the following applies:

883 - Parent Elements are marked with a dot at the end of the name:

884 <parent Element>.

885 E.g.:

886 value.

887 - If there are sub-Elements, they are described in own rows with the name of the parent
 888 Element as prefix, separated by a dot and a blank space:

<parent Element>. <sub-Element>

E.g.:

value. number

3.1.4 Rules for "Feature Types and Functions..." tables

3.1.4.1 Presence indications for "Feature Types and Functions..." tables

The following presence indications are used:

Abbreviation	Meaning	Link to requirement keywords
M	Mandatory	SHALL
R	Recommended	SHOULD
O	Optional	MAY

Table 12: Presence indication of Feature Types and Functions support

If at least one Function of a Feature has the presence indication "M", it is mandatory to support the Feature.

3.1.4.2 Rules for "Possible operations" column

Within the "Feature Types and Functions..." tables the column "Possible operations" state whether the Function is read- or writeable (as defined in the detailed discovery mechanism, see [ProtocolSpecification]).

If the "partial" concept (also called "restricted function exchange") SHALL be supported, the following notation is used (separated for read and write access):

read (M). partial (M)

write (M). partial (M)

If the "partial" concept SHOULD be supported, the following notation is used:

read (M). partial (R)

write (M). partial (R)

If the "partial" concept MAY be supported, the following notation is used:

read (M). partial (O)

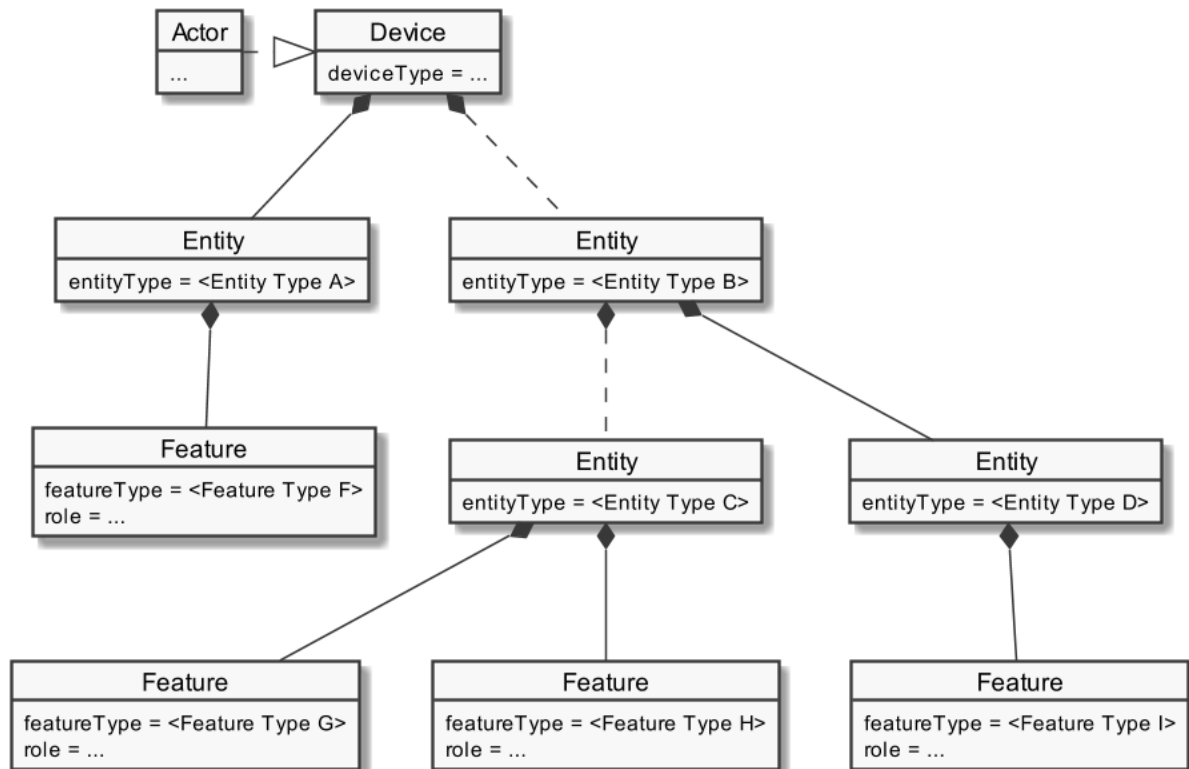
write (M). partial (O)

The server can decide whether a notification is submitted complete or partial (as described in [ProtocolSpecification]) if not defined differently within this Use Case Specification.

3.1.5 "Actor ... overview" diagram rules

Within the "Actor [...] overview" diagrams in the "Actors" sub-sections the complete functionality of this Use Case is provided, including optional Scenarios. Which Scenarios are optional can be found in Table 1. The Actor MAY have more functionality implemented than needed for this Use Case.

921 For the following Actor overview example, a brief description of the graphical symbols will be
 922 described.



923
 924 *Figure 4: Actor overview example*

925 The solid lines in the figure represent an immediate parent-childhood relation: The Entity with
 926 "<Entity Type A>" is a direct child of "Device". The Entity with "<Entity Type D>" is a direct child of the
 927 Entity with "<Entity Type B>". All Features are immediate child of the respective Entity.

928 The dashed lines in the figure express that there MAY be additional Entities between the shown
 929 Entities: A vendor's implementation MAY have one or more Entities between "Device" and the Entity
 930 with "<Entity Type B>". Likewise, a vendor's implementation MAY have one or more Entities between
 931 the Entity with "<Entity Type B>" and the Entity with "<Entity Type C>".

932

933 3.1.6 Specializations

934 Within the "Actors" sub-sections, Specializations are referenced. A Specialization describes a dataset
 935 necessary to fulfil the specific requirements of a High-Level Use Case and its Scenarios. Often, data
 936 from multiple different Features and Functions are needed to fulfil the requirements. Therefore, a
 937 Specialization defines a dataset that may encompass multiple related Functions from one or more
 938 different Features.

939 As different Use Cases sometimes share similar requirements, Specializations are also important
 940 from a re-usability perspective. This approach is used to improve consistency across Use Cases and
 941 avoid multiple variances of basically the same dataset. This is especially important in the case when
 942 an implementation supports multiple Use Cases. E.g. if a power measurement is necessary in two
 943 different Use Cases, both Use Cases could define slightly different datasets. In this case, the server as

well as the client functionality would have to implement both variances if both Use Cases are supported. This means, depending on the number of Use Cases, two or more datasets need to be generated, transmitted and stored instead of one. Therefore, common Specializations are defined to enable consolidated implementations. Specializations are detailed per Actor, as distinct Actors may use different parts of a Specialization or even have different permissions to change data. In general, this may also depend on the respective Scenario.

If a Feature server can provide the data of a Specialization, the data does not necessarily always need to be available at the Feature server. There might be situations where the user deactivates a Use Case. There may also be other reasons why Use Case data cannot be provided currently. Therefore, a client always needs to be subscribed (as described in section 3.3.4) on the corresponding dataset to stay updated.

The SPINE resource descriptions given in the "SPINE resources of the Actor" sections are derived from the Specializations given in the Actor's overview diagram.

3.1.7 Order of messages within the sequence diagrams

There are several sequence diagrams in this document describing message flows. The order of the messages SHOULD be kept by the communications partners, but there might be cases where a different order makes sense. The communications partners SHALL be able to handle the Scenario functionalities even if the messages are transmitted in a different order by the other Actor(s). The sequence diagrams can be seen as examples.

3.1.8 Further information and rules

3.1.8.1 Frequently used Element rules for the Resource and Specialization tables

This section serves as a collection of rules frequently used by Resource and Specialization tables of the subsequent sections. Each rule applies only where referenced explicitly in the tables.

Note: The purpose of this collection is just to reduce the size of the tables. As such, no rule has a meaning without a reference indicating the required rule. A reference looks like "See [Measurement value rules]", e.g.

[Measurement value rules]:

SHALL be set if a value is available. Otherwise the whole list entry SHALL be omitted or the Element *valueState* SHALL be set to "error".

If *valueState* is set to "error", but *value* is set, the content of *value* SHALL be ignored by the client.

If *valueState* is set to "outOfRange", but *value* is set, the content of *value* SHALL be interpreted as being out of range.

If *valueState* is set to "outOfRange", *measurementConstraintsListData.valueRangeMax* is set and *value* is equal or bigger than *valueRangeMax*, *value* SHALL be interpreted as above *valueRangeMax*.

981 If *valueState* is set to "outOfRange", *measurementConstraintsListData.valueRangeMin* is set and
982 *value* is equal or smaller than *valueRangeMin*, *value* SHALL be interpreted as below *valueRangeMin*.

983 If set, *measurementDescriptionListData.measurementType* SHALL be set, too.

984

985 **[Value state rules]:**

986 The Element *valueState* SHALL be set if its content differs from "normal" ([MOI-005]). This means, if
987 the state of the value is "outOfRange" or "error" this SHALL be denoted in the *valueState* Element. A
988 client SHALL always consider the content of *valueState*, if set. If omitted, a value of "normal" is
989 assumed.

990

991 **[String-length rules]:**

992 The string-length SHOULD NOT be longer than 256 characters. If it is longer, the sender SHALL
993 consider the possibility that the receiver will shorten the string to 256 characters.

994

995 **[Scaled number rules]:**

996 The sub-Elements "number" and "scale" represent a value according to the following formula:
997 $\text{number} * 10^{\text{scale}}$

998

999 **[Evaluation period rules]:**

1000 The "evaluationPeriod" expresses the analysis period for the according value.

1001

1002 **3.1.8.2 Rules regarding the usage of time-related information**

1003 Timestamps used within this Use Case SHALL be presented as absolute times. This means relative
1004 times SHALL NOT be used for timestamps.

1005 Durations used within this Use Case SHALL be presented as relative times.

1006 Note: This means the Actors need to have a common understanding of absolute times. It is strongly
1007 recommended that each Actor is "sufficiently synchronised" with the official "Coordinated Universal
1008 Time" (UTC).

1009

1010 **3.1.8.3 Applied sign convention**

1011 For the applied sign convention, please see section 2.6.1.

1012

3.2 Actors

3.2.1 Monitoring Appliance

3.2.1.1 Resource hierarchy

If Use Case discovery is supported (see section 3.1.2) this Actor SHALL be denoted as "MonitoringAppliance" in the Element "nodeManagementUseCaseData. useCaseInformation. actor".

The following diagram provides an overview of the Actor Monitoring Appliance's resource hierarchy.

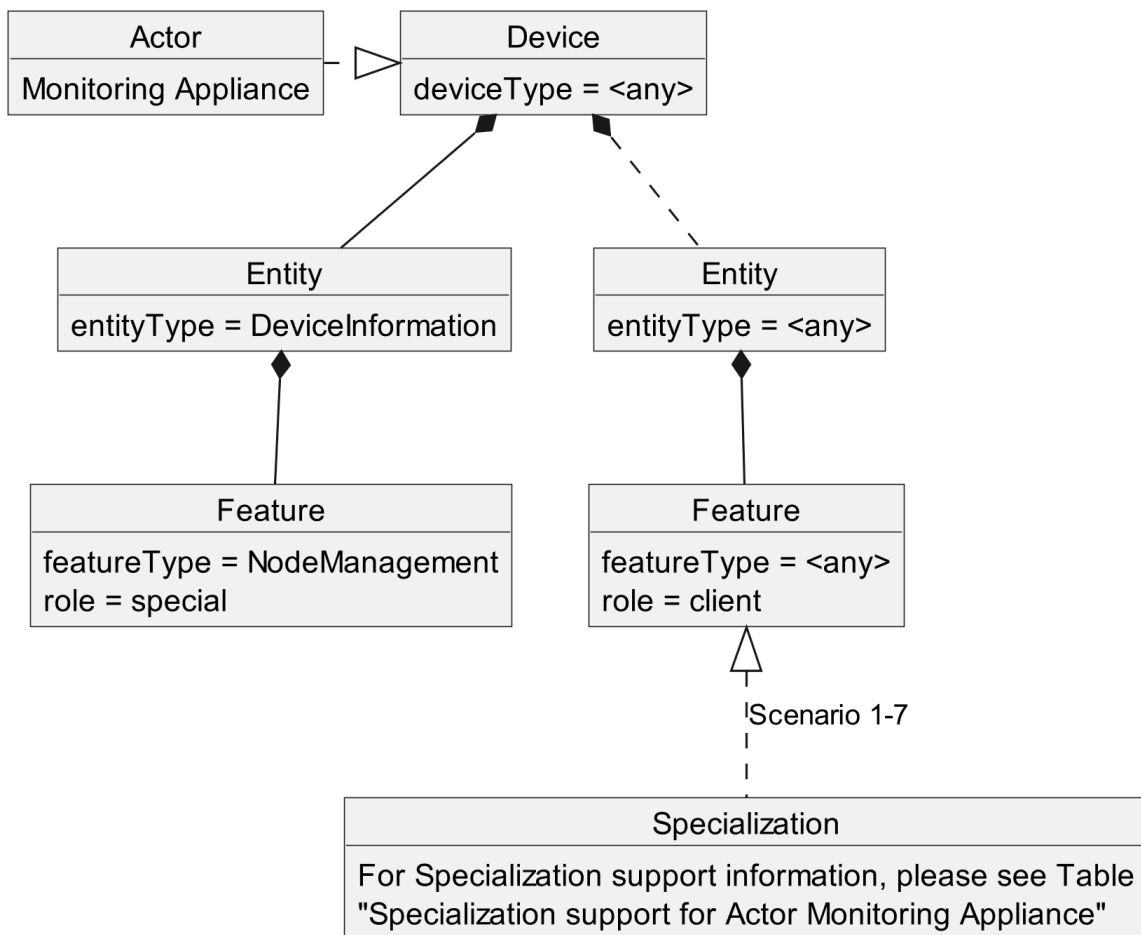


Figure 5: Actor "Monitoring Appliance" overview

The "Actor ... overview" diagram rules" section describes how to interpret the diagram above. See the "Specializations" section for more information regarding the Specializations given in the diagram above.

Note: The entityType "DeviceInformation" with the featureType "NodeManagement" is required by the SPINE protocol and therefore SHALL be supported. Both types are added in the figure for completeness but are not directly linked to the Use Case.

The Use Case specific data follow behind any entityType. The Specializations represent the Scenario specific data that must be supported for each Scenario and are realized through the corresponding featureTypes.

If a Specialization is connected to a Feature with the role "client", the Actor has a client role for this data. This means that the Actor accesses the data set described by the Specialization at a

corresponding server Feature. Further details are described in the sub-section "Client data - Specializations".

If a Specialization is connected to a Feature with the role "server", the Actor has the server role for this data. This means that the Actor must provide the corresponding data set of the Specialization as part of its Features. Further details are described in the sub-section "Server data - Resources".

Specialization name	Scenario	Described in table
DeviceClassification_ManufacturerData_MOI	1	Table 14
DeviceConfiguration_InverterGridCode	1	Table 15
DeviceConfiguration_PvStringAvailabilityStatus	2	Table 16
DeviceConfiguration_BatteryAvailabilityStatus	2	Table 17
DeviceConfiguration_GridConnectionStatus	2	Table 18
DeviceConfiguration_TimeToAcChargePowerMax	6	Table 19
DeviceConfiguration_TimeToAcDischargePowerMax	6	Table 20
DeviceDiagnosis_OperatingState	2	Table 21
StateInformation_InverterStateInformation	2	Table 22
Measurement_AcPowerApparentTotal	3	Table 23
Measurement_AcPowerApparentPhaseSpecific	3	Table 24
Measurement_AcPowerReactiveTotal	3	Table 25
Measurement_AcPowerReactivePhaseSpecific	3	Table 26
Measurement_AcYieldTotal	4	Table 27
Measurement_AcYieldDay	4	Table 28
Measurement_AcYieldDayHistorical	4	Table 29
Measurement_AcYieldMonth	4	Table 30
Measurement_AcYieldMonthHistorical	4	Table 31
Measurement_AcYieldYear	4	Table 32
Measurement_AcYieldYearHistorical	4	Table 33
Measurement_AcCosPhi	5	Table 34
Measurement_ComponentTemperature	7	Table 35
ElectricalConnection_EntityPowerProductionNominalMax	6	Table 36
ElectricalConnection_EntityApparentPowerProductionNominalMax	6	Table 37
ElectricalConnection_EntityPowerConsumptionNominalMax	6	Table 38
ElectricalConnection_EntityApparentPowerConsumptionNominalMax	6	Table 39
ElectricalConnection_BatteryPowerConsumptionNominalMin	6	Table 40
ElectricalConnection_BatteryPowerConsumptionNominalMax	6	Table 41
ElectricalConnection_BatteryPowerConsumptionMax	6	Table 42
ElectricalConnection_BatteryPowerProductionNominalMin	6	Table 43
ElectricalConnection_BatteryPowerProductionNominalMax	6	Table 44
ElectricalConnection_BatteryPowerProductionMax	6	Table 45

Table 13: Specialization support for Actor Monitoring Appliance

3.2.1.2 Server data - Resources

As this Actor has only client functionality, no resources are described within this section.

1042 **3.2.1.3 Client data - Specializations**

1043 3.2.1.3.1 Topic "DeviceClassification"

1044 3.2.1.3.1.1 Specialization "DeviceClassification_ManufacturerData_MOI"

Scenario [...]: M/R/O [W][C]	Element	Value	[High-Level Mapping] Element and value rules
1: M	DeviceClassification. deviceClassificationManufacturerData.		
1: M	deviceName		[MOI-011] [String-length rules]
1: M	deviceCode		[MOI-012] [String-length rules]
1: M	serialNumber		[MOI-013] [String-length rules]
1: M	softwareRevision		[MOI-014] [String-length rules]
1: M	hardwareRevision		[MOI-015] [String-length rules]
1: M	vendorName		[MOI-016a] [String-length rules]
1: O	vendorCode		[String-length rules]
1: M	brandName		[MOI-016b] [String-length rules]
1: O	manufacturerLabel		[MOI-017] [String-length rules]
1: O	manufacturerDescription		[MOI-018] The string-length SHOULD NOT be longer than 4096 characters

1045 Table 14: Content of Specialization "DeviceClassification_ManufacturerData_MOI" at Actor Monitoring Appliance

1046

1047 3.2.1.3.2 Topic "DeviceConfiguration"

1048 3.2.1.3.2.1 Specialization "DeviceConfiguration_InverterGridCode"

Scenario [...]: M/R/O [W][C]	Element	Value	[High-Level Mapping] Element and value rules
1: M	DeviceConfiguration. deviceConfigurationKeyValueDescriptionListData. deviceConfigurationKeyValueDescriptionData.		
1: M	keyId	<k1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
1: M	keyName	"inverterGridCode"	
1: M	valueType	"string"	

1: M	DeviceConfiguration. deviceConfigurationKeyValueListData. deviceConfigurationKeyValueData.		
1: M	keyId	<k1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
1: M	value.		[MOI-019] Exactly one of the child Elements SHALL be set. This SHALL match with the content of <i>valueType</i> Element within the key value description part (see above).
1: M	value. string		The format SHALL be: "[country;]technical connection rule[;year][;ancillary]" Items in brackets are optional. Examples: - "AT;TORD4;2016;LV" - "DE;VDE-AR-N4105;2019" - "SE;EN50438;2013" - "UK;G98/1-3" - "VDE0126-1-1"

Table 15: Content of Specialization "DeviceConfiguration_InverterGridCode" at Actor Monitoring Appliance

3.2.1.3.2.2 Specialization "DeviceConfiguration_PvStringAvailabilityStatus"

Scenario {...}: M/R/O [W][C]	Element	Value	[High-Level Mapping] Element and value rules
2: M	DeviceConfiguration. deviceConfigurationKeyValueDescriptionListData. deviceConfigurationKeyValueDescriptionData.		
2: M	keyId	<k2#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M	keyName	"pvStringAvailabilityStatus"	
2: M	valueType	"string"	
2: M	DeviceConfiguration. deviceConfigurationKeyValueListData. deviceConfigurationKeyValueData.		
2: M	keyId	<k2#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M	value.		[MOI-023] Exactly one of the child Elements SHALL be set. This SHALL match with the content of <i>valueType</i> Element within the key value description part (see above).
2: M	value. string	"unknown"	[MOI-023/1]
		"disconnected"	[MOI-023/2]
		"connected"	[MOI-023/3]
		"available"	[MOI-023/4]
		"inOperation"	[MOI-023/5]

Table 16: Content of Specialization "DeviceConfiguration_PvStringAvailabilityStatus" at Actor Monitoring Appliance

1054 3.2.1.3.2.3 Specialization "DeviceConfiguration_BatteryAvailabilityStatus"

Scenario [...]: M/R/O [W][\C]	Element	Value	[High-Level Mapping] Element and value rules
2: M	DeviceConfiguration. deviceConfigurationKeyValueDescriptionListData. deviceConfigurationKeyValueDescriptionData.		
2: M	keyId	<k3#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M	keyName	"batteryAvailabilityStatus"	
2: M	valueType	"string"	
2: M	DeviceConfiguration. deviceConfigurationKeyValueListData. deviceConfigurationKeyValueData.		
2: M	keyId	<k3#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M	value.		[MOI-024] Exactly one of the child Elements SHALL be set. This SHALL match with the content of <i>valueType</i> Element within the key value description part (see above).
2: M	value. string	"unknown"	[MOI-024/1]
		"disconnected"	[MOI-024/2]
		"connected"	[MOI-024/3]
		"available"	[MOI-024/4]
		"inOperation"	[MOI-024/5]

1055 Table 17: Content of Specialization "DeviceConfiguration_BatteryAvailabilityStatus" at Actor Monitoring Appliance

1056

1057 3.2.1.3.2.4 Specialization "DeviceConfiguration_GridConnectionStatus"

Scenario [...]: M/R/O [W][\C]	Element	Value	[High-Level Mapping] Element and value rules
2: M	DeviceConfiguration. deviceConfigurationKeyValueDescriptionListData. deviceConfigurationKeyValueDescriptionData.		
2: M	keyId	<k4#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M	keyName	"gridConnectionStatus"	
2: M	valueType	"string"	
2: M	DeviceConfiguration. deviceConfigurationKeyValueListData. deviceConfigurationKeyValueData.		
2: M	keyId	<k4#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M	value.		[MOI-025] Exactly one of the child Elements SHALL be set. This SHALL match with the content of <i>valueType</i> Element within the key value description part (see above).
2: M	value. string	"unknown"	[MOI-025/1]
		"disconnected"	[MOI-025/2]

		"connected"	[MOI-025/3]
--	--	-------------	-------------

Table 18: Content of Specialization "DeviceConfiguration_GridConnectionStatus" at Actor Monitoring Appliance

3.2.1.3.2.5 Specialization "DeviceConfiguration_TimeToAcChargePowerMax"

Scenario [{...}]: M/R/O [\W][\C]	Element	Value	[High-Level Mapping] Element and value rules
6: M	DeviceConfiguration. deviceConfigurationKeyValueDescriptionListData. deviceConfigurationKeyValueDescriptionData.		
6: M	keyId	<k5#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
6: M	keyName	"timeToAcChargePowerMax"	
6: M	valueType	"scaledNumber"	
6: M	unit	"s"	The unit SHALL be applied to the value of the key.
6: M	DeviceConfiguration. deviceConfigurationKeyValueListData. deviceConfigurationKeyValueData.		
6: M	keyId	<k5#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
6: M	value.		[MOI-067a] Exactly one of the child Elements SHALL be set. This SHALL match with the content of <i>valueType</i> Element within the key value description part (see above).
6: M	value. scaledNumber.		SHALL be used. [Scaled number rules]
6: M	value. scaledNumber. number		SHALL be used.
6: M	value. scaledNumber. scale		SHALL be interpreted. If absent, a default value of "0" applies.

Table 19: Content of Specialization "DeviceConfiguration_TimeToAcChargePowerMax" at Actor Monitoring Appliance

3.2.1.3.2.6 Specialization "DeviceConfiguration_TimeToAcDischargePowerMax"

Scenario [{...}]: M/R/O [\W][\C]	Element	Value	[High-Level Mapping] Element and value rules
6: M	DeviceConfiguration. deviceConfigurationKeyValueDescriptionListData. deviceConfigurationKeyValueDescriptionData.		
6: M	keyId	<k6#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.

6: M	keyName	"timeToAcDischargePowerMax"	
6: M	valueType	"scaledNumber"	
6: M	unit	"s"	The unit SHALL be applied to the value of the key.
6: M	DeviceConfiguration. deviceConfigurationKeyValueListData. deviceConfigurationKeyValueData.		
6: M	keyId	<k6#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
6: M	value.		[MOI-067b] Exactly one of the child Elements SHALL be set. This SHALL match with the content of <i>valueType</i> Element within the key value description part (see above).
6: M	value. scaledNumber.		SHALL be used. The sub-Elements "number" and "scale" represent a value according to the following formula: $\text{number} * 10^{\text{scale}}$
6: M	value. scaledNumber. number		SHALL be used.
6: M	value. scaledNumber. scale		SHALL be interpreted. If absent, a default value of "0" applies.

Table 20: Content of Specialization "DeviceConfiguration_TimeToAcDischargePowerMax" at Actor Monitoring Appliance

3.2.1.3.3 Topic "DeviceDiagnosis"

3.2.1.3.3.1 Specialization "DeviceDiagnosis_OperatingState"

Scenario [{...}]: M/R/O [W][C]	Element	Value	[High-Level Mapping] Element and value rules
2: M	DeviceDiagnosis. deviceDiagnosisStateData.		
2: O	timestamp		
2: M	operatingState	"normalOperation" [MOI-021/1]	[MOI-021]
		"standby" [MOI-021/2]	
		"failure" [MOI-021/5]	
		"temporarilyNotReady" [MOI-021/3]	
		"off" [MOI-021/4]	
2: R	vendorStateCode		[MOI-022]

Table 21: Content of Specialization "DeviceDiagnosis_OperatingState" at Actor Monitoring Appliance

1070 3.2.1.3.4 Topic "StateInformation"

1071 3.2.1.3.4.1 Specialization "StateInformation_InverterStateInformation"

Scenario [{...}]: M/R/O [W][C]	Element	Value	[High-Level Mapping] Element and value rules
2: M	StateInformation. stateInformationListData. stateInformationData.		
2: M	stateInformationId	<si1#{1..31}>	SHALL be set as PRIMARY IDENTIFIER.
2: M	stateInformation	"externalOverrideFromGrid" (-><si1#1>) [MOI-026/1/1]	[MOI-026]
		"autonomousGridSupport" (-><si1#2>) [MOI-026/1/2]	[MOI-026]
		"islandingMode" (-><si1#3>) [MOI-026/1/3]	[MOI-026]
		"balancing" (-><si1#4>) [MOI-026/1/4]	[MOI-026]
		"trickleCharging" (-><si1#5>) [MOI-026/1/5]	[MOI-026]
		"calibration" (-><si1#6>) [MOI-026/1/6]	[MOI-026]
		"commissioningMissing" (-><si1#7>) [MOI-026/1/7]	[MOI-026]
		"sleeping" (-><si1#8>) [MOI-026/1/8]	[MOI-026]
		"starting" (-><si1#9>) [MOI-026/1/9]	[MOI-026]
		"mppt" (-><si1#10>) [MOI-026/1/10]	[MOI-026]
		"throttled" (-><si1#11>) [MOI-026/1/11]	[MOI-026]
		"shuttingDown" (-><si1#12>) [MOI-026/1/12]	[MOI-026]
		"manualShutdown" (-><si1#13>) [MOI-026/1/13]	[MOI-026]
		"inverterDefective" (-><si1#14>) [MOI-026/2/1]	[MOI-026]
		"batteryOvercurrentProtection" (-><si1#15>) [MOI-026/2/2]	[MOI-026]
		"pvStringOvercurrentProtection" (-><si1#16>) [MOI-026/2/3]	[MOI-026]
		"gridFault" (-><si1#17>) [MOI-026/2/4]	[MOI-026]
		"groundFault" (-><si1#18>) [MOI-026/2/5]	[MOI-026]
		"acDisconnected" (-><si1#19>) [MOI-026/2/6]	[MOI-026]
		"dcDisconnected" (-><si1#20>)	[MOI-026]

		[MOI-026/2/7]	
		"cabinetOpen" (-><si1#21>) [MOI-026/2/8]	[MOI-026]
		"overTemperature" (-><si1#22>) [MOI-026/2/9]	[MOI-026]
		"underTemperature" (-><si1#23>) [MOI-026/2/10]	[MOI-026]
		"frequencyAboveLimit" (-><si1#24>) [MOI-026/2/11]	[MOI-026]
		"frequencyBelowLimit" (-><si1#25>) [MOI-026/2/12]	[MOI-026]
		"acVoltageAboveLimit" (-><si1#26>) [MOI-026/2/13]	[MOI-026]
		"acVoltageBelowLimit" (-><si1#27>) [MOI-026/2/14]	[MOI-026]
		"dcVoltageAboveLimit" (-><si1#28>) [MOI-026/2/15]	[MOI-026]
		"dcVoltageBelowLimit" (-><si1#29>) [MOI-026/2/16]	[MOI-026]
		"hardwareTestFailure" (-><si1#30>) [MOI-026/2/17]	[MOI-026]
		"genericInternalError" (-><si1#31>) [MOI-026/2/18]	[MOI-026]
2: M	isActive	"true" "false"	
2: M	category	"functionality" (-><si1#1-13>) [MOI-026/1]	
		"failure" (-><si1#14-31>) [MOI-026/2]	
2: O	timeOfLastChange		

Table 22: Content of Specialization "StateInformation_InverterStateInformation" at Actor Monitoring Appliance

3.2.1.3.5 Topic "Measurement"

3.2.1.3.5.1 Specialization "Measurement_AcPowerApparentTotal"

Scenario [{...}]: M/R/O [W][V]	Element	Value	[High-Level Mapping] Element and value rules
3: M	Measurement. measurementDescriptionListData. measurementDescriptionData.		
3: M	measurementId	<m1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
3: M	measurementType	"power"	
3: M	commodityType	"electricity"	
3: M	unit	"VA"	
3: M	scopeType	"acPowerApparent Total"	
3: R	Measurement. measurementConstraintsListData. measurementConstraintsData.		
3: M	measurementId	<m1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.

3: R	valueRangeMin.		[MOI-002/1] SHOULD be used. [Scaled number rules]
3: M	valueRangeMin. number		SHALL be used.
3: M	valueRangeMin. scale		SHALL be interpreted. If absent, a default value of "0" applies.
3: R	valueRangeMax.		[MOI-002/1] SHOULD be used. [Scaled number rules]
3: M	valueRangeMax. number		SHALL be used.
3: M	valueRangeMax. scale		SHALL be interpreted. If absent, a default value of "0" applies.
3: R	valueStepSize.		SHOULD be used. [Scaled number rules]
3: M	valueStepSize. number		SHALL be used.
3: M	valueStepSize. scale		SHALL be interpreted. If absent, a default value of "0" applies.
3: M	Measurement. measurementListData. measurementData.		
3: M	measurementId	<m1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
3: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
3: O	timestamp	<t#(1..1)->m1#1>	MAY be used as SUB IDENTIFIER. Only the newest measurement value SHALL be stated. Additional historical values are forbidden.
3: M	value.		[MOI-002/1], [MOI-031] [Measurement value rules] [Scaled number rules]
3: M	value. number		SHALL be used.
3: M	value. scale		SHALL be interpreted. If absent, a default value of "0" applies.
3: R	valueSource	"measuredValue"	
		"calculatedValue"	
		"empiricalValue"	
3: M	valueState		[MOI-005] [Value state rules]
3: M	ElectricalConnection. electricalConnectionDescriptionListData. electricalConnectionDescriptionData.		
3: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
3: M	powerSupplyType	"ac"	
3: M	positiveEnergyDirection	"consume"	
3: M	ElectricalConnection. electricalConnectionParameterDescriptionListData. electricalConnectionParameterDescriptionData.		
3: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
3: M	parameterId	<p1#(1..1)->ec1#1>	SHALL be set as SUB IDENTIFIER.
3: M	measurementId	<m1#(1..1)->p1#1>	SHALL be set as FOREIGN IDENTIFIER.
3: M	voltageType	"ac"	

3: M	acMeasuredPhases	"abc" "ab" "bc" "ac" "a" "b" "c"	If the Inverter is connected to less than three phases, one of the other combinations like "a" or "ab" are allowed instead of "abc". The values "a", "b", and "c" are permitted if and only if only one phase is measured by the server.
3: M	acMeasuredInReference To	"neutral"	
3: M	acMeasurementType	"apparent"	
3: M	acMeasurementVariant	"rms"	

Table 23: Content of Specialization "Measurement_AcPowerApparentTotal" at Actor Monitoring Appliance

3.2.1.3.5.2 Specialization "Measurement_AcPowerApparentPhaseSpecific"

Scenario [{...}]: M/R/O [W][\C]	Element	Value	[High-Level Mapping] Element and value rules
3: M	Measurement. measurementDescriptionListData. measurementDescriptionData.		
3: M	measurementId	<m2#{1..3}>	SHALL be set as PRIMARY IDENTIFIER.
3: M	measurementType	"power"	
3: M	commodityType	"electricity"	
3: M	unit	"VA"	
3: M	scopeType	"acPowerApparent"	
3: R	Measurement. measurementConstraintsListData. measurementConstraintsData.		
3: M	measurementId	<m2#{1..3}>	SHALL be set as PRIMARY IDENTIFIER.
3: R	valueRangeMin.		[MOI-002/1] SHOULD be used. [Scaled number rules]
3: M	valueRangeMin. number		SHALL be used.
3: M	valueRangeMin. scale		SHALL be interpreted. If absent, a default value of "0" applies.
3: R	valueRangeMax.		[MOI-002/1] SHOULD be used. [Scaled number rules]
3: M	valueRangeMax. number		SHALL be used.
3: M	valueRangeMax. scale		SHALL be interpreted. If absent, a default value of "0" applies.
3: R	valueStepSize.		SHOULD be used. [Scaled number rules]
3: M	valueStepSize. number		SHALL be used.

3: M	valueStepSize. scale		SHALL be interpreted. If absent, a default value of "0" applies.
3: M	Measurement. measurementListData. measurementData.		
3: M	measurementId	<m2#{1..3}>	SHALL be set as PRIMARY IDENTIFIER.
3: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
3: O	timestamp	<t#{1..1}->m2#{1..3}>	MAY be used as SUB IDENTIFIER. Only the newest measurement value SHALL be stated. Additional historical values are forbidden.
3: M	value.		[MOI-002/1], [MOI-032] [Measurement value rules] [Scaled number rules]
3: M	value. number		SHALL be used.
3: M	value. scale		SHALL be interpreted. If absent, a default value of "0" applies.
3: R	valueSource	"measuredValue"	
		"calculatedValue"	
		"empiricalValue"	
3: M	valueState		[MOI-005] [Value state rules]
3: M	ElectricalConnection. electricalConnectionDescriptionListData. electricalConnectionDescriptionData.		
3: M	electricalConnectionId	<ec1#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
3: M	powerSupplyType	"ac"	
3: M	positiveEnergyDirection	"consume"	
3: M	ElectricalConnection. electricalConnectionParameterDescriptionListData. electricalConnectionParameterDescriptionData.		
3: M	electricalConnectionId	<ec1#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
3: M	parameterId	<p2#{1..3}->ec1#1>	SHALL be set as SUB IDENTIFIER.
3: M	measurementId	<m2#{1..3}->p2#{1..3}>	SHALL be set as FOREIGN IDENTIFIER.
3: M	voltageType	"ac"	
3: M	acMeasuredPhases	"a" (->p2#1)	[MOI-032/1]
		"b" (->p2#2)	[MOI-032/2]
		"c" (->p2#3)	[MOI-032/3]
3: M	acMeasuredInReference To	"neutral"	
3: M	acMeasurementType	"apparent"	
3: M	acMeasurementVariant	"rms"	

Table 24: Content of Specialization "Measurement_AcPowerApparentPhaseSpecific" at Actor Monitoring Appliance

1081 3.2.1.3.5.3 Specialization "Measurement_AcPowerReactiveTotal"

Scenario [{...}]: M/R/O [W][C]	Element	Value	[High-Level Mapping] Element and value rules
3: M	Measurement. measurementDescriptionListData. measurementDescriptionData.		
3: M	measurementId	<m3#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
3: M	measurementType	"power"	
3: M	commodityType	"electricity"	
3: M	unit	"var"	
3: M	scopeType	"acPowerReactiveTotal"	
3: R	Measurement. measurementConstraintsListData. measurementConstraintsData.		
3: M	measurementId	<m3#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
3: R	valueRangeMin.		[MOI-002/2] SHOULD be used. [Scaled number rules]
3: M	valueRangeMin. number		SHALL be used.
3: M	valueRangeMin. scale		SHALL be interpreted. If absent, a default value of "0" applies.
3: R	valueRangeMax.		[MOI-002/2] SHOULD be used. [Scaled number rules]
3: M	valueRangeMax. number		SHALL be used.
3: M	valueRangeMax. scale		SHALL be interpreted. If absent, a default value of "0" applies.
3: R	valueStepSize.		SHOULD be used. [Scaled number rules]
3: M	valueStepSize. number		SHALL be used.
3: M	valueStepSize. scale		SHALL be interpreted. If absent, a default value of "0" applies.
3: M	Measurement. measurementListData. measurementData.		
3: M	measurementId	<m3#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
3: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
3: O	timestamp	<t#(1..1)->m3#1>	MAY be used as SUB IDENTIFIER. Only the newest measurement value SHALL be stated. Additional historical values are forbidden.
3: M	value.		[MOI-002/2], [MOI-033] [Measurement value rules] [Scaled number rules]
3: M	value. number		SHALL be used.
3: M	value. scale		SHALL be interpreted. If absent, a default value of "0" applies.
3: R	valueSource	"measuredValue"	

		"calculatedValue"	
		"empiricalValue"	
3: M	valueState		[MOI-005] [Value state rules]
3: M	ElectricalConnection. electricalConnectionDescriptionListData. electricalConnectionDescriptionData.		
3: M	electricalConnectionId	<ec1#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
3: M	powerSupplyType	"ac"	
3: M	positiveEnergyDirection	"consume"	
3: M	ElectricalConnection. electricalConnectionParameterDescriptionListData. electricalConnectionParameterDescriptionData.		
3: M	electricalConnectionId	<ec1#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
3: M	parameterId	<p3#{1..1}->ec1#1>	SHALL be set as SUB IDENTIFIER.
3: M	measurementId	<m3#{1..1}->p3#{1..1}>	SHALL be set as FOREIGN IDENTIFIER.
3: M	voltageType	"ac"	
3: M	acMeasuredPhases	"abc" "ab" "bc" "ac" "a" "b" "c" (->p5#1)	If the Inverter is connected to less than three phases, one of the other combinations like "a" or "ab" are allowed instead of "abc". The values "a", "b", and "c" are permitted if and only if only one phase is measured by the server.
3: M	acMeasuredInReference To	"neutral"	
3: M	acMeasurementType	"reactive"	
3: M	acMeasurementVariant	"rms"	

Table 25: Content of Specialization "Measurement_AcPowerReactiveTotal" at Actor Monitoring Appliance

3.2.1.3.5.4 Specialization "Measurement_AcPowerReactivePhaseSpecific"

Scenario [{...}]: M/R/O [W][C]	Element	Value	[High-Level Mapping] Element and value rules
3: M	Measurement. measurementDescriptionListData. measurementDescriptionData.		
3: M	measurementId	<m4#{1..3}>	SHALL be set as PRIMARY IDENTIFIER.
3: M	measurementType	"power"	
3: M	commodityType	"electricity"	
3: M	unit	"var"	
3: M	scopeType	"acPowerReactive"	

3: R	Measurement. measurementConstraintsListData. measurementConstraintsData.		
3: M	measurementId	<m4#(1..3)>	SHALL be set as PRIMARY IDENTIFIER.
3: R	valueRangeMin.		[MOI-002/2] SHOULD be used. [Scaled number rules]
3: M	valueRangeMin. number		SHALL be used.
3: M	valueRangeMin. scale		SHALL be interpreted. If absent, a default value of "0" applies.
3: R	valueRangeMax.		[MOI-002/2] SHOULD be used. [Scaled number rules]
3: M	valueRangeMax. number		SHALL be used.
3: M	valueRangeMax. scale		SHALL be interpreted. If absent, a default value of "0" applies.
3: R	valueStepSize.		SHOULD be used. [Scaled number rules]
3: M	valueStepSize. number		SHALL be used.
3: M	valueStepSize. scale		SHALL be interpreted. If absent, a default value of "0" applies.
3: M	Measurement. measurementListData. measurementData.		
3: M	measurementId	<m4#(1..3)>	SHALL be set as PRIMARY IDENTIFIER.
3: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
3: O	timestamp	<t#(1..1)->m4#(1..3)>	MAY be used as SUB IDENTIFIER. Only the newest measurement value SHALL be stated. Additional historical values are forbidden.
3: M	value.		[MOI-002/2], [MOI-034] [Measurement value rules] [Scaled number rules]
3: M	value. number		SHALL be used.
3: M	value. scale		SHALL be interpreted. If absent, a default value of "0" applies.
3: R	valueSource	"measuredValue"	
		"calculatedValue"	
		"empiricalValue"	
3: M	valueState		[MOI-005] [Value state rules]
3: M	ElectricalConnection. electricalConnectionDescriptionListData. electricalConnectionDescriptionData.		
3: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
3: M	powerSupplyType	"ac"	
3: M	positiveEnergyDirection	"consume"	
3: M	ElectricalConnection. electricalConnectionParameterDescriptionListData. electricalConnectionParameterDescriptionData.		
3: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
3: M	parameterId	<p4#(1..3)->ec1#1>	SHALL be set as SUB IDENTIFIER.

3: M	measurementId	<m4#(1..3)->p4#(1..3)>	SHALL be set as FOREIGN IDENTIFIER.
3: M	voltageType	"ac"	
3: M	acMeasuredPhases	"a" (->p4#1)	[MOI-034/1]
		"b" (->p4#2)	[MOI-034/2]
		"c" (->p4#3)	[MOI-034/3]
3: M	acMeasuredInReference To	"neutral"	
3: M	acMeasurementType	"reactive"	
3: M	acMeasurementVariant	"rms"	

Table 26: Content of Specialization "Measurement_AcPowerReactivePhaseSpecific" at Actor Monitoring Appliance

3.2.1.3.5.5 Specialization "Measurement_AcYieldTotal"

Scenario [{...}]: M/R/O [W][C]	Element	Value	[High-Level Mapping] Element and value rules
4: M	Measurement. measurementDescriptionListData. measurementDescriptionData.		
4: M	measurementId	<m5#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
4: M	measurementType	"energy"	
4: M	commodityType	"electricity"	
4: M	unit	"Wh"	
4: M	scopeType	"acYieldTotal"	
4: R	Measurement. measurementConstraintsListData. measurementConstraintsData.		
4: M	measurementId	<m5#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
4: R	valueRangeMin.		[MOI-001] SHOULD be used. [Scaled number rules]
4: M	valueRangeMin. number		SHALL be used.
4: M	valueRangeMin. scale		SHALL be interpreted. If absent, a default value of "0" applies.
4: R	valueRangeMax.		[MOI-001] SHOULD be used. [Scaled number rules]
4: M	valueRangeMax. number		SHALL be used.
4: M	valueRangeMax. scale		SHALL be interpreted. If absent, a default value of "0" applies.
4: R	valueStepSize.		SHOULD be used. [Scaled number rules]
4: M	valueStepSize. number		SHALL be used.
4: M	valueStepSize. scale		SHALL be interpreted. If absent, a default value of "0" applies.
4: M	Measurement. measurementListData. measurementData.		
4: M	measurementId	<m5#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
4: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
4: O	timestamp	<t#(1..1)->m5#1>	MAY be used as SUB IDENTIFIER. Only the newest measurement value SHALL

			be stated. Additional historical values are forbidden.
4: M	value.		[MOI-001], [MOI-041] [Measurement value rules] [Scaled number rules]
4: M	value. number		SHALL be used.
4: M	value. scale		SHALL be interpreted. If absent, a default value of "0" applies.
4: R	evaluationPeriod.		[Evaluation period rules]
4: M	evaluationPeriod. startTime		
4: M	evaluationPeriod. endTime		
4: R	valueSource	"measuredValue"	
		"calculatedValue"	
		"empiricalValue"	
4: M	valueState		[MOI-005] [Value state rules]
4: M	ElectricalConnection. electricalConnectionDescriptionListData. electricalConnectionDescriptionData.		
4: M	electricalConnectionId	<ec1#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
4: M	powerSupplyType	"ac"	
4: M	positiveEnergyDirection	"consume"	
4: M	ElectricalConnection. electricalConnectionParameterDescriptionListData. electricalConnectionParameterDescriptionData.		
4: M	electricalConnectionId	<ec1#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
4: M	parameterId	<p5#{1..1}->ec1#1>	SHALL be set as SUB IDENTIFIER.
4: M	measurementId	<m5#{1..1}->p5#1>	SHALL be set as FOREIGN IDENTIFIER.
4: M	voltageType	"ac"	
4: M	acMeasurementType	"real"	

Table 27: Content of Specialization "Measurement_AcYieldTotal" at Actor Monitoring Appliance

3.2.1.3.5.6 Specialization "Measurement_AcYieldDay"

Scenario {...}: M/R/O [W]/[C]	Element	Value	[High-Level Mapping] Element and value rules
4: M	Measurement. measurementDescriptionListData. measurementDescriptionData.		
4: M	measurementId	<m6#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
4: M	measurementType	"energy"	
4: M	commodityType	"electricity"	
4: M	unit	"Wh"	
4: M	scopeType	"acYieldDay"	
4: R	Measurement. measurementConstraintsListData. measurementConstraintsData.		
4: M	measurementId	<m6#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
4: R	valueRangeMin.		[MOI-001] SHOULD be used.

			[Scaled number rules]
4: M	valueRangeMin. number		SHALL be used.
4: M	valueRangeMin. scale		SHALL be interpreted. If absent, a default value of "0" applies.
4: R	valueRangeMax.		[MOI-001] SHOULD be used. [Scaled number rules]
4: M	valueRangeMax. number		SHALL be used.
4: M	valueRangeMax. scale		SHALL be interpreted. If absent, a default value of "0" applies.
4: R	valueStepSize.		SHOULD be used. [Scaled number rules]
4: M	valueStepSize. number		SHALL be used.
4: M	valueStepSize. scale		SHALL be interpreted. If absent, a default value of "0" applies.
4: M	Measurement. measurementListData. measurementData.		
4: M	measurementId	<m6#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
4: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
4: O	timestamp	<t#(1..1)->m6#1>	MAY be used as SUB IDENTIFIER. Only the newest measurement value SHALL be stated. Additional historical values are forbidden.
4: M	value.		[MOI-001], [MOI-042a] [Measurement value rules] [Scaled number rules]
4: M	value. number		SHALL be used.
4: M	value. scale		SHALL be interpreted. If absent, a default value of "0" applies.
4: R	evaluationPeriod.		[Evaluation period rules]
4: M	evaluationPeriod. startTime		
4: M	evaluationPeriod. endTime		
4: R	valueSource	"measuredValue"	
		"calculatedValue"	
		"empiricalValue"	
4: M	valueState		[MOI-005] [Value state rules]
4: M	ElectricalConnection. electricalConnectionDescriptionListData. electricalConnectionDescriptionData.		
4: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
4: M	powerSupplyType	"ac"	
4: M	positiveEnergyDirection	"consume"	
4: M	ElectricalConnection. electricalConnectionParameterDescriptionListData. electricalConnectionParameterDescriptionData.		
4: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
4: M	parameterId	<p6#(1..1)->ec1#1>	SHALL be set as SUB IDENTIFIER.
4: M	measurementId	<m6#(1..1)->p6#1>	SHALL be set as FOREIGN IDENTIFIER.
4: M	voltageType	"ac"	

4: M	acMeasurementType	"real"	
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Table 28: Content of Specialization "Measurement_AcYieldDay" at Actor Monitoring Appliance

3.2.1.3.5.7 Specialization "Measurement_AcYieldDayHistorical"

Scenario [...]: M/R/O [W][C]	Element	Value	[High-Level Mapping] Element and value rules
4: M	Measurement. measurementDescriptionListData. measurementDescriptionData.		
4: M	measurementId	<m6#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
4: M	measurementType	"energy"	
4: M	commodityType	"electricity"	
4: M	unit	"Wh"	
4: M	scopeType	"acYieldDay"	
4: R	Measurement. measurementConstraintsListData. measurementConstraintsData.		
4: M	measurementId	<m6#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
4: R	valueRangeMin.		[MOI-001] SHOULD be used. [Scaled number rules]
4: M	valueRangeMin. number		SHALL be used.
4: M	valueRangeMin. scale		SHALL be interpreted. If absent, a default value of "0" applies.
4: R	valueRangeMax.		[MOI-001] SHOULD be used. [Scaled number rules]
4: M	valueRangeMax. number		SHALL be used.
4: M	valueRangeMax. scale		SHALL be interpreted. If absent, a default value of "0" applies.
4: R	valueStepSize.		SHOULD be used. [Scaled number rules]
4: M	valueStepSize. number		SHALL be used.
4: M	valueStepSize. scale		SHALL be interpreted. If absent, a default value of "0" applies.
4: M	Measurement. measurementSeriesListData. measurementSeriesData.		
4: M	measurementId	<m6#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
4: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
4: M	timestamp	<t#(1..)->m6#1>	SHALL be set as SUB IDENTIFIER.
4: M	value.		[MOI-001], [MOI-042b] [Measurement value rules] [Scaled number rules]
4: M	value. number		SHALL be used.
4: M	value. scale		SHALL be interpreted. If absent, a default value of "0" applies.
4: R	evaluationPeriod.		[Evaluation period rules]
4: M	evaluationPeriod. startTime		

4: M	evaluationPeriod. endTime		
4: R	valueSource	"measuredValue"	
		"calculatedValue"	
		"empiricalValue"	
4: M	valueState		[MOI-005] [Value state rules]
4: M	ElectricalConnection. electricalConnectionDescriptionListData. electricalConnectionDescriptionData.		
4: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
4: M	powerSupplyType	"ac"	
4: M	positiveEnergyDirection	"consume"	
4: M	ElectricalConnection. electricalConnectionParameterDescriptionListData. electricalConnectionParameterDescriptionData.		
4: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
4: M	parameterId	<p6#(1..1)->ec1#1>	SHALL be set as SUB IDENTIFIER.
4: M	measurementId	<m6#(1..1)->p6#1>	SHALL be set as FOREIGN IDENTIFIER.
4: M	voltageType	"ac"	
4: M	acMeasurementType	"real"	

Table 29: Content of Specialization "Measurement_AcYieldDayHistorical" at Actor Monitoring Appliance

3.2.1.3.5.8 Specialization "Measurement_AcYieldMonth"

Scenario [...]: M/R/O [W]/[C]	Element	Value	[High-Level Mapping] Element and value rules
4: M	Measurement. measurementDescriptionListData. measurementDescriptionData.		
4: M	measurementId	<m7#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
4: M	measurementType	"energy"	
4: M	commodityType	"electricity"	
4: M	unit	"Wh"	
4: M	scopeType	"acYieldMonth"	
4: R	Measurement. measurementConstraintsListData. measurementConstraintsData.		
4: M	measurementId	<m7#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
4: R	valueRangeMin.		[MOI-001] SHOULD be used. [Scaled number rules]
4: M	valueRangeMin. number		SHALL be used.
4: M	valueRangeMin. scale		SHALL be interpreted. If absent, a default value of "0" applies.
4: R	valueRangeMax.		[MOI-001] SHOULD be used. [Scaled number rules]
4: M	valueRangeMax. number		SHALL be used.

4: M	valueRangeMax. scale		SHALL be interpreted. If absent, a default value of "0" applies.
4: R	valueStepSize.		SHOULD be used. [Scaled number rules]
4: M	valueStepSize. number		SHALL be used.
4: M	valueStepSize. scale		SHALL be interpreted. If absent, a default value of "0" applies.
4: M	Measurement. measurementListData. measurementData.		
4: M	measurementId	<m7#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
4: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
4: O	timestamp	<t#(1..1)->m7#1>	MAY be used as SUB IDENTIFIER. Only the newest measurement value SHALL be stated. Additional historical values are forbidden.
4: M	value.		[MOI-001], [MOI-043a] [Measurement value rules] [Scaled number rules]
4: M	value. number		SHALL be used.
4: M	value. scale		SHALL be interpreted. If absent, a default value of "0" applies.
4: R	evaluationPeriod.		[Evaluation period rules]
4: M	evaluationPeriod. startTime		
4: M	evaluationPeriod. endTime		
4: R	valueSource	"measuredValue"	
		"calculatedValue"	
		"empiricalValue"	
4: M	valueState		[MOI-005] [Value state rules]
4: M	ElectricalConnection. electricalConnectionDescriptionListData. electricalConnectionDescriptionData.		
4: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
4: M	powerSupplyType	"ac"	
4: M	positiveEnergyDirection	"consume"	
4: M	ElectricalConnection. electricalConnectionParameterDescriptionListData. electricalConnectionParameterDescriptionData.		
4: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
4: M	parameterId	<p7#(1..1)->ec1#1>	SHALL be set as SUB IDENTIFIER.
4: M	measurementId	<m7#(1..1)->p7#1>	SHALL be set as FOREIGN IDENTIFIER.
4: M	voltageType	"ac"	
4: M	acMeasurementType	"real"	

Table 30: Content of Specialization "Measurement_AcYieldMonth" at Actor Monitoring Appliance

1099 3.2.1.3.5.9 Specialization "Measurement_AcYieldMonthHistorical"

Scenario [{...}]: M/R/O [W][C]	Element	Value	[High-Level Mapping] Element and value rules
4: M	Measurement. measurementDescriptionListData. measurementDescriptionData.		
4: M	measurementId	<m7#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
4: M	measurementType	"energy"	
4: M	commodityType	"electricity"	
4: M	unit	"Wh"	
4: M	scopeType	"acYieldMonth"	
4: R	Measurement. measurementConstraintsListData. measurementConstraintsData.		
4: M	measurementId	<m7#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
4: R	valueRangeMin.		[MOI-001] SHOULD be used. [Scaled number rules]
4: M	valueRangeMin. number		SHALL be used.
4: M	valueRangeMin. scale		SHALL be interpreted. If absent, a default value of "0" applies.
4: R	valueRangeMax.		[MOI-001] SHOULD be used. [Scaled number rules]
4: M	valueRangeMax. number		SHALL be used.
4: M	valueRangeMax. scale		SHALL be interpreted. If absent, a default value of "0" applies.
4: R	valueStepSize.		SHOULD be used. [Scaled number rules]
4: M	valueStepSize. number		SHALL be used.
4: M	valueStepSize. scale		SHALL be interpreted. If absent, a default value of "0" applies.
4: M	Measurement. measurementSeriesListData. measurementSeriesData.		
4: M	measurementId	<m7#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
4: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
4: M	timestamp	<t#{1..}->m7#1>	SHALL be set as SUB IDENTIFIER.
4: M	value.		[MOI-001], [MOI-043b] [Measurement value rules] [Scaled number rules]
4: M	value. number		SHALL be used.
4: M	value. scale		SHALL be interpreted. If absent, a default value of "0" applies.
4: R	evaluationPeriod.		[Evaluation period rules]
4: M	evaluationPeriod. startTime		
4: M	evaluationPeriod. endTime		
4: R	valueSource	"measuredValue"	
		"calculatedValue"	

		"empiricalValue"	
4: M	valueState		[MOI-005] [Value state rules]
4: M	ElectricalConnection. electricalConnectionDescriptionListData. electricalConnectionDescriptionData.		
4: M	electricalConnectionId	<ec1#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
4: M	powerSupplyType	"ac"	
4: M	positiveEnergyDirection	"consume"	
4: M	ElectricalConnection. electricalConnectionParameterDescriptionListData. electricalConnectionParameterDescriptionData.		
4: M	electricalConnectionId	<ec1#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
4: M	parameterId	<p7#{1..1}->ec1#1>	SHALL be set as SUB IDENTIFIER.
4: M	measurementId	<m7#{1..1}->p7#1>	SHALL be set as FOREIGN IDENTIFIER.
4: M	voltageType	"ac"	
4: M	acMeasurementType	"real"	

Table 31: Content of Specialization "Measurement_AcYieldMonthHistorical" at Actor Monitoring Appliance

3.2.1.3.5.10 Specialization "Measurement_AcYieldYear"

Scenario [{...}]: M/R/O [W][C]	Element	Value	[High-Level Mapping] Element and value rules
4: M	Measurement. measurementDescriptionListData. measurementDescriptionData.		
4: M	measurementId	<m8#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
4: M	measurementType	"energy"	
4: M	commodityType	"electricity"	
4: M	unit	"Wh"	
4: M	scopeType	"acYieldYear"	
4: R	Measurement. measurementConstraintsListData. measurementConstraintsData.		
4: M	measurementId	<m8#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
4: R	valueRangeMin.		[MOI-001] SHOULD be used. [Scaled number rules]
4: M	valueRangeMin. number		SHALL be used.
4: M	valueRangeMin. scale		SHALL be interpreted. If absent, a default value of "0" applies.
4: R	valueRangeMax.		[MOI-001] SHOULD be used. [Scaled number rules]
4: M	valueRangeMax. number		SHALL be used.
4: M	valueRangeMax. scale		SHALL be interpreted. If absent, a default value of "0" applies.
4: R	valueStepSize.		SHOULD be used. [Scaled number rules]
4: M	valueStepSize. number		SHALL be used.
4: M	valueStepSize. scale		SHALL be interpreted. If absent, a default value of "0" applies.

4: M	Measurement. measurementListData. measurementData.		
4: M	measurementId	<m8#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
4: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
4: O	timestamp	<t#(1..1)->m8#1>	MAY be used as SUB IDENTIFIER. Only the newest measurement value SHALL be stated. Additional historical values are forbidden.
4: M	value.		[MOI-001], [MOI-044a] [Measurement value rules] [Scaled number rules]
4: M	value. number		SHALL be used.
4: M	value. scale		SHALL be interpreted. If absent, a default value of "0" applies.
4: R	evaluationPeriod.		[Evaluation period rules]
4: M	evaluationPeriod. startTime		
4: M	evaluationPeriod. endTime		
4: R	valueSource	"measuredValue"	
		"calculatedValue"	
		"empiricalValue"	
4: M	valueState		[MOI-005] [Value state rules]
4: M	ElectricalConnection. electricalConnectionDescriptionListData. electricalConnectionDescriptionData.		
4: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
4: M	powerSupplyType	"ac"	
4: M	positiveEnergyDirection	"consume"	
4: M	ElectricalConnection. electricalConnectionParameterDescriptionListData. electricalConnectionParameterDescriptionData.		
4: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
4: M	parameterId	<p8#(1..1)->ec1#1>	SHALL be set as SUB IDENTIFIER.
4: M	measurementId	<m8#(1..1)->p8#1>	SHALL be set as FOREIGN IDENTIFIER.
4: M	voltageType	"ac"	
4: M	acMeasurementType	"real"	

Table 32: Content of Specialization "Measurement_AcYieldYear" at Actor Monitoring Appliance

3.2.1.3.5.11 Specialization "Measurement_AcYieldYearHistorical"

Scenario [{...}]: M/R/O [W][C]	Element	Value	[High-Level Mapping] Element and value rules
4: M	Measurement. measurementDescriptionListData. measurementDescriptionData.		
4: M	measurementId	<m8#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
4: M	measurementType	"energy"	
4: M	commodityType	"electricity"	
4: M	unit	"Wh"	

4: M	scopeType	"acYieldYear"	
4: R	Measurement. measurementConstraintsListData. measurementConstraintsData.		
4: M	measurementId	<m8#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
4: R	valueRangeMin.		[MOI-001] SHOULD be used. [Scaled number rules]
4: M	valueRangeMin. number		SHALL be used.
4: M	valueRangeMin. scale		SHALL be interpreted. If absent, a default value of "0" applies.
4: R	valueRangeMax.		[MOI-001] SHOULD be used. [Scaled number rules]
4: M	valueRangeMax. number		SHALL be used.
4: M	valueRangeMax. scale		SHALL be interpreted. If absent, a default value of "0" applies.
4: R	valueStepSize.		SHOULD be used. [Scaled number rules]
4: M	valueStepSize. number		SHALL be used.
4: M	valueStepSize. scale		SHALL be interpreted. If absent, a default value of "0" applies.
4: M	Measurement. measurementSeriesListData. measurementSeriesData.		
4: M	measurementId	<m8#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
4: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
4: M	timestamp	<t#{1..}->m8#1>	SHALL be set as SUB IDENTIFIER.
4: M	value.		[MOI-001], [MOI-044b] [Measurement value rules] [Scaled number rules]
4: M	value. number		SHALL be used.
4: M	value. scale		SHALL be interpreted. If absent, a default value of "0" applies.
4: R	evaluationPeriod.		[Evaluation period rules]
4: M	evaluationPeriod. startTime		
4: M	evaluationPeriod. endTime		
4: R	valueSource	"measuredValue" "calculatedValue" "empiricalValue"	
4: M	valueState		[MOI-005] [Value state rules]
4: M	ElectricalConnection. electricalConnectionDescriptionListData. electricalConnectionDescriptionData.		
4: M	electricalConnectionId	<ec1#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
4: M	powerSupplyType	"ac"	
4: M	positiveEnergyDirection	"consume"	
4: M	ElectricalConnection. electricalConnectionParameterDescriptionListData. electricalConnectionParameterDescriptionData.		
4: M	electricalConnectionId	<ec1#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
4: M	parameterId	<p8#{1..1}->ec1#1>	SHALL be set as SUB IDENTIFIER.

4: M	measurementId	<m8#(1..1)->p8#1>	SHALL be set as FOREIGN IDENTIFIER.
4: M	voltageType	"ac"	
4: M	acMeasurementType	"real"	

Table 33: Content of Specialization "Measurement_AcYieldYearHistorical" at Actor Monitoring Appliance

3.2.1.3.5.12 Specialization "Measurement_AcCosPhi"

Scenario [{...}]: M/R/O [W][\C]	Element	Value	[High-Level Mapping] Element and value rules
5: M	Measurement. measurementDescriptionListData. measurementDescriptionData.		
5: M	measurementId	<m9#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
5: M	measurementType	"powerFactor"	
5: M	commodityType	"electricity"	
5: M	unit	"1"	
5: M	scopeType	"acCosPhi"	
5: R	Measurement. measurementConstraintsListData. measurementConstraintsData. * ¹		
5: M	measurementId	<m9#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
5: R	valueRangeMin.		[MOI-002/3] SHOULD be used. [Scaled number rules]
5: M	valueRangeMin. number		SHALL be used.
5: M	valueRangeMin. scale		SHALL be interpreted. If absent, a default value of "0" applies.
5: R	valueRangeMax.		[MOI-002/3] SHOULD be used. [Scaled number rules]
5: M	valueRangeMax. number		SHALL be used.
5: M	valueRangeMax. scale		SHALL be interpreted. If absent, a default value of "0" applies.
5: R	valueStepSize.		SHOULD be used. [Scaled number rules]
5: M	valueStepSize. number		SHALL be used.
5: M	valueStepSize. scale		SHALL be interpreted. If absent, a default value of "0" applies.
5: M	Measurement. measurementListData. measurementData.		
5: M	measurementId	<m9#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
5: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
5: O	timestamp	<t#(1..1)->m9#1>	MAY be used as SUB IDENTIFIER. Only the newest measurement value SHALL be stated. Additional historical values are forbidden.
5: M	value.		[MOI-002/3], [MOI-051] [Measurement value rules] [Scaled number rules]

5: M	value. number		SHALL be used.
5: M	value. scale		SHALL be interpreted. If absent, a default value of "0" applies.
5: R	valueSource	"measuredValue"	
		"calculatedValue"	
		"empiricalValue"	
5: M	valueState		[MOI-005] [Value state rules]
5: M	ElectricalConnection. electricalConnectionDescriptionListData. electricalConnectionDescriptionData.		
5: M	electricalConnectionId	<ec1#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
5: M	powerSupplyType	"ac"	
5: M	positiveEnergyDirection	"consume"	
5: M	ElectricalConnection. electricalConnectionParameterDescriptionListData. electricalConnectionParameterDescriptionData.		
5: M	electricalConnectionId	<ec1#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
5: M	parameterId	<p9#{1..1}->ec1#1>	SHALL be set as SUB IDENTIFIER.
5: M	measurementId	<m9#{1..1}->p9#1>	SHALL be set as FOREIGN IDENTIFIER.
5: M	voltageType	"ac"	
5: M	acMeasurementType	"phase"	
5: M	acMeasurementVariant	"cosPhi"	

1109 Table 34: Content of Specialization "Measurement_AcCosPhi" at Actor Monitoring Appliance

1110 *¹: The constraints for the power factor $\cos \varphi$ define a range for the minimum and maximum
 1111 possible value. For Inverters that are capable of consuming and producing power, a range of [-1.0,
 1112 1.0] must be chosen, although this might not define the actual value range as (e.g.) the range
 1113 between [-0.9, 0.9] might not be possible for the Inverter.

1114

1115 3.2.1.3.5.13 Specialization "Measurement_ComponentTemperature"

Scenario [...]: M/R/O [W][\C]	Element	Value	[High-Level Mapping] Element and value rules
7: M	Measurement. measurementDescriptionListData. measurementDescriptionData.		
7: M	measurementId	<m10#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
7: M	measurementType	"temperature"	
7: M	unit	"degC" "degF" "K"	
7: M	scopeType	"componentTempe rature"	
7: R	Measurement. measurementConstraintsListData. measurementConstraintsData.		
7: M	measurementId	<m10#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
7: R	valueRangeMin.		SHOULD be used. [Scaled number rules]
7: M	valueRangeMin. number		SHALL be used.

7: M	valueRangeMin. scale		SHALL be interpreted. If absent, a default value of "0" applies.
7: R	valueRangeMax.		SHOULD be used. [Scaled number rules]
7: M	valueRangeMax. number		SHALL be used.
7: M	valueRangeMax. scale		SHALL be interpreted. If absent, a default value of "0" applies.
7: R	valueStepSize.		SHOULD be used. [Scaled number rules]
7: M	valueStepSize. number		SHALL be used.
7: M	valueStepSize. scale		SHALL be interpreted. If absent, a default value of "0" applies.
7: M	Measurement. measurementListData. measurementData.		
7: M	measurementId	<m10#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
7: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
7: O	timestamp	<t#{1..1}->m10#1>	MAY be used as SUB IDENTIFIER. Only the newest measurement value SHALL be stated. Additional historical values are forbidden.
7: M	value.		[MOI-071] [Measurement value rules] [Scaled number rules]
7: M	value. number		SHALL be used.
7: M	value. scale		SHALL be interpreted. If absent, a default value of "0" applies.
7: R	valueSource	"measuredValue"	
		"calculatedValue"	
		"empiricalValue"	
7: M	valueState		[MOI-005] [Value state rules]

Table 35: Content of Specialization "Measurement_ComponentTemperature" at Actor Monitoring Appliance

3.2.1.3.6 Topic "ElectricalConnection"

3.2.1.3.6.1 Specialization "ElectricalConnection_EntityPowerProductionNominalMax"

Scenario [...]: M/R/O [W]/[C]	Element	Value	[High-Level Mapping] Element and value rules
6: M	ElectricalConnection. electricalConnectionCharacteristicListData. electricalConnectionCharacteristicData.		
6: M	electricalConnectionId	<ec1#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
6: M	parameterId	<p10#{1..1}>	SHALL be set as SUB IDENTIFIER. The same parameterId SHALL be used for <p10#1> of this

			Use Case and <p1#1> of the Use Case "Monitoring of Power Consumption". E.g. if <p1#1> (in the Use Case MPC) is set to "3", <p10#1> (in this Use Case) SHALL be set to "3", too.
6: M	characteristicId	<cc1#(1..1)>	SHALL be set as SUB IDENTIFIER.
6: M	characteristicContext	"entity"	
6: M	characteristicType	"powerProductionNominalMax"	
6: M	value.		[MOI-003], [MOI-061a] [Scaled number rules]
6: M	value. number		SHALL be used.
6: M	value. scale		SHALL be interpreted. If absent, a default value of "0" applies.
6: M	unit	"W"	

Table 36: Content of Specialization "ElectricalConnection_EntityPowerProductionNominalMax" at Actor Monitoring Appliance

3.2.1.3.6.2 Specialization "ElectricalConnection_EntityApparentPowerProductionNominalMax"

Scenario [...]: M/R/O [W][C]	Element	Value	[High-Level Mapping] Element and value rules
6: M	ElectricalConnection. electricalConnectionCharacteristicListData. electricalConnectionCharacteristicData.		
6: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
6: M	parameterId	<p1#(1..1)>	SHALL be set as SUB IDENTIFIER.
6: M	characteristicId	<cc2#(1..1)>	SHALL be set as SUB IDENTIFIER.
6: M	characteristicContext	"entity"	
6: M	characteristicType	"apparentPowerProductionNominalMax"	
6: M	value.		[MOI-003], [MOI-062a] [Scaled number rules]
6: M	value. number		SHALL be used.
6: M	value. scale		SHALL be interpreted. If absent, a default value of "0" applies.
6: M	unit	"VA"	

Table 37: Content of Specialization "ElectricalConnection_EntityApparentPowerProductionNominalMax" at Actor Monitoring Appliance

1127 3.2.1.3.6.3 Specialization "ElectricalConnection_EntityPowerConsumptionNominalMax"

Scenario [...]: M/R/O [W][C]	Element	Value	[High-Level Mapping] Element and value rules
6: M	ElectricalConnection. electricalConnectionCharacteristicListData. electricalConnectionCharacteristicData.		
6: M	electricalConnectionId	<ec1#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
6: M	parameterId	<p10#{1..1}>	SHALL be set as SUB IDENTIFIER. The same parameterId SHALL be used for <p10#1> of this Use Case and <p1#1> of the Use Case "Monitoring of Power Consumption". E.g. if <p1#1> (in the Use Case MPC) is set to "3", <p10#1> (in this Use Case) SHALL be set to "3", too.
6: M	characteristicId	<cc3#{1..1}>	SHALL be set as SUB IDENTIFIER.
6: M	characteristicContext	"entity"	
6: M	characteristicType	"powerConsumptionNominalMax"	
6: M	value.		[MOI-003], [MOI-061b] [Scaled number rules]
6: M	value. number		SHALL be used.
6: M	value. scale		SHALL be interpreted. If absent, a default value of "0" applies.
6: M	unit	"W"	

Table 38: Content of Specialization "ElectricalConnection_EntityPowerConsumptionNominalMax" at Actor Monitoring Appliance

1130

1131 3.2.1.3.6.4 Specialization "ElectricalConnection_EntityApparentPowerConsumptionNominalMax"

Scenario [...]: M/R/O [W][C]	Element	Value	[High-Level Mapping] Element and value rules
6: M	ElectricalConnection. electricalConnectionCharacteristicListData. electricalConnectionCharacteristicData.		
6: M	electricalConnectio nId	<ec1#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.

6: M	parameterId	<p1#(1..1)>	SHALL be set as SUB IDENTIFIER.
6: M	characteristicId	<cc4#(1..1)>	SHALL be set as SUB IDENTIFIER.
6: M	characteristicContext	"entity"	
6: M	characteristicType	"apparentPowerConsumptionNominalMax"	
6: M	value.		[MOI-003], [MOI-062b] [Scaled number rules]
6: M	value. number		SHALL be used.
6: M	value. scale		SHALL be interpreted. If absent, a default value of "0" applies.
6: M	unit	"VA"	

Table 39: Content of Specialization "ElectricalConnection_EntityApparentPowerConsumptionNominalMax" at Actor Monitoring Appliance

3.2.1.3.6.5 Specialization "ElectricalConnection_BatteryPowerConsumptionNominalMin"

Scenario [...]: M/R/O [W][C]	Element	Value	[High-Level Mapping] Element and value rules
6: M	ElectricalConnection. electricalConnectionCharacteristicListData. electricalConnectionCharacteristicData.		
6: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
6: M	parameterId	<p10#(1..1)>	SHALL be set as SUB IDENTIFIER. The same parameterId SHALL be used for <p10#1> of this Use Case and <p1#1> of the Use Case "Monitoring of Power Consumption". E.g. if <p1#1> (in the Use Case MPC) is set to "3", <p10#1> (in this Use Case) SHALL be set to "3", too.
6: M	characteristicId	<cc5#(1..1)>	SHALL be set as SUB IDENTIFIER.
6: M	characteristicContext	"battery"	
6: M	characteristicType	"powerConsumptionNominalMin"	
6: M	value.		[MOI-003], [MOI-063a] [Scaled number rules]
6: M	value. number		SHALL be used.
6: M	value. scale		SHALL be interpreted. If absent, a default value of "0" applies.
6: M	unit	"W"	

Table 40: Content of Specialization "ElectricalConnection_BatteryPowerConsumptionNominalMin" at Actor Monitoring Appliance

3.2.1.3.6.6 Specialization "ElectricalConnection_BatteryPowerConsumptionNominalMax"

Scenario {...}: M/R/O [W][\C]	Element	Value	[High-Level Mapping] Element and value rules
6: M	ElectricalConnection. electricalConnectionCharacteristicListData. electricalConnectionCharacteristicData.		
6: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
6: M	parameterId	<p10#(1..1)>	SHALL be set as SUB IDENTIFIER. The same parameterId SHALL be used for <p10#1> of this Use Case and <p1#1> of the Use Case "Monitoring of Power Consumption". E.g. if <p1#1> (in the Use Case MPC) is set to "3", <p10#1> (in this Use Case) SHALL be set to "3", too.
6: M	characteristicId	<cc6#(1..1)>	SHALL be set as SUB IDENTIFIER.
6: M	characteristicContext	"battery"	
6: M	characteristicType	"powerConsumptionNominalMax"	
6: M	value.		[MOI-003], [MOI-063b] [Scaled number rules]
6: M	value. number		SHALL be used.
6: M	value. scale		SHALL be interpreted. If absent, a default value of "0" applies.
6: M	unit	"W"	

Table 41: Content of Specialization "ElectricalConnection_BatteryPowerConsumptionNominalMax" at Actor Monitoring Appliance

3.2.1.3.6.7 Specialization "ElectricalConnection_BatteryPowerConsumptionMax"

Scenario {...}: M/R/O [W][\C]	Element	Value	[High-Level Mapping] Element and value rules
----------------------------------	---------	-------	---

6: M	ElectricalConnection. electricalConnectionCharacteristicListData. electricalConnectionCharacteristicData.		
6: M	electricalConnectionId	<ec1#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
6: M	parameterId	<p10#{1..1}>	SHALL be set as SUB IDENTIFIER. The same parameterId SHALL be used for <p10#1> of this Use Case and <p1#1> of the Use Case "Monitoring of Power Consumption". E.g. if <p1#1> (in the Use Case MPC) is set to "3", <p10#1> (in this Use Case) SHALL be set to "3", too.
6: M	characteristicId	<cc7#{1..1}>	SHALL be set as SUB IDENTIFIER.
6: M	characteristicContext	"battery"	
6: M	characteristicType	"powerConsumptionMax"	
6: M	value.		[MOI-003], [MOI-064] [Scaled number rules]
6: M	value. number		SHALL be used.
6: M	value. scale		SHALL be interpreted. If absent, a default value of "0" applies.
6: M	unit	"W"	

Table 42: Content of Specialization "ElectricalConnection_BatteryPowerConsumptionMax" at Actor Monitoring Appliance

3.2.1.3.6.8 Specialization "ElectricalConnection_BatteryPowerProductionNominalMin"

Scenario [{...}]: M/R/O [W][\C]	Element	Value	[High-Level Mapping] Element and value rules
6: M	ElectricalConnection. electricalConnectionCharacteristicListData. electricalConnectionCharacteristicData.		
6: M	electricalConnectionId	<ec1#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
6: M	parameterId	<p10#{1..1}>	SHALL be set as SUB IDENTIFIER. The same parameterId SHALL be used for <p10#1> of this Use Case and <p1#1> of the Use Case "Monitoring of Power Consumption". E.g. if <p1#1> (in the Use Case MPC) is set to "3", <p10#1> (in this Use Case) SHALL be set to "3", too.
6: M	characteristicId	<cc8#{1..1}>	SHALL be set as SUB IDENTIFIER.
6: M	characteristicContext	"battery"	
6: M	characteristicType	"powerProductionNominalMin"	
6: M	value.		[MOI-003], [MOI-065a] [Scaled number rules]

6: M	value. number		SHALL be used.
6: M	value. scale		SHALL be interpreted. If absent, a default value of "0" applies.
6: M	unit	"W"	

Table 43: Content of Specialization "ElectricalConnection_BatteryPowerProductionNominalMin" at Actor Monitoring Appliance

3.2.1.3.6.9 Specialization "ElectricalConnection_BatteryPowerProductionNominalMax"

Scenario [...]: M/R/O [W][C]	Element	Value	[High-Level Mapping] Element and value rules
6: M	ElectricalConnection. electricalConnectionCharacteristicListData. electricalConnectionCharacteristicData.		
6: M	electricalConnectionId	<ec1#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
6: M	parameterId	<p10#{1..1}>	SHALL be set as SUB IDENTIFIER. The same parameterId SHALL be used for <p10#1> of this Use Case and <p1#1> of the Use Case "Monitoring of Power Consumption". E.g. if <p1#1> (in the Use Case MPC) is set to "3", <p10#1> (in this Use Case) SHALL be set to "3", too.
6: M	characteristicId	<cc9#{1..1}>	SHALL be set as SUB IDENTIFIER.
6: M	characteristicContext	"battery"	
6: M	characteristicType	"powerProductionNominalMax"	
6: M	value.		[MOI-003], [MOI-065b] [Scaled number rules]
6: M	value. number		SHALL be used.
6: M	value. scale		SHALL be interpreted. If absent, a default value of "0" applies.
6: M	unit	"W"	

Table 44: Content of Specialization "ElectricalConnection_BatteryPowerProductionNominalMax" at Actor Monitoring Appliance

1154 3.2.1.3.6.10 Specialization "ElectricalConnection_BatteryPowerProductionMax"

Scenario [{...}]: M/R/O [W][C]	Element	Value	[High-Level Mapping] Element and value rules
6: M	ElectricalConnection. electricalConnectionCharacteristicListData. electricalConnectionCharacteristicData.		
6: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
6: M	parameterId	<p10#(1..1)>	SHALL be set as SUB IDENTIFIER. The same parameterId SHALL be used for <p10#1> of this Use Case and <p1#1> of the Use Case "Monitoring of Power Consumption". E.g. if <p1#1> (in the Use Case MPC) is set to "3", <p10#1> (in this Use Case) SHALL be set to "3", too.
6: M	characteristicId	<cc10#(1..1)>	SHALL be set as SUB IDENTIFIER.
6: M	characteristicContext	"battery"	
6: M	characteristicType	"powerProductionMax"	
6: M	value.		[MOI-003], [MOI-066] [Scaled number rules]
6: M	value. number		SHALL be used.
6: M	value. scale		SHALL be interpreted. If absent, a default value of "0" applies.
6: M	unit	"W"	

1155 Table 45: Content of Specialization "ElectricalConnection_BatteryPowerProductionMax" at Actor Monitoring Appliance

1156

1157 3.2.2 Inverter

1158 3.2.2.1 Resource hierarchy

1159 If Use Case discovery is supported (see section 3.1.2) this Actor SHALL be denoted as "Inverter" in the
 1160 Element "nodeManagementUseCaseData. useCaseInformation. actor".

1161 The following diagram provides an overview of the Actor Inverter resource hierarchy.

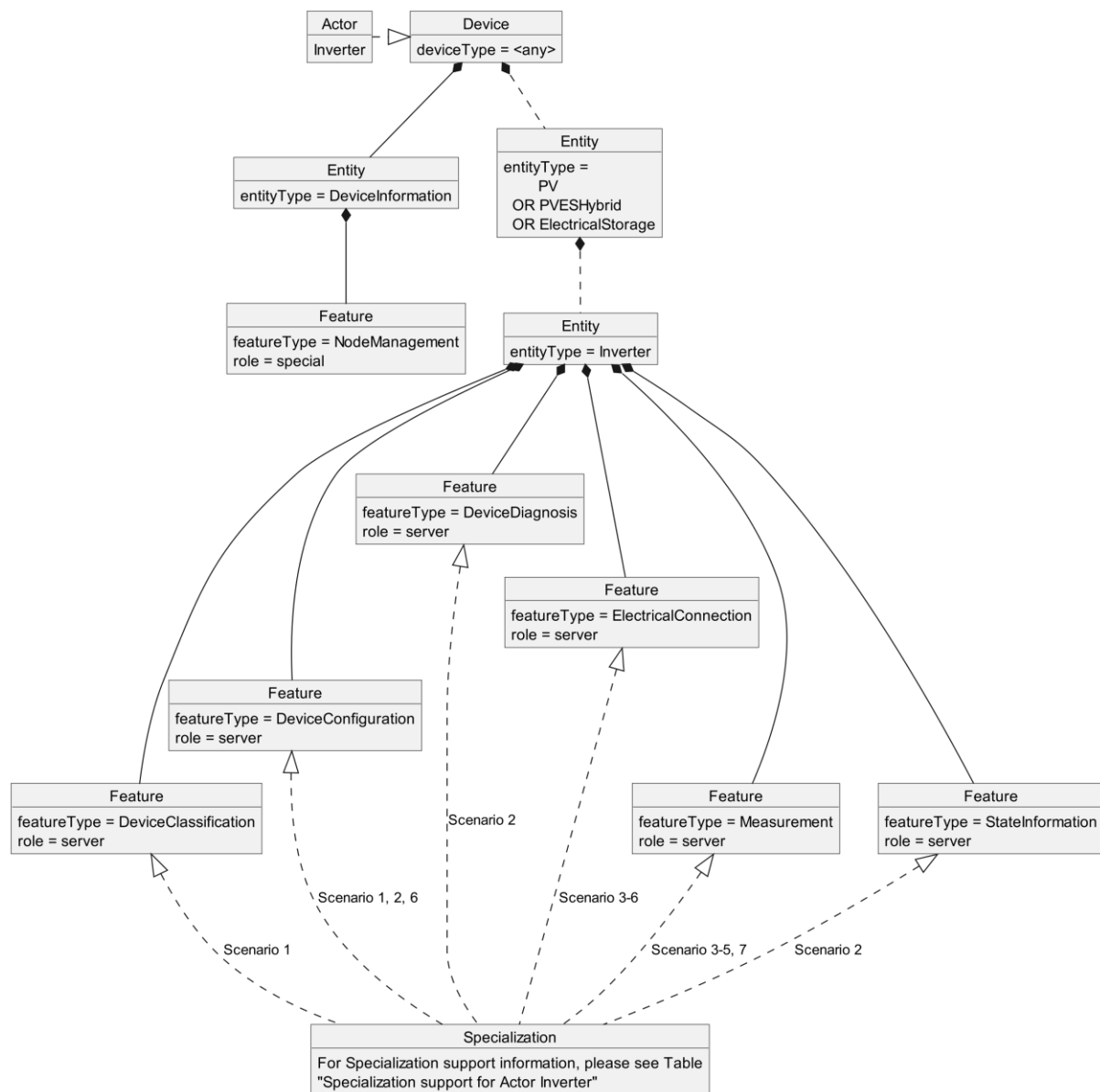


Figure 6: Actor "Inverter" overview

The "Actor ... overview" diagram rules" section describes how to interpret the diagram above. See the "Specializations" section for more information regarding the Specializations given in the diagram above.

Note: The entityType "DeviceInformation" with the featureType "NodeManagement" is required by the SPINE protocol and therefore SHALL be supported. Both types are added in the figure for completeness but are not directly linked to the Use Case.

The Use Case specific data follow behind the entityType "Inverter" which is a sub-Entity of one of the following Entities: "PV" ([MOI-004/1]), "PVESHybrid" ([MOI-004/2]) or "ElectricalStorage" ([MOI-004/3]) (see section 2.2). The Specializations represent the Scenario specific data that must be supported for each Scenario and are realized with the according featureTypes.

If a Specialization is connected to a Feature with the role "client", the Actor has a client role for this data. This means that the Actor accesses the data set described by the Specialization at a

1176 corresponding server Feature. Further details are described in the sub-section "Client data -
1177 Specializations".

1178 If a Specialization is connected to a Feature with the role "server", the Actor has the server role for
1179 this data. This means that the Actor must provide the corresponding data set of the Specialization as
1180 part of its Features. Further details are described in the sub-section "Server data - Resources".

Specialization name	Scenario	Used Feature...	...in tables
DeviceClassification_ManufacturerData_MOI	1	DeviceClassification	Table 48
DeviceConfiguration_InverterGridCode	1	DeviceConfiguration	Table 49
DeviceConfiguration_PvStringAvailabilityStatus	2		Table 50
DeviceConfiguration_BatteryAvailabilityStatus	2		
DeviceConfiguration_GridConnectionStatus	2		
DeviceConfiguration_TimeToAcChargePowerMax	6		
DeviceConfiguration_TimeToAcDischargePowerMax	6		
DeviceDiagnosis_OperatingState	2	DeviceDiagnosis	Table 51
StateInformation_InverterStateInformation	2	StateInformation	Table 59
Measurement_AcPowerApparentTotal	3	ElectricalConnection	Table 52
Measurement_AcPowerApparentPhaseSpecific	3		Table 53
Measurement_AcPowerReactiveTotal	3		Table 55
Measurement_AcPowerReactivePhaseSpecific	3		Table 56
Measurement_AcYieldTotal	4		Table 57
Measurement_AcYieldDay	4		
Measurement_AcYieldMonth	4		
Measurement_AcYieldYear	4		
Measurement_AcCosPhi	5		
Measurement_AcYieldDayHistorical	4		Table 52
Measurement_AcYieldMonthHistorical	4	Measurement	Table 53
Measurement_AcYieldYearHistorical	4		Table 55 Table 56 Table 58
Measurement_ComponentTemperature	7	Measurement	Table 55 Table 56 Table 57
ElectricalConnection_EntityPowerProductionNominalMax	6	ElectricalConnection	Table 54
ElectricalConnection_EntityApparentPowerProductionNominalMax	6		
ElectricalConnection_EntityPowerConsumptionNominalMax	6		
ElectricalConnection_EntityApparentPowerConsumptionNominalMax	6		
ElectricalConnection_BatteryPowerConsumptionNominalMin	6		
ElectricalConnection_BatteryPowerConsumptionNominalMax	6		
ElectricalConnection_BatteryPowerConsumptionMax	6		

ElectricalConnection_BatteryPowerProductionNominalMin	6		
ElectricalConnection_BatteryPowerProductionNominalMax	6		
ElectricalConnection_BatteryPowerProductionMax	6		

Table 46: Specialization support for Actor Inverter

3.2.2.2 Server data - Resources

3.2.2.2.1 Overview

Behind the entityType "Inverter" the Actor Inverter SHALL offer the Feature Types and Functions given in the table below.

Feature Type	Scenario: M/R/O	Function	Possible operations
DeviceClassification	1: M	deviceClassificationManufacturerData	read (M)
DeviceConfiguration	1: M 2: M 6: M	deviceConfigurationKeyValueDescriptionListData	read (M). partial (R)
	1: M 2: M 6: M	deviceConfigurationKeyValueListData	read (M). partial (R)
DeviceDiagnosis	2: M	deviceDiagnosisStateData	read (M)
ElectricalConnection	3: M 4: M 5: M	electricalConnectionDescriptionListData	read (M). partial (R)
	3: M 4: M 5: M	electricalConnectionParameterDescriptionListData	read (M). partial (R)
	6: M	electricalConnectionCharacteristicListData	read (M). partial (R)
Measurement	3: M 4: M 5: M 7: M	measurementDescriptionListData	read (M). partial (R)
	3: R 4: R 5: R 7: R	measurementConstraintsListData	read (M). partial (R)
	3: M 4: M 5: M 7: M	measurementListData	read (M). partial (R)
	4: M	measurementSeriesListData	read (M). partial (R)
StateInformation	2: M	stateInformationListData	read (M). partial (R)

Table 47: Feature Types and Functions used within this Use Case by the Actor Inverter

1188 For each of these Feature Types, the following rule applies: There SHALL be at maximum one Feature
1189 with the Feature Type in the Entity.

1190 Note: As a consequence of the previous rule, an implementation may need to have Feature data
1191 from different Scenarios/Specializations or even Use Cases in a given Feature.

1192 The Scenario number shows in which Scenarios the Inverter acts as server and which Feature Types
1193 and Functions are relevant in each Scenario.

1194 A detailed definition of the Elements and values that shall be supported in each Function is given in
1195 the following sub-sections.

1196 Note: If in the table above "partial" read is not mentioned or is only optional, it still might be
1197 mandatory to support partial notifications. The details of "partial" support are described within the
1198 Scenario sections.

1199 Note: The presence indications stated above are meant relative to the ones of the according Scenario
1200 stated in Table 1. I.e., if a Scenario is optional ("O") and a Feature Type is mandatory ("M"), the
1201 Feature Type need only be supported if the Scenario is supported, too.

1202 Note: Further Features MAY be implemented on the same Entities; also, further Functions MAY be
1203 implemented in the used Entities.

1204

1205 3.2.2.2.2 Feature Type "DeviceClassification"

1206 3.2.2.2.2.1 Function "deviceClassificationManufacturerData"

Scenario [{...}]: M/R/O [W][C]	Element	Value	[High-Level Mapping] Element and value rules
1: M	DeviceClassification. deviceClassificationManufacturerData.		
1: M	deviceName		[MOI-011] [String-length rules]
1: M	deviceCode		[MOI-012] [String-length rules]
1: M	serialNumber		[MOI-013] [String-length rules]
1: M	softwareRevision		[MOI-014] [String-length rules]
1: M	hardwareRevision		[MOI-015] [String-length rules]
1: M	vendorName		[MOI-016a] [String-length rules]
1: O	vendorCode		[String-length rules]
1: M	brandName		[MOI-016b] [String-length rules]
1: O	manufacturerLabel		[MOI-017]

			[String-length rules]
1: O	manufacturerDescription		[MOI-018] The string-length SHOULD NOT be longer than 4096 characters

Table 48: Content of Function "deviceClassificationManufacturerData" at Actor Inverter

3.2.2.2.3 Feature Type "DeviceConfiguration"

3.2.2.2.3.1 Function "deviceConfigurationKeyValueDescriptionListData"

Scenario [...]: M/R/O [W][\C]	Element	Value	[High-Level Mapping] Element and value rules
1: M	DeviceConfiguration. deviceConfigurationKeyValueDescriptionListData. deviceConfigurationKeyValueDescriptionData.		
1: M	keyId	<k1#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
1: M	keyName	"inverterGridCode"	
1: M	valueType	"string"	
2: M	DeviceConfiguration. deviceConfigurationKeyValueDescriptionListData. deviceConfigurationKeyValueDescriptionData.		
2: M	keyId	<k2#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
2: M	keyName	"pvStringAvailabilityStatus"	
2: M	valueType	"string"	
2: M	DeviceConfiguration. deviceConfigurationKeyValueDescriptionListData. deviceConfigurationKeyValueDescriptionData.		
2: M	keyId	<k3#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
2: M	keyName	"batteryAvailabilityStatus"	
2: M	valueType	"string"	
2: M	DeviceConfiguration. deviceConfigurationKeyValueDescriptionListData. deviceConfigurationKeyValueDescriptionData.		
2: M	keyId	<k4#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
2: M	keyName	"gridConnectionStatus"	
2: M	valueType	"string"	
6: M	DeviceConfiguration. deviceConfigurationKeyValueDescriptionListData. deviceConfigurationKeyValueDescriptionData.		
6: M	keyId	<k5#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
6: M	keyName	"timeToAcChargePowerMax"	
6: M	valueType	"scaledNumber"	
6: M	unit	"s"	The unit SHALL be applied to the value of the key.
6: M	DeviceConfiguration. deviceConfigurationKeyValueDescriptionListData. deviceConfigurationKeyValueDescriptionData.		

6: M	keyId	<k6#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
6: M	keyName	"timeToAcDischargePowerMax"	
6: M	valueType	"scaledNumber"	
6: M	unit	"s"	The unit SHALL be applied to the value of the key.

Table 49: Content of Function "deviceConfigurationKeyValueDescriptionListData" at Actor Inverter

3.2.2.2.3.2 Function "deviceConfigurationKeyValueListData"

Scenario [...]: M/R/O [W]/[C]	Element	Value	[High-Level Mapping] Element and value rules
1: M	DeviceConfiguration. deviceConfigurationKeyValueListData. deviceConfigurationKeyValueData.		
1: M	keyId	<k1#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
1: M	value.		[MOI-019] Exactly one of the child Elements SHALL be set. This SHALL match with the content of <i>valueType</i> Element within the key value description part (see above).
1: M	value. string		The format SHALL be: "[country;]technical connection rule[;year][;ancillary]" Items in brackets are optional. Examples: - "AT;TORD4;2016;LV" - "DE;VDE-AR-N4105;2019" - "SE;EN50438;2013" - "UK;G98/1-3" - "VDE0126-1-1"
2: M	DeviceConfiguration. deviceConfigurationKeyValueListData. deviceConfigurationKeyValueData.		
2: M	keyId	<k2#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
2: M	value.		[MOI-023] Exactly one of the child Elements SHALL be set. This SHALL match with the content of <i>valueType</i> Element within the key value description part (see above).
2: M	value. string	"unknown"	[MOI-023/1]
		"disconnected"	[MOI-023/2]
		"connected"	[MOI-023/3]
		"available"	[MOI-023/4]
		"inOperation"	[MOI-023/5]
2: M	DeviceConfiguration. deviceConfigurationKeyValueListData. deviceConfigurationKeyValueData.		
2: M	keyId	<k3#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
2: M	value.		[MOI-024]

			Exactly one of the child Elements SHALL be set. This SHALL match with the content of <i>valueType</i> Element within the key value description part (see above).
2: M	value. string	"unknown"	[MOI-024/1]
		"disconnected"	[MOI-024/2]
		"connected"	[MOI-024/3]
		"available"	[MOI-024/4]
		"inOperation"	[MOI-024/5]
2: M	DeviceConfiguration. deviceConfigurationKeyValueListData. deviceConfigurationKeyValueData.		
2: M	keyId	<k4#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M	value.		[MOI-025] Exactly one of the child Elements SHALL be set. This SHALL match with the content of <i>valueType</i> Element within the key value description part (see above).
2: M	value. string	"unknown"	[MOI-025/1]
		"disconnected"	[MOI-025/2]
		"connected"	[MOI-025/3]
6: M	DeviceConfiguration. deviceConfigurationKeyValueListData. deviceConfigurationKeyValueData.		
6: M	keyId	<k5#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
6: M	value.		[MOI-067a] Exactly one of the child Elements SHALL be set. This SHALL match with the content of <i>valueType</i> Element within the key value description part (see above).
6: M	value. scaledNumber.		SHALL be used. [Scaled number rules]
6: M	value. scaledNumber. number		SHALL be used.
6: O	value. scaledNumber. scale		MAY be used. If absent, a default value of "0" applies.
6: M	DeviceConfiguration. deviceConfigurationKeyValueListData. deviceConfigurationKeyValueData.		
6: M	keyId	<k6#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
6: M	value.		[MOI-067b] Exactly one of the child Elements SHALL be set. This SHALL match with the content of <i>valueType</i> Element within the key value description part (see above).
6: M	value. scaledNumber.		SHALL be used. The sub-Elements "number" and "scale" represent a value according to the following formula: $\text{number} * 10^{\text{scale}}$
6: M	value. scaledNumber. number		SHALL be used.
6: O	value. scaledNumber. scale		MAY be used. If absent, a default value of "0" applies.

1214

Table 50: Content of Function "deviceConfigurationKeyValueListData" at Actor Inverter

1215

1216 3.2.2.2.4 Feature Type "DeviceDiagnosis"

1217 3.2.2.2.4.1 Function "deviceDiagnosisStateData"

Scenario {...}: M/R/O [W][C]	Element	Value	[High-Level Mapping] Element and value rules
2: M	DeviceDiagnosis. deviceDiagnosisStateData.		
2: O	timestamp		
2: M	operatingState	"normalOperation" [MOI-021/1]	[MOI-021]
		"standby" [MOI-021/2]	
		"failure" [MOI-021/5]	
		"temporarilyNotReady" [MOI-021/3]	
		"off" [MOI-021/4]	
2: R	vendorStateCode		[MOI-022]

1218 Table 51: Content of Function "deviceDiagnosisStateData" at Actor Inverter

1219

1220 3.2.2.2.5 Feature Type "ElectricalConnection"

1221 3.2.2.2.5.1 Function "electricalConnectionDescriptionListData"

Scenario {...}: M/R/O [W][C]	Element	Value	[High-Level Mapping] Element and value rules
3: M 4: M 5: M	ElectricalConnection. electricalConnectionDescriptionListData. electricalConnectionDescriptionData.		
3: M 4: M 5: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
3: M 4: M 5: M	powerSupplyType	"ac"	
3: M 4: M 5: M	positiveEnergyDirection	"consume"	

1222 Table 52: Content of Function "electricalConnectionDescriptionListData" at Actor Inverter

1223

1224 3.2.2.2.5.2 Function "electricalConnectionParameterDescriptionListData"

Scenario [{...}]: M/R/O [W][C]	Element	Value	[High-Level Mapping] Element and value rules
3: M	ElectricalConnection. electricalConnectionParameterDescriptionListData. electricalConnectionParameterDescriptionData.		
3: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
3: M	parameterId	<p1#(1..1)->ec1#1>	SHALL be set as SUB IDENTIFIER.
3: M	measurementId	<m1#(1..1)->p1#1>	SHALL be set as FOREIGN IDENTIFIER.
3: M	voltageType	"ac"	
3: M	acMeasuredPhases	"abc" "ab" "bc" "ac" "a" "b" "c"	[MOI-031] If the Inverter is connected to less than three phases, one of the other combinations like "a" or "ab" are allowed instead of "abc". The values "a", "b", and "c" are permitted if and only if only one phase is measured by the server.
3: M	acMeasuredInReferenceTo	"neutral"	
3: M	acMeasurementType	"apparent"	
3: M	acMeasurementVariant	"rms"	
3: M	ElectricalConnection. electricalConnectionParameterDescriptionListData. electricalConnectionParameterDescriptionData.		
3: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
3: M	parameterId	<p2#(1..3)->ec1#1>	SHALL be set as SUB IDENTIFIER.
3: M	measurementId	<m2#(1..3)->p2#(1..3)>	SHALL be set as FOREIGN IDENTIFIER.
3: M	voltageType	"ac"	
3: M	acMeasuredPhases	"a" (->p2#1)	[MOI-032/1]
		"b" (->p2#2)	[MOI-032/2]
		"c" (->p2#3)	[MOI-032/3]
3: M	acMeasuredInReferenceTo	"neutral"	
3: M	acMeasurementType	"apparent"	
3: M	acMeasurementVariant	"rms"	
3: M	ElectricalConnection. electricalConnectionParameterDescriptionListData. electricalConnectionParameterDescriptionData.		
3: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
3: M	parameterId	<p3#(1..1)->ec1#1>	SHALL be set as SUB IDENTIFIER.

3: M	measurementId	<m3#(1..1)->p3#1>	SHALL be set as FOREIGN IDENTIFIER.
3: M	voltageType	"ac"	
3: M	acMeasuredPhases	"abc" "ab" "bc" "ac" "a" "b" "c"	[MOI-033] If the Inverter is connected to less than three phases, one of the other combinations like "a" or "ab" are allowed instead of "abc". The values "a", "b", and "c" are permitted if and only if only one phase is measured by the server.
3: M	acMeasuredInReferenceTo	"neutral"	
3: M	acMeasurementType	"reactive"	
3: M	acMeasurementVariant	"rms"	
3: M	ElectricalConnection. electricalConnectionParameterDescriptionListData. electricalConnectionParameterDescriptionData.		
3: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
3: M	parameterId	<p4#(1..3)->ec1#1>	SHALL be set as SUB IDENTIFIER.
3: M	measurementId	<m4#(1..3)->p4#(1..3)>	SHALL be set as FOREIGN IDENTIFIER.
3: M	voltageType	"ac"	
3: M	acMeasuredPhases	"a" (->p4#1)	[MOI-034/1]
		"b" (->p4#2)	[MOI-034/2]
		"c" (->p4#3)	[MOI-034/3]
3: M	acMeasuredInReferenceTo	"neutral"	
3: M	acMeasurementType	"reactive"	
3: M	acMeasurementVariant	"rms"	
4: M	ElectricalConnection. electricalConnectionParameterDescriptionListData. electricalConnectionParameterDescriptionData.		
4: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
4: M	parameterId	<p5#(1..1)->ec1#1>	SHALL be set as SUB IDENTIFIER.
4: M	measurementId	<m5#(1..1)->p5#1>	SHALL be set as FOREIGN IDENTIFIER.
4: M	voltageType	"ac"	
4: M	acMeasurementType	"real"	
4: M	ElectricalConnection. electricalConnectionParameterDescriptionListData. electricalConnectionParameterDescriptionData.		
4: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
4: M	parameterId	<p6#(1..1)->ec1#1>	SHALL be set as SUB IDENTIFIER.
4: M	measurementId	<m6#(1..1)->p6#1>	SHALL be set as FOREIGN IDENTIFIER.
4: M	voltageType	"ac"	
4: M	acMeasurementType	"real"	

4: M	ElectricalConnection. electricalConnectionParameterDescriptionListData. electricalConnectionParameterDescriptionData.		
4: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
4: M	parameterId	<p7#(1..1)->ec1#1>	SHALL be set as SUB IDENTIFIER.
4: M	measurementId	<m7#(1..1)->p7#1>	SHALL be set as FOREIGN IDENTIFIER.
4: M	voltageType	"ac"	
4: M	acMeasurementType	"real"	
4: M	ElectricalConnection. electricalConnectionParameterDescriptionListData. electricalConnectionParameterDescriptionData.		
4: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
4: M	parameterId	<p8#(1..1)->ec1#1>	SHALL be set as SUB IDENTIFIER.
4: M	measurementId	<m8#(1..1)->p8#1>	SHALL be set as FOREIGN IDENTIFIER.
4: M	voltageType	"ac"	
4: M	acMeasurementType	"real"	
5: M	ElectricalConnection. electricalConnectionParameterDescriptionListData. electricalConnectionParameterDescriptionData.		
5: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
5: M	parameterId	<p9#(1..1)->ec1#1>	SHALL be set as SUB IDENTIFIER.
5: M	measurementId	<m9#(1..1)->p9#1>	SHALL be set as FOREIGN IDENTIFIER.
5: M	voltageType	"ac"	
5: M	acMeasurementType	"phase"	
5: M	acMeasurementVariant	"cosPhi"	

Table 53: Content of Function "electricalConnectionParameterDescriptionListData" at Actor Inverter

3.2.2.2.5.3 Function "electricalConnectionCharacteristicListData"

Scenario {...}: M/R/O [W][C]	Element	Value	[High-Level Mapping] Element and value rules
6: M	ElectricalConnection. electricalConnectionCharacteristicListData. electricalConnectionCharacteristicData.		
6: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
6: M	parameterId	<p10#(1..1)>	SHALL be set as SUB IDENTIFIER. The same parameterId SHALL be used for <p10#1> of this Use Case and <p1#1> of the Use Case "Monitoring of Power

			Consumption". E.g. if <p1#1> (in the Use Case MPC) is set to "3", <p10#1> (in this Use Case) SHALL be set to "3", too.
6: M	characteristicId	<cc1#(1..1)>	SHALL be set as SUB IDENTIFIER.
6: M	characteristicContext	"entity"	
6: M	characteristicType	"powerProductionNominalMax"	
6: M	value.		[MOI-003], [MOI-061a] [Scaled number rules]
6: M	value. number		SHALL be used.
6: O	value. scale		MAY be used. If absent, a default value of "0" applies.
6: M	unit	"W"	
6: M	ElectricalConnection. electricalConnectionCharacteristicListData. electricalConnectionCharacteristicData.		
6: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
6: M	parameterId	<p1#(1..1)>	SHALL be set as SUB IDENTIFIER.
6: M	characteristicId	<cc2#(1..1)>	SHALL be set as SUB IDENTIFIER.
6: M	characteristicContext	"entity"	
6: M	characteristicType	"apparentPowerProductionNominalMax"	
6: M	value.		[MOI-003], [MOI-062a] [Scaled number rules]
6: M	value. number		SHALL be used.
6: O	value. scale		MAY be used. If absent, a default value of "0" applies.
6: M	unit	"VA"	
6: M	ElectricalConnection. electricalConnectionCharacteristicListData. electricalConnectionCharacteristicData.		
6: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
6: M	parameterId	<p10#(1..1)>	SHALL be set as SUB IDENTIFIER. The same parameterId SHALL be used for <p10#1> of this Use Case and <p1#1> of the Use Case "Monitoring of Power Consumption". E.g. if <p1#1> (in the Use Case MPC) is set to "3", <p10#1> (in this Use Case) SHALL be set to "3", too.
6: M	characteristicId	<cc3#(1..1)>	SHALL be set as SUB IDENTIFIER.
6: M	characteristicContext	"entity"	
6: M	characteristicType	"powerConsumptionNominalMax"	
6: M	value.		[MOI-003], [MOI-061b] [Scaled number rules]
6: M	value. number		SHALL be used.
6: O	value. scale		MAY be used. If absent, a default value of "0" applies.
6: M	unit	"W"	

6: M	ElectricalConnection. electricalConnectionCharacteristicListData. electricalConnectionCharacteristicData.		
6: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
6: M	parameterId	<p1#(1..1)>	SHALL be set as SUB IDENTIFIER.
6: M	characteristicId	<cc4#(1..1)>	SHALL be set as SUB IDENTIFIER.
6: M	characteristicContext	"entity"	
6: M	characteristicType	"apparentPowerConsumptionNominalMax"	
6: M	value.		[MOI-003], [MOI-062b] [Scaled number rules]
6: M	value. number		SHALL be used.
6: O	value. scale		MAY be used. If absent, a default value of "0" applies.
6: M	unit	"VA"	
6: M	ElectricalConnection. electricalConnectionCharacteristicListData. electricalConnectionCharacteristicData.		
6: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
6: M	parameterId	<p10#(1..1)>	SHALL be set as SUB IDENTIFIER. The same parameterId SHALL be used for <p10#1> of this Use Case and <p1#1> of the Use Case "Monitoring of Power Consumption". E.g. if <p1#1> (in the Use Case MPC) is set to "3", <p10#1> (in this Use Case) SHALL be set to "3", too.
6: M	characteristicId	<cc5#(1..1)>	SHALL be set as SUB IDENTIFIER.
6: M	characteristicContext	"battery"	
6: M	characteristicType	"powerConsumptionNominalMin"	
6: M	value.		[MOI-003], [MOI-063a] [Scaled number rules]
6: M	value. number		SHALL be used.
6: O	value. scale		MAY be used. If absent, a default value of "0" applies.
6: M	unit	"W"	
6: M	ElectricalConnection. electricalConnectionCharacteristicListData. electricalConnectionCharacteristicData.		
6: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
6: M	parameterId	<p10#(1..1)>	SHALL be set as SUB IDENTIFIER. The same parameterId SHALL be used for <p10#1> of this Use Case and <p1#1> of the Use Case "Monitoring of Power Consumption". E.g. if <p1#1> (in the Use Case MPC) is set to "3", <p10#1> (in this Use Case) SHALL be set to "3", too.
6: M	characteristicId	<cc6#(1..1)>	SHALL be set as SUB IDENTIFIER.

6: M	characteristicContext	"battery"	
6: M	characteristicType	"powerConsumptionNominalMax"	
6: M	value.		[MOI-003], [MOI-063b] [Scaled number rules]
6: M	value. number		SHALL be used.
6: O	value. scale		MAY be used. If absent, a default value of "0" applies.
6: M	unit	"W"	
6: M	ElectricalConnection. electricalConnectionCharacteristicListData. electricalConnectionCharacteristicData.		
6: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
6: M	parameterId	<p10#(1..1)>	SHALL be set as SUB IDENTIFIER. The same parameterId SHALL be used for <p10#1> of this Use Case and <p1#1> of the Use Case "Monitoring of Power Consumption". E.g. if <p1#1> (in the Use Case MPC) is set to "3", <p10#1> (in this Use Case) SHALL be set to "3", too.
6: M	characteristicId	<cc7#(1..1)>	SHALL be set as SUB IDENTIFIER.
6: M	characteristicContext	"battery"	
6: M	characteristicType	"powerConsumptionMax"	
6: M	value.		[MOI-003], [MOI-064] [Scaled number rules]
6: M	value. number		SHALL be used.
6: O	value. scale		MAY be used. If absent, a default value of "0" applies.
6: M	unit	"W"	
6: M	ElectricalConnection. electricalConnectionCharacteristicListData. electricalConnectionCharacteristicData.		
6: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
6: M	parameterId	<p10#(1..1)>	SHALL be set as SUB IDENTIFIER. The same parameterId SHALL be used for <p10#1> of this Use Case and <p1#1> of the Use Case "Monitoring of Power Consumption". E.g. if <p1#1> (in the Use Case MPC) is set to "3", <p10#1> (in this Use Case) SHALL be set to "3", too.
6: M	characteristicId	<cc8#(1..1)>	SHALL be set as SUB IDENTIFIER.
6: M	characteristicContext	"battery"	
6: M	characteristicType	"powerProductionNominalMin"	
6: M	value.		[MOI-003], [MOI-065a] [Scaled number rules]
6: M	value. number		SHALL be used.
6: O	value. scale		MAY be used. If absent, a default value of "0" applies.

6: M	unit	"W"	
6: M	ElectricalConnection. electricalConnectionCharacteristicListData. electricalConnectionCharacteristicData.		
6: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
6: M	parameterId	<p10#(1..1)>	SHALL be set as SUB IDENTIFIER. The same parameterId SHALL be used for <p10#1> of this Use Case and <p1#1> of the Use Case "Monitoring of Power Consumption". E.g. if <p1#1> (in the Use Case MPC) is set to "3", <p10#1> (in this Use Case) SHALL be set to "3", too.
6: M	characteristicId	<cc9#(1..1)>	SHALL be set as SUB IDENTIFIER.
6: M	characteristicContext	"battery"	
6: M	characteristicType	"powerProductionNominalMax"	
6: M	value.		[MOI-003], [MOI-065b] [Scaled number rules]
6: M	value. number		SHALL be used.
6: O	value. scale		MAY be used. If absent, a default value of "0" applies.
6: M	unit	"W"	
6: M	ElectricalConnection. electricalConnectionCharacteristicListData. electricalConnectionCharacteristicData.		
6: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
6: M	parameterId	<p10#(1..1)>	SHALL be set as SUB IDENTIFIER. The same parameterId SHALL be used for <p10#1> of this Use Case and <p1#1> of the Use Case "Monitoring of Power Consumption". E.g. if <p1#1> (in the Use Case MPC) is set to "3", <p10#1> (in this Use Case) SHALL be set to "3", too.
6: M	characteristicId	<cc10#(1..1)>	SHALL be set as SUB IDENTIFIER.
6: M	characteristicContext	"battery"	
6: M	characteristicType	"powerProductionMax"	
6: M	value.		[MOI-003], [MOI-066] [Scaled number rules]
6: M	value. number		SHALL be used.
6: O	value. scale		MAY be used. If absent, a default value of "0" applies.
6: M	unit	"W"	

Table 54: Content of Function "electricalConnectionCharacteristicListData" at Actor Inverter

1230 3.2.2.2.6 Feature Type "Measurement"

1231 3.2.2.2.6.1 Function "measurementDescriptionListData"

Scenario [{...}]: M/R/O [W][C]	Element	Value	[High-Level Mapping] Element and value rules
3: M	Measurement. measurementDescriptionListData. measurementDescriptionData.		
3: M	measurementId	<m1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
3: M	measurementType	"power"	
3: M	commodityType	"electricity"	
3: M	unit	"VA"	
3: M	scopeType	"acPowerApparentTotal"	
3: M	Measurement. measurementDescriptionListData. measurementDescriptionData.		
3: M	measurementId	<m2#(1..3)>	SHALL be set as PRIMARY IDENTIFIER.
3: M	measurementType	"power"	
3: M	commodityType	"electricity"	
3: M	unit	"VA"	
3: M	scopeType	"acPowerApparent"	
3: M	Measurement. measurementDescriptionListData. measurementDescriptionData.		
3: M	measurementId	<m3#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
3: M	measurementType	"power"	
3: M	commodityType	"electricity"	
3: M	unit	"var"	
3: M	scopeType	"acPowerReactiveTotal"	
3: M	Measurement. measurementDescriptionListData. measurementDescriptionData.		
3: M	measurementId	<m4#(1..3)>	SHALL be set as PRIMARY IDENTIFIER.
3: M	measurementType	"power"	
3: M	commodityType	"electricity"	
3: M	unit	"var"	
3: M	scopeType	"acPowerReactive"	
4: M	Measurement. measurementDescriptionListData. measurementDescriptionData.		
4: M	measurementId	<m5#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
4: M	measurementType	"energy"	
4: M	commodityType	"electricity"	
4: M	unit	"Wh"	
4: M	scopeType	"acYieldTotal"	
4: M	Measurement. measurementDescriptionListData. measurementDescriptionData.		
4: M	measurementId	<m6#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
4: M	measurementType	"energy"	
4: M	commodityType	"electricity"	
4: M	unit	"Wh"	
4: M	scopeType	"acYieldDay"	
4: M	Measurement. measurementDescriptionListData. measurementDescriptionData.		
4: M	measurementId	<m7#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
4: M	measurementType	"energy"	
4: M	commodityType	"electricity"	
4: M	unit	"Wh"	

4: M	scopeType	"acYieldMonth"	
4: M	Measurement. measurementDescriptionListData. measurementDescriptionData.		
4: M	measurementId	<m8#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
4: M	measurementType	"energy"	
4: M	commodityType	"electricity"	
4: M	unit	"Wh"	
4: M	scopeType	"acYieldYear"	
5: M	Measurement. measurementDescriptionListData. measurementDescriptionData.		
5: M	measurementId	<m9#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
5: M	measurementType	"powerFactor"	
5: M	commodityType	"electricity"	
5: M	unit	"1"	
5: M	scopeType	"acCosPhi"	
7: M	Measurement. measurementDescriptionListData. measurementDescriptionData.		
7: M	measurementId	<m10#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
7: M	measurementType	"temperature"	
7: M	unit	"degC" "degF" "K"	
7: M	scopeType	"componentTemperature"	

Table 55: Content of Function "measurementDescriptionListData" at Actor Inverter

3.2.2.2.6.2 Function "measurementConstraintsListData"

Scenario [...]: M/R/O [W][C]	Element	Value	[High-Level Mapping] Element and value rules
3: R	Measurement. measurementConstraintsListData. measurementConstraintsData.		
3: M	measurementId	<m1#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
3: R	valueRangeMin.		[MOI-002/1] SHOULD be used. [Scaled number rules]
3: M	valueRangeMin. number		SHALL be used.
3: O	valueRangeMin. scale		MAY be used. If absent, a default value of "0" applies.
3: R	valueRangeMax.		[MOI-002/1] SHOULD be used. [Scaled number rules]
3: M	valueRangeMax. number		SHALL be used.
3: O	valueRangeMax. scale		MAY be used. If absent, a default value of "0" applies.
3: R	valueStepSize.		SHOULD be used. [Scaled number rules]
3: M	valueStepSize. number		SHALL be used.
3: O	valueStepSize. scale		MAY be used. If absent, a default value of "0" applies.
3: R	Measurement. measurementConstraintsListData. measurementConstraintsData.		
3: M	measurementId	<m2#{1..3}>	SHALL be set as PRIMARY IDENTIFIER.
3: R	valueRangeMin.		[MOI-002/1]

			SHOULD be used. [Scaled number rules]
3: M	valueRangeMin. number		SHALL be used.
3: O	valueRangeMin. scale		MAY be used. If absent, a default value of "0" applies.
3: R	valueRangeMax.		[MOI-002/1] SHOULD be used. [Scaled number rules]
3: M	valueRangeMax. number		SHALL be used.
3: O	valueRangeMax. scale		MAY be used. If absent, a default value of "0" applies.
3: R	valueStepSize.		SHOULD be used. [Scaled number rules]
3: M	valueStepSize. number		SHALL be used.
3: O	valueStepSize. scale		MAY be used. If absent, a default value of "0" applies.
3: R	Measurement. measurementConstraintsListData. measurementConstraintsData.		
3: M	measurementId	<m3#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
3: R	valueRangeMin.		[MOI-002/2] SHOULD be used. [Scaled number rules]
3: M	valueRangeMin. number		SHALL be used.
3: O	valueRangeMin. scale		MAY be used. If absent, a default value of "0" applies.
3: R	valueRangeMax.		[MOI-002/2] SHOULD be used. [Scaled number rules]
3: M	valueRangeMax. number		SHALL be used.
3: O	valueRangeMax. scale		MAY be used. If absent, a default value of "0" applies.
3: R	valueStepSize.		SHOULD be used. [Scaled number rules]
3: M	valueStepSize. number		SHALL be used.
3: O	valueStepSize. scale		MAY be used. If absent, a default value of "0" applies.
3: R	Measurement. measurementConstraintsListData. measurementConstraintsData.		
3: M	measurementId	<m4#(1..3)>	SHALL be set as PRIMARY IDENTIFIER.
3: R	valueRangeMin.		[MOI-002/2] SHOULD be used. [Scaled number rules]
3: M	valueRangeMin. number		SHALL be used.
3: O	valueRangeMin. scale		MAY be used. If absent, a default value of "0" applies.
3: R	valueRangeMax.		[MOI-002/2] SHOULD be used. [Scaled number rules]
3: M	valueRangeMax. number		SHALL be used.
3: O	valueRangeMax. scale		MAY be used. If absent, a default value of "0" applies.
3: R	valueStepSize.		SHOULD be used. [Scaled number rules]
3: M	valueStepSize. number		SHALL be used.

3: O	valueStepSize. scale		MAY be used. If absent, a default value of "0" applies.
4: R	Measurement. measurementConstraintsListData. measurementConstraintsData.		
4: M	measurementId	<m5#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
4: R	valueRangeMin.		[MOI-001] SHOULD be used. [Scaled number rules]
4: M	valueRangeMin. number		SHALL be used.
4: O	valueRangeMin. scale		MAY be used. If absent, a default value of "0" applies.
4: R	valueRangeMax.		[MOI-001] SHOULD be used. [Scaled number rules]
4: M	valueRangeMax. number		SHALL be used.
4: O	valueRangeMax. scale		MAY be used. If absent, a default value of "0" applies.
4: R	valueStepSize.		SHOULD be used. [Scaled number rules]
4: M	valueStepSize. number		SHALL be used.
4: O	valueStepSize. scale		MAY be used. If absent, a default value of "0" applies.
4: R	Measurement. measurementConstraintsListData. measurementConstraintsData.		
4: M	measurementId	<m6#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
4: R	valueRangeMin.		[MOI-001] SHOULD be used. [Scaled number rules]
4: M	valueRangeMin. number		SHALL be used.
4: O	valueRangeMin. scale		MAY be used. If absent, a default value of "0" applies.
4: R	valueRangeMax.		[MOI-001] SHOULD be used. [Scaled number rules]
4: M	valueRangeMax. number		SHALL be used.
4: O	valueRangeMax. scale		MAY be used. If absent, a default value of "0" applies.
4: R	valueStepSize.		SHOULD be used. [Scaled number rules]
4: M	valueStepSize. number		SHALL be used.
4: O	valueStepSize. scale		MAY be used. If absent, a default value of "0" applies.
4: R	Measurement. measurementConstraintsListData. measurementConstraintsData.		
4: M	measurementId	<m7#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
4: R	valueRangeMin.		[MOI-001] SHOULD be used. [Scaled number rules]
4: M	valueRangeMin. number		SHALL be used.
4: O	valueRangeMin. scale		MAY be used. If absent, a default value of "0" applies.
4: R	valueRangeMax.		[MOI-001] SHOULD be used. [Scaled number rules]
4: M	valueRangeMax. number		SHALL be used.

4: O	valueRangeMax. scale		MAY be used. If absent, a default value of "0" applies.
4: R	valueStepSize.		SHOULD be used. [Scaled number rules]
4: M	valueStepSize. number		SHALL be used.
4: O	valueStepSize. scale		MAY be used. If absent, a default value of "0" applies.
4: R	Measurement. measurementConstraintsListData. measurementConstraintsData.		
4: M	measurementId	<m8#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
4: R	valueRangeMin.		[MOI-001] SHOULD be used. [Scaled number rules]
4: M	valueRangeMin. number		SHALL be used.
4: O	valueRangeMin. scale		MAY be used. If absent, a default value of "0" applies.
4: R	valueRangeMax.		[MOI-001] SHOULD be used. [Scaled number rules]
4: M	valueRangeMax. number		SHALL be used.
4: O	valueRangeMax. scale		MAY be used. If absent, a default value of "0" applies.
4: R	valueStepSize.		SHOULD be used. [Scaled number rules]
4: M	valueStepSize. number		SHALL be used.
4: O	valueStepSize. scale		MAY be used. If absent, a default value of "0" applies.
5: R	Measurement. measurementConstraintsListData. measurementConstraintsData. * ¹		
5: M	measurementId	<m9#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
5: R	valueRangeMin.		[MOI-002/3] SHOULD be used. [Scaled number rules]
5: M	valueRangeMin. number		SHALL be used.
5: O	valueRangeMin. scale		MAY be used. If absent, a default value of "0" applies.
5: R	valueRangeMax.		[MOI-002/3] SHOULD be used. [Scaled number rules]
5: M	valueRangeMax. number		SHALL be used.
5: O	valueRangeMax. scale		MAY be used. If absent, a default value of "0" applies.
5: R	valueStepSize.		SHOULD be used. [Scaled number rules]
5: M	valueStepSize. number		SHALL be used.
5: O	valueStepSize. scale		MAY be used. If absent, a default value of "0" applies.
7: R	Measurement. measurementConstraintsListData. measurementConstraintsData.		
7: M	measurementId	<m10#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
7: R	valueRangeMin.		SHOULD be used. [Scaled number rules]
7: M	valueRangeMin. number		SHALL be used.
7: O	valueRangeMin. scale		MAY be used. If absent, a default value of "0" applies.

7: R	valueRangeMax.		SHOULD be used. [Scaled number rules]
7: M	valueRangeMax. number		SHALL be used.
7: O	valueRangeMax. scale		MAY be used. If absent, a default value of "0" applies.
7: R	valueStepSize.		SHOULD be used. [Scaled number rules]
7: M	valueStepSize. number		SHALL be used.
7: O	valueStepSize. scale		MAY be used. If absent, a default value of "0" applies.

Table 56: Content of Function "measurementConstraintsListData" at Actor Inverter

*1: The constraints for the power factor $\cos \varphi$ define a range for the minimum and maximum possible value. For Inverters that are capable of consuming and producing power, a range of [-1.0, 1.0] must be chosen, although this might not define the actual value range as (e.g.) the range between [-0.9, 0.9] might not be possible for the Inverter.

3.2.2.2.6.3 Function "measurementListData"

Scenario {...}: M/R/O [W]/[C]	Element	Value	[High-Level Mapping] Element and value rules
3: M	Measurement. measurementListData. measurementData.		
3: M	measurementId	<m1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
3: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
3: O	timestamp	<t#(1..1)->m1#1>	MAY be used as SUB IDENTIFIER. Only the newest measurement value SHALL be stated. Additional historical values are forbidden.
3: M	value.		[MOI-002/1], [MOI-031] [Measurement value rules] [Scaled number rules]
3: M	value. number		SHALL be used.
3: O	value. scale		MAY be used. If absent, a default value of "0" applies.
3: M	valueSource	"measuredValue"	
		"calculatedValue"	
		"empiricalValue"	
3: R	valueState		[MOI-005] [Value state rules]
3: M	Measurement. measurementListData. measurementData.		
3: M	measurementId	<m2#(1..3)>	SHALL be set as PRIMARY IDENTIFIER.
3: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
3: O	timestamp	<t#(1..1)->m2#(1..3)>	MAY be used as SUB IDENTIFIER. Only the newest measurement value SHALL be stated. Additional historical values are forbidden.
3: M	value.		[MOI-002/1], [MOI-032] [Measurement value rules]

			[Scaled number rules]
3: M	value. number		SHALL be used.
3: O	value. scale		MAY be used. If absent, a default value of "0" applies.
3: M	valueSource	"measuredValue"	
		"calculatedValue"	
		"empiricalValue"	
3: R	valueState		[MOI-005] [Value state rules]
3: M	Measurement. measurementListData. measurementData.		
3: M	measurementId	<m3#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
3: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
3: O	timestamp	<t#{1..1}->m3#1>	MAY be used as SUB IDENTIFIER. Only the newest measurement value SHALL be stated. Additional historical values are forbidden.
3: M	value.		[MOI-002/2], [MOI-033] [Measurement value rules] [Scaled number rules]
3: M	value. number		SHALL be used.
3: O	value. scale		MAY be used. If absent, a default value of "0" applies.
3: M	valueSource	"measuredValue"	
		"calculatedValue"	
		"empiricalValue"	
3: R	valueState		[MOI-005] [Value state rules]
3: M	Measurement. measurementListData. measurementData.		
3: M	measurementId	<m4#{1..3}>	SHALL be set as PRIMARY IDENTIFIER.
3: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
3: O	timestamp	<t#{1..1}->m4#{1..3}>	MAY be used as SUB IDENTIFIER. Only the newest measurement value SHALL be stated. Additional historical values are forbidden.
3: M	value.		[MOI-002/2], [MOI-034] [Measurement value rules] [Scaled number rules]
3: M	value. number		SHALL be used.
3: O	value. scale		MAY be used. If absent, a default value of "0" applies.
3: M	valueSource	"measuredValue"	
		"calculatedValue"	
		"empiricalValue"	
3: R	valueState		[MOI-005] [Value state rules]
4: M	Measurement. measurementListData. measurementData.		
4: M	measurementId	<m5#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
4: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
4: O	timestamp	<t#{1..1}->m5#1>	MAY be used as SUB IDENTIFIER. Only the newest measurement value SHALL be stated. Additional historical values are forbidden.
4: M	value.		[MOI-001], [MOI-041] [Measurement value rules]

			[Scaled number rules]
4: M	value. number		SHALL be used.
4: O	value. scale		MAY be used. If absent, a default value of "0" applies.
4: O	evaluationPeriod.		[Evaluation period rules]
4: M	evaluationPeriod.startTime		
4: M	evaluationPeriod.endTime		
4: M	valueSource	"measuredValue"	
		"calculatedValue"	
		"empiricalValue"	
4: R	valueState		[MOI-005] [Value state rules]
4: M	Measurement. measurementListData. measurementData.		
4: M	measurementId	<m6#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
4: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
4: O	timestamp	<t#{1..1}->m6#1>	MAY be used as SUB IDENTIFIER. Only the newest measurement value SHALL be stated. Additional historical values are forbidden.
4: M	value.		[MOI-001], [MOI-042a] [Measurement value rules] [Scaled number rules]
4: M	value. number		SHALL be used.
4: O	value. scale		MAY be used. If absent, a default value of "0" applies.
4: R	evaluationPeriod.		[Evaluation period rules]
4: M	evaluationPeriod.startTime		
4: M	evaluationPeriod.endTime		
4: M	valueSource	"measuredValue"	
		"calculatedValue"	
		"empiricalValue"	
4: R	valueState		[MOI-005] [Value state rules]
4: M	Measurement. measurementListData. measurementData.		
4: M	measurementId	<m7#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
4: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
4: O	timestamp	<t#{1..1}->m7#1>	MAY be used as SUB IDENTIFIER. Only the newest measurement value SHALL be stated. Additional historical values are forbidden.
4: M	value.		[MOI-001], [MOI-043a] [Measurement value rules] [Scaled number rules]
4: M	value. number		SHALL be used.
4: O	value. scale		MAY be used. If absent, a default value of "0" applies.

4: R	evaluationPeriod.		[Evaluation period rules]
4: M	evaluationPeriod.startTime		
4: M	evaluationPeriod.endTime		
4: M	valueSource	"measuredValue"	
		"calculatedValue"	
		"empiricalValue"	
4: R	valueState		[MOI-005] [Value state rules]
4: M	Measurement. measurementListData. measurementData.		
4: M	measurementId	<m8#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
4: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
4: O	timestamp	<t#{1..1}->m8#1>	MAY be used as SUB IDENTIFIER. Only the newest measurement value SHALL be stated. Additional historical values are forbidden.
4: M	value.		[MOI-001], [MOI-044a] [Measurement value rules] [Scaled number rules]
4: M	value. number		SHALL be used.
4: O	value. scale		MAY be used. If absent, a default value of "0" applies.
4: R	evaluationPeriod.		[Evaluation period rules]
4: M	evaluationPeriod.startTime		
4: M	evaluationPeriod.endTime		
4: M	valueSource	"measuredValue"	
		"calculatedValue"	
		"empiricalValue"	
4: R	valueState		[MOI-005] [Value state rules]
5: M	Measurement. measurementListData. measurementData.		
5: M	measurementId	<m9#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
5: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
5: O	timestamp	<t#{1..1}->m9#1>	MAY be used as SUB IDENTIFIER. Only the newest measurement value SHALL be stated. Additional historical values are forbidden.
5: M	value.		[MOI-002/3], [MOI-051] [Measurement value rules] [Scaled number rules]
5: M	value. number		SHALL be used.
5: O	value. scale		MAY be used. If absent, a default value of "0" applies.
5: M	valueSource	"measuredValue"	
		"calculatedValue"	
		"empiricalValue"	
5: R	valueState		[MOI-005] [Value state rules]

7: M	Measurement. measurementListData. measurementData.		
7: M	measurementId	<m10#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
7: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
7: O	timestamp	<t#{1..1}->m10#1>	MAY be used as SUB IDENTIFIER. Only the newest measurement value SHALL be stated. Additional historical values are forbidden.
7: M	value.		[MOI-071] [Measurement value rules] [Scaled number rules]
7: M	value. number		SHALL be used.
7: O	value. scale		MAY be used. If absent, a default value of "0" applies.
7: M	valueSource	"measuredValue"	
		"calculatedValue"	
		"empiricalValue"	
7: R	valueState		[MOI-005] [Value state rules]

Table 57: Content of Function "measurementListData" at Actor Inverter

3.2.2.2.6.4 Function "measurementSeriesListData"

Scenario {...}: M/R/O [W][\C]	Element	Value	[High-Level Mapping] Element and value rules
4: M	Measurement. measurementSeriesListData. measurementSeriesData.		
4: M	measurementId	<m6#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
4: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
4: M	timestamp	<t#{1..1}->m6#1>	SHALL be set as SUB IDENTIFIER.
4: M	value.		[MOI-001], [MOI-042b] [Measurement value rules] [Scaled number rules]
4: M	value. number		SHALL be used.
4: O	value. scale		MAY be used. If absent, a default value of "0" applies.
4: R	evaluationPeriod.		[Evaluation period rules]
4: M	evaluationPeriod. startTime		
4: M	evaluationPeriod. endTime		
4: M	valueSource	"measuredValue"	
		"calculatedValue"	
		"empiricalValue"	
4: R	valueState		[MOI-005] [Value state rules]
4: M	Measurement. measurementSeriesListData. measurementSeriesData.		
4: M	measurementId	<m7#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
4: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.

4: M	timestamp	<t#{1..}->m7#1>	SHALL be set as SUB IDENTIFIER.
4: M	value.		[MOI-001], [MOI-043b] [Measurement value rules] [Scaled number rules]
4: M	value. number		SHALL be used.
4: O	value. scale		MAY be used. If absent, a default value of "0" applies.
4: R	evaluationPeriod.		[Evaluation period rules]
4: M	evaluationPeriod. startTime		
4: M	evaluationPeriod. endTime		
4: M	valueSource	"measuredValue"	
		"calculatedValue"	
		"empiricalValue"	
4: R	valueState		[MOI-005] [Value state rules]
4: M	Measurement. measurementSeriesListData. measurementSeriesData.		
4: M	measurementId	<m8#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
4: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
4: M	timestamp	<t#{1..}->m8#1>	SHALL be set as SUB IDENTIFIER.
4: M	value.		[MOI-001], [MOI-044b] [Measurement value rules] [Scaled number rules]
4: M	value. number		SHALL be used.
4: O	value. scale		MAY be used. If absent, a default value of "0" applies.
4: R	evaluationPeriod.		[Evaluation period rules]
4: M	evaluationPeriod. startTime		
4: M	evaluationPeriod. endTime		
4: M	valueSource	"measuredValue"	
		"calculatedValue"	
		"empiricalValue"	
4: R	valueState		[MOI-005] [Value state rules]

Table 58: Content of Function "measurementSeriesListData" at Actor Inverter

3.2.2.2.7 Feature Type "StateInformation"

3.2.2.2.7.1 Function "stateInformationListData"

Scenario {...}]: M/R/O [W]/[C]	Element	Value	[High-Level Mapping] Element and value rules
2: M	StateInformation. stateInformationListData. stateInformationData.		

2: M	stateInformationId	<si1#(1..31)>	SHALL be set as PRIMARY IDENTIFIER.
2: M	stateInformation	"externalOverrideFromGrid" (-><si1#1>) [MOI-026/1/1]	[MOI-026]
		"autonomousGridSupport" (-><si1#2>) [MOI-026/1/2]	[MOI-026]
		"islandingMode" (-><si1#3>) [MOI-026/1/3]	[MOI-026]
		"balancing" (-><si1#4>) [MOI-026/1/4]	[MOI-026]
		"trickleCharging" (-><si1#5>) [MOI-026/1/5]	[MOI-026]
		"calibration" (-><si1#6>) [MOI-026/1/6]	[MOI-026]
		"commissioningMissing" (-><si1#7>) [MOI-026/1/7]	[MOI-026]
		"sleeping" (-><si1#8>) [MOI-026/1/8]	[MOI-026]
		"starting" (-><si1#9>) [MOI-026/1/9]	[MOI-026]
		"mppt" (-><si1#10>) [MOI-026/1/10]	[MOI-026]
		"throttled" (-><si1#11>) [MOI-026/1/11]	[MOI-026]
		"shuttingDown" (-><si1#12>) [MOI-026/1/12]	[MOI-026]
		"manualShutdown" (-><si1#13>) [MOI-026/1/13]	[MOI-026]
		"inverterDefective" (-><si1#14>) [MOI-026/2/1]	[MOI-026]
		"batteryOvercurrentProtection" (-><si1#15>) [MOI-026/2/2]	[MOI-026]
		"pvStringOvercurrentProtection" (-><si1#16>) [MOI-026/2/3]	[MOI-026]
		"gridFault" (-><si1#17>) [MOI-026/2/4]	[MOI-026]
		"groundFault" (-><si1#18>) [MOI-026/2/5]	[MOI-026]
		"acDisconnected" (-><si1#19>) [MOI-026/2/6]	[MOI-026]
		"dcDisconnected" (-><si1#20>) [MOI-026/2/7]	[MOI-026]
		"cabinetOpen" (-><si1#21>) [MOI-026/2/8]	[MOI-026]
		"overTemperature" (-><si1#22>) [MOI-026/2/9]	[MOI-026]
		"underTemperature" (-><si1#23>) [MOI-026/2/10]	[MOI-026]
		"frequencyAboveLimit" (-><si1#24>) [MOI-026/2/11]	[MOI-026]
		"frequencyBelowLimit" (-><si1#25>) [MOI-026/2/12]	[MOI-026]

		"acVoltageAboveLimit" (-><si1#26>) [MOI-026/2/13]	[MOI-026]
		"acVoltageBelowLimit" (-><si1#27>) [MOI-026/2/14]	[MOI-026]
		"dcVoltageAboveLimit" (-><si1#28>) [MOI-026/2/15]	[MOI-026]
		"dcVoltageBelowLimit" (-><si1#29>) [MOI-026/2/16]	[MOI-026]
		"hardwareTestFailure" (-><si1#30>) [MOI-026/2/17]	[MOI-026]
		"genericInternalError" (-><si1#31>) [MOI-026/2/18]	[MOI-026]
2: M	isActive	"true" "false"	
2: M	category	"functionality" (-><si1#1-13>) [MOI-026/1]	
		"failure" (-><si1#14-31>) [MOI-026/2]	
2: O	timeOfLastChange		

Table 59: Content of Function "stateInformationListData" at Actor Inverter

3.2.2.3 Client data - Specializations

As this Actor has only server functionality, no Specializations are described within this section.

3.3 Pre-Scenario communication

3.3.1 General information

The Pre-Scenario communication is needed if a client does not know the corresponding addresses on the server or if the required subscriptions or bindings are not active. In this case certain general communication mechanisms SHALL be used within SPINE:

- a) Detailed discovery: allows to discover resource addresses.
- b) Binding: allows to bind to resource address, which is frequently necessary to obtain write privileges.
- c) Subscription: allows to subscribe to resource addresses, which is necessary to receive unsolicited notifications if a resource changes during runtime.

It is possible to combine those steps for multiple Scenarios or also multiple Use Cases:

- E.g. if multiple Scenarios in multiple Use Cases use the same Feature, only one subscription needs to occur.
- E.g. a complete detailed discovery or a subscription to the NodeManagement Feature needs to occur only once for all Use Cases.

Depending on which Entity, Feature and Functions are used within a Scenario the payload of the corresponding messages may slightly differ, but the basic principles and messages used stay the same.

1272 The subsequent messages SHALL be exchanged for those parts that have not already been performed
1273 since the current connection is established or if those parts are outdated for another reason (e.g. if
1274 the detailed discovery is needed, but the bindings and subscriptions are still active from a previous
1275 connection only the detailed discovery messages need to be exchanged). If all Pre-Scenario
1276 communication parts are up to date, this section MAY be skipped, and the implementation can
1277 proceed as described in the corresponding "Scenario communication" sections.

1278 After the connection is re-established (e.g. due to a power loss or a firmware update) the Pre-
1279 Scenario communication SHALL be performed as well. There may be circumstances where messages
1280 from the Pre-Scenario communication may be exchanged again.

1281 Often the necessary messages of different Scenarios can be combined, so that only one single
1282 message is needed instead of multiple messages for the different Scenarios. This also is the case for
1283 the Pre-Scenario communication. In most cases only one "read" operation on the detailed discovery
1284 is necessary, as well as only one subscription request or binding request is needed for each Feature.
1285 Often multiple Scenarios within a Use Case access the same Feature, so only one subscription or
1286 binding is necessary.

1287

1288 3.3.2 Detailed discovery

1289 For the functionality where a client already has current detailed discovery information (i.e.
1290 independent of this Use Case or any Scenario of it) the remainder of this section SHOULD be skipped.

1291 Otherwise, the following procedure SHALL be performed in the given order:

- 1292 1. If a client is not subscribed to the primary NodeManagement instance, the client SHALL
1293 acquire a subscription according to the figure provided within this sub-section.
- 1294 2. A client SHALL read the detailed discovery information according to the figure provided
1295 within this sub-section. It SHALL keep the received information as far as it concerns
1296 mandatory and supported optional Entity Types, Feature Types and Functions of this Use
1297 Case that are needed by the client. This means that a client may choose how to store the
1298 necessary information. E.g. a client Actor can store the information how to address the
1299 necessary Features of the implemented Scenarios but may discard the Entity information.
- 1300 3. If and as long as a client has a subscription to the detailed discovery information of an Actor
1301 and receives proper notifications, it SHALL consider (i.e. integrate into the kept detailed
1302 discovery information) the received information as far as it concerns mandatory and
1303 supported optional Entity Types, Feature Types and Functions of this Use Case.

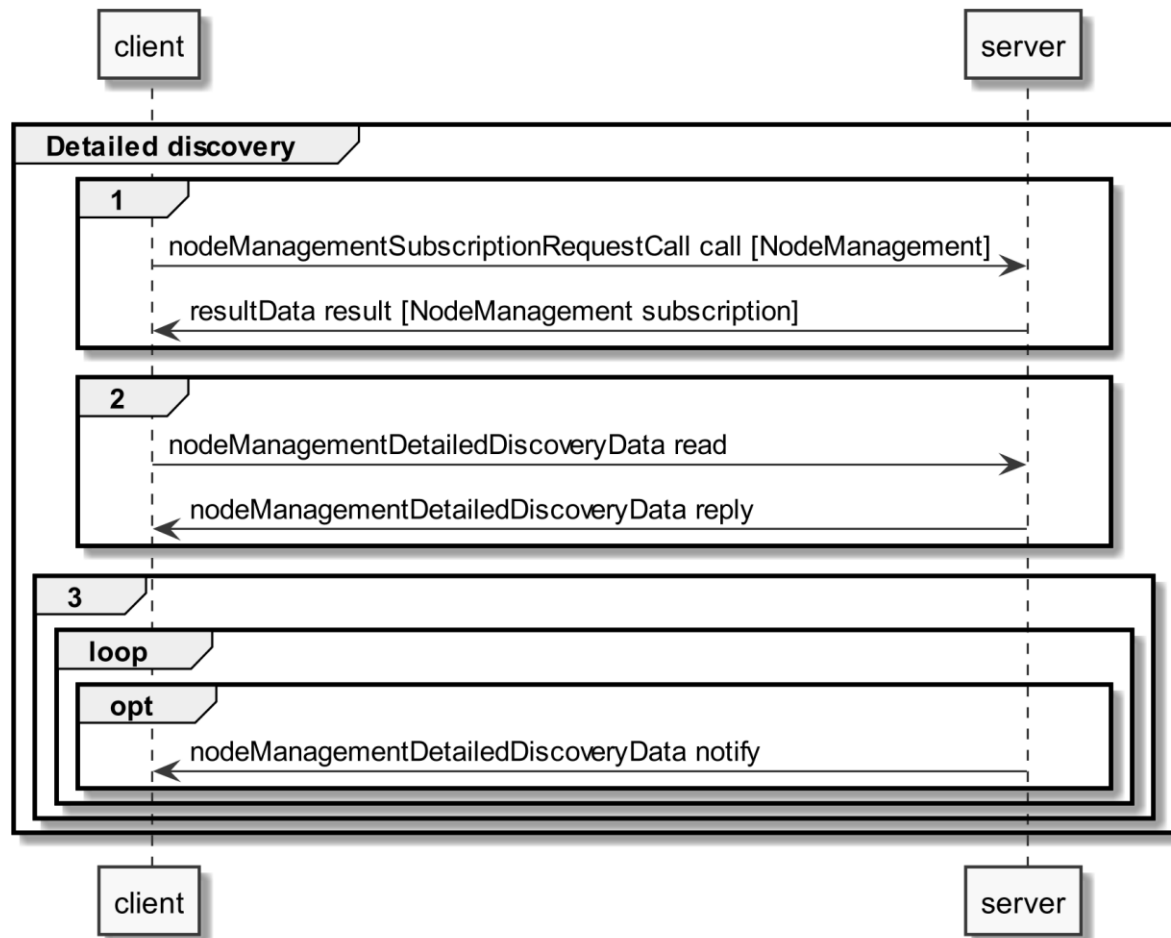


Figure 7: Pre-Scenario communication - Detailed discovery sequence diagram

If the "nodeManagementDetailedDiscoveryData read" fails, the client SHOULD retry to read the detailed discovery information until the "nodeManagementDetailedDiscoveryData reply" message was received successfully.

If all functionality is present at all times: The "nodeManagementDetailedDiscoveryData reply" message contains at least the mandatory Entities and Features given in the "Actor [...] overview" diagrams as well as the used Functions and their "possible operations" described in section 3.2 and its sub-sections.

If functionality is added or removed dynamically: The "nodeManagementDetailedDiscoveryData reply" message does not need to contain all mandatory Entities and Features given in the "Actor [...] overview" diagrams as well as all needed Functions and their "possible operations" described in section 3.2 and its sub-sections. However, as soon as the functionality is available it will be announced via a "nodeManagementDetailedDiscoveryData notify" message.

For the nodeManagementDetailedDiscoveryData read Function it is recommended to use a partial read with separated Selectors that may use one of the following Elements:

- entityType
- featureType

Note: Even with the usage of Selectors Features and Entities that are not relevant for this Use Case may be discovered. However, only Features and Entities that fulfil the hierarchical order as described within the Actors' sections shall be considered for this Use Case.

A "partial" notify SHALL be supported without using Selectors and Elements. Partial "delete" notify SHOULD also be supported with separated Selectors that may use one of the following Elements:

- entityAddress
- featureAddress

3.3.3 Binding

If binding is required by a Scenario that uses Features with writeable or changeable data, the server SHALL support binding for the respective Features. Before a write on a Function of a Feature occurs, the client SHALL create a binding to the corresponding Feature. For this the nodeManagementBindingRequestCall Function is used as shown in the following sequence diagram:

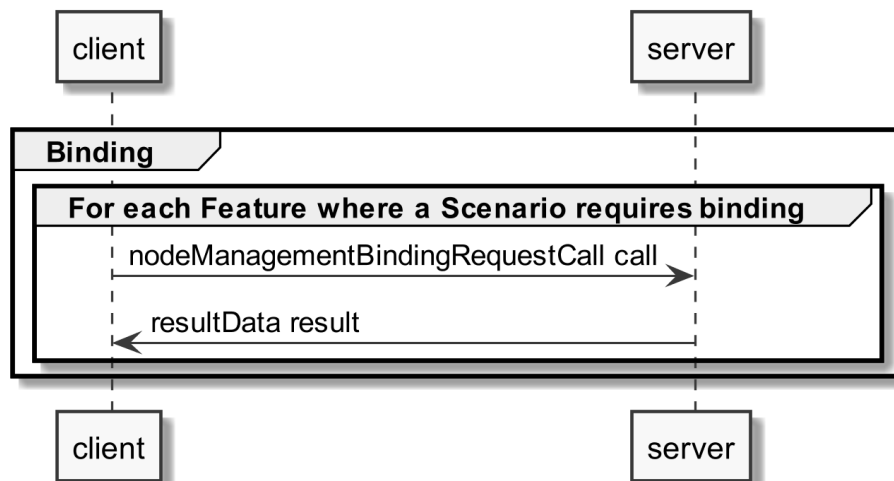


Figure 8: Pre-Scenario communication - Binding sequence diagram

If functionality is added or removed dynamically, binding may not be possible at all times on the required Functions. A client SHALL retry to create a binding again when receiving according updated detailed discovery information.

3.3.4 Subscription

A server SHALL support subscription for all Features that contain readable data that may change during runtime. The client SHALL create a subscription for all Features that the client wants to read. For this the nodeManagementSubscriptionRequestCall Function is used as shown in the following sequence diagram:

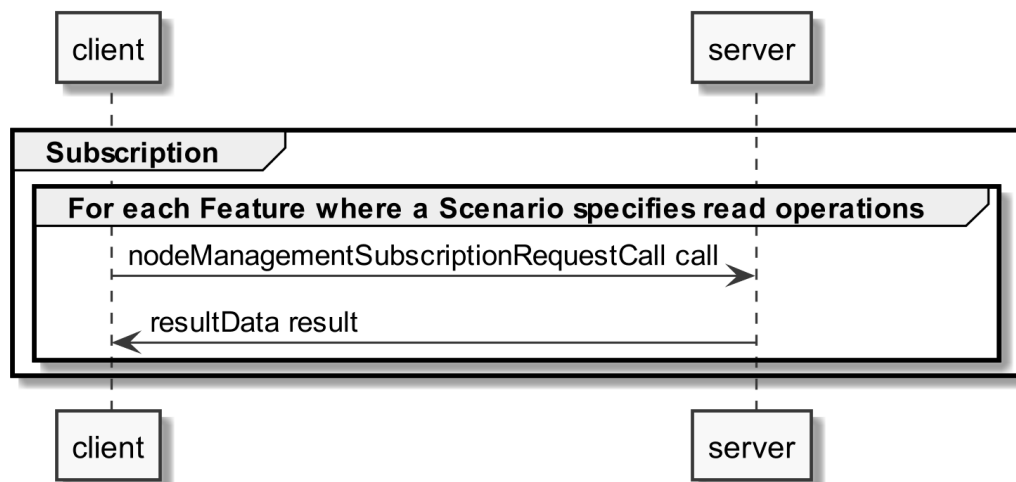


Figure 9: Pre-Scenario communication - Subscription sequence diagram

If the subscription request fails (e.g. because it is not supported by the server or the maximum number of possible subscriptions is reached), the client SHOULD read the data periodically (so-called "polling").

If functionality is added or removed dynamically, subscription may not be possible at all times on the required Functions. A client SHALL retry its subscription procedure again when receiving according updated detailed discovery information.

3.3.5 Dynamic behaviour

In case Entities or Features are removed, a nodeManagementDetailedDiscoveryData "notify" is transmitted that informs about the deleted Entities and Features. All existing binding or subscription entries on the deleted Features SHALL be deleted by each device.

In case Entities or Features are added the Pre-Scenario communication starts with transmitting a nodeManagementDetailedDiscoveryData "notify" that contains the added Entities and Features.

3.4 Scenarios

3.4.1 Scenario 1-7 - Overall Scenario information

3.4.1.1 Pre-Scenario communication

1. **Detailed discovery:** Actors that act as client within the Scenarios of this Use Case, need to know the addresses of the server Features used in the Initial Scenario communication. If the address of a particular server Feature is not known, the detailed discovery must be used, as described in section 3.3.2.
2. **Binding:** Binding SHOULD NOT be used for the Scenarios of this Use Case.
3. **Subscription:** Actors SHALL create a subscription for each server Feature that is relevant for the corresponding Actor within the Scenarios of this Use Case, as described in section 3.3.4.

The Initial Scenario communication SHALL start at the latest when the required resources on an Actor are known and the necessary binding and subscription procedures have been finished. However, as

1374 soon as the address of a required resource is known, the Initial Scenario communication for this
1375 resource MAY start already, even if the addresses of other required resources are not known yet.

1376 If required resources are removed and added again, they are re-discovered, and the Initial Scenario
1377 communication is triggered again for those resources.

1378 Note: This section holds for all Scenarios of this Use Case.

1379

1380 **3.4.1.2 Initial Scenario communication**

1381 Each time a (re-)connection is established, even if the Pre-Scenario communication phase is skipped,
1382 the messages shown in the following sequence diagram SHALL be exchanged (if a supported Scenario
1383 requires the respective message), as the corresponding resources may have changed in the
1384 meantime:

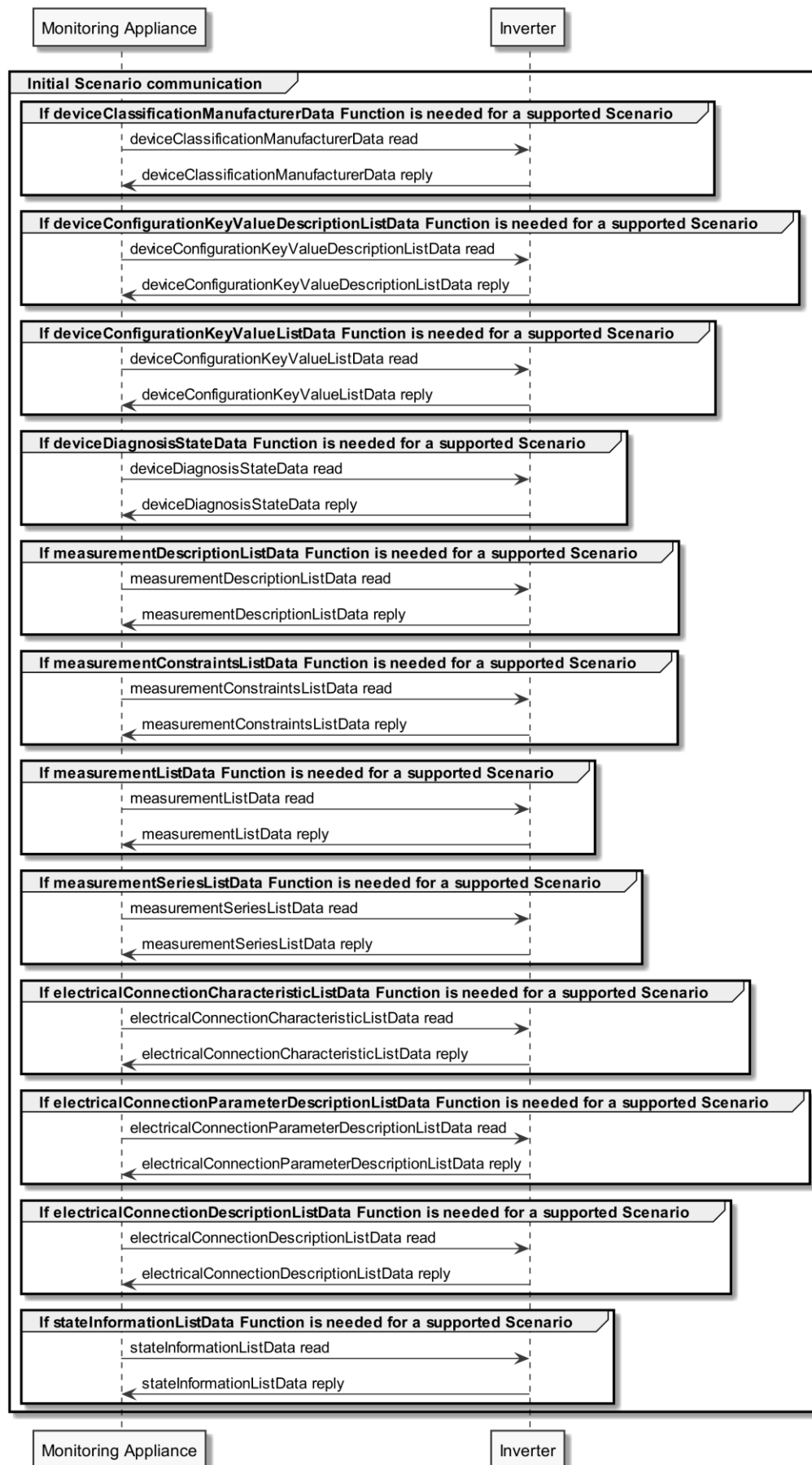


Figure 10: Scenario 1-7 - Initial Scenario communication sequence diagram

1387

1388 The following table shows where the required content of the messages from the sequence diagram is
 1389 described:

Message name from sequence diagram	Content description in table	Scenario number in table
deviceClassificationManufacturerData reply	Table 48	1
deviceConfigurationKeyValueDescriptionListData reply	Table 49	1
		2
		6
deviceConfigurationKeyValueListData reply	Table 50	1
		2
		6
deviceDiagnosisStateData reply	Table 51	2
electricalConnectionDescriptionListData reply	Table 52	3
		4
		5
electricalConnectionParameterDescriptionListData reply	Table 53	3
		4
		5
electricalConnectionCharacteristicListData reply	Table 54	6
measurementDescriptionListData reply	Table 55	3
		4
		5
		7
measurementConstraintsListData reply	Table 56	3
		4
		5
		7
measurementListData reply	Table 57	3
		4
		5
		7
measurementSeriesListData reply	Table 58	4
stateInformationListData reply	Table 59	2

1390 *Table 60: Initial Scenario communication content references for Scenario 1-7*

1391 Note: Within the Initial Scenario communication the content required by the Scenarios of this Use
 1392 Case MAY not be provided completely but later during Runtime Scenario communication.

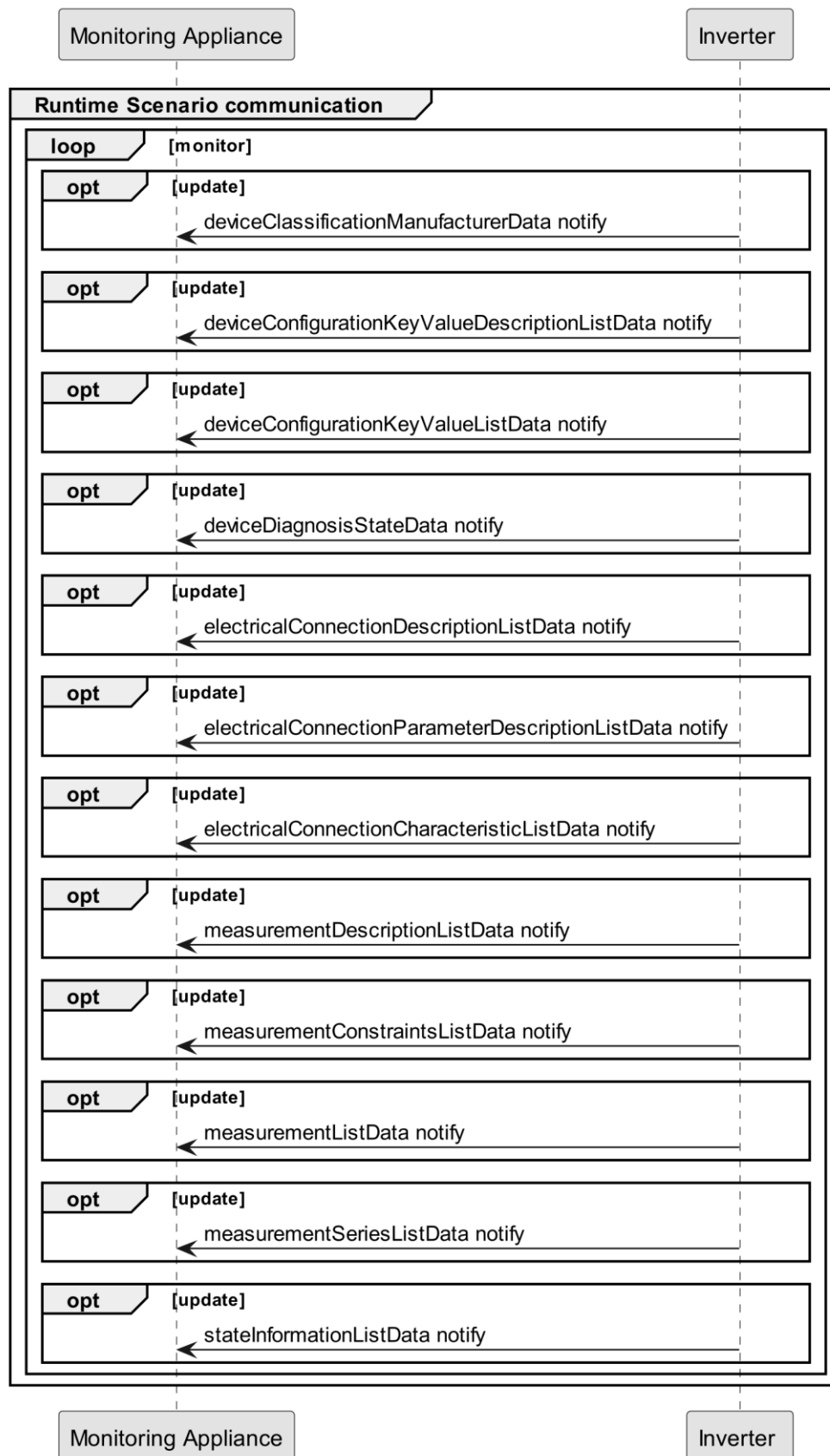
1393 Note: This section holds for all Scenarios of this Use Case.

1394

1395 **3.4.1.3 Runtime Scenario communication**

1396 Based on the Initial Scenario communication the Runtime Scenario communication provides updates
 1397 during runtime.

1398 If one of the referenced server Functions' data change, the server SHALL submit the change as shown
 1399 in the following figure:



1400

1401 *Figure 11: Scenario 1-7 - Runtime Scenario communication sequence diagram*

1402 Note: To interpret partial notification messages correctly the information obtained during the Initial
 1403 Scenario communication phase is required.

1404 Note: A read operation ("polling") on all Functions is possible at any time, e.g. if a notification could
 1405 not be evaluated.

1406

1407 The following table shows where the required content of the messages of the sequence diagram is
 1408 described:

Message name from sequence diagram	Content description in table	Scenario number in table
deviceClassificationManufacturerData notify	Table 48	1
deviceConfigurationKeyValueDescriptionListData notify	Table 49	1
		2
		6
deviceConfigurationKeyValueListData notify	Table 50	1
		2
		6
deviceDiagnosisStateData notify	Table 51	2
electricalConnectionDescriptionListData notify	Table 52	3
		4
		5
electricalConnectionParameterDescriptionListData notify	Table 53	3
		4
		5
electricalConnectionCharacteristicListData notify	Table 54	6
measurementDescriptionListData notify	Table 55	3
		4
		5
		7
measurementConstraintsListData notify	Table 56	3
		4
		5
		7
measurementListData notify	Table 57	3
		4
		5
		7
measurementSeriesListData notify	Table 58	4
stateInformationListData notify	Table 59	2

1409 *Table 61: Runtime Scenario communication content references for Scenario 1-7*

1410 Note: This section holds for all Scenarios of this Use Case.

1411

1412 3.4.2 Scenario 1 - Monitor Inverter identification

1413 3.4.2.1 Additional information

1414 None.

1415

1416 3.4.2.2 Information access**1417 Partial data information regarding [MOI-019]:**

1418 The deviceConfigurationKeyValueDescriptionListData read SHOULD be a "partial" read operation
1419 with the following Selectors:

- 1420 - keyName = "inverterGridCode"

1421 The deviceConfigurationKeyValueListData read SHOULD be a "partial" read operation with the
1422 following Selectors:

- 1423 - keyId (derived from the deviceConfigurationKeyValueDescriptionListData reply)

1424 Note: If partial read is not supported a full read SHALL be performed.

1425

1426 3.4.3 Scenario 2 - Monitor Inverter state**1427 3.4.3.1 Additional information**

1428 None.

1429

1430 3.4.3.2 Information access**1431 Partial data information regarding [MOI-023]:**

1432 The deviceConfigurationKeyValueDescriptionListData read SHOULD be a "partial" read operation
1433 with the following Selectors:

- 1434 - keyName = "pvStringAvailabilityStatus"

1435 The deviceConfigurationKeyValueListData read SHOULD be a "partial" read operation with the
1436 following Selectors:

- 1437 - keyId (derived from the deviceConfigurationKeyValueDescriptionListData reply)

1438 Note: If partial read is not supported a full read SHALL be performed.

1439

1440 Partial data information regarding [MOI-024]:

1441 The deviceConfigurationKeyValueDescriptionListData read SHOULD be a "partial" read operation
1442 with the following Selectors:

- 1443 - keyName = "batteryAvailabilityStatus"

1444 The deviceConfigurationKeyValueListData read SHOULD be a "partial" read operation with the
1445 following Selectors:

- 1446 - keyId (derived from the deviceConfigurationKeyValueDescriptionListData reply)

1447 Note: If partial read is not supported a full read SHALL be performed.

1448

1449 **Partial data information regarding [MOI-025]:**

1450 The deviceConfigurationKeyValueDescriptionListData read SHOULD be a "partial" read operation
1451 with the following Selectors:

1452 - keyName = "gridConnectionStatus"

1453 The deviceConfigurationKeyValueListData read SHOULD be a "partial" read operation with the
1454 following Selectors:

1455 - keyId (derived from the deviceConfigurationKeyValueDescriptionListData reply)

1456 Note: If partial read is not supported a full read SHALL be performed.

1457

1458 **Partial data information regarding [MOI-026]:**

1459 The stateInformationListData read SHOULD be a "partial" read operation with (a selection of) the
1460 following Selectors (depending on the needs of the client):

1461 - stateInformation = <String of state information of interest> *or*

1462 - isActive = "true" (if only active states shall be retrieved) *or*

1463 - category = <String of category of interest>

1464

1465 **3.4.4 Scenario 3 - Monitor Inverter AC power details**1466 **3.4.4.1 Additional information**

1467 None.

1468

1469 **3.4.4.2 Information access**1470 **Partial data information regarding [MOI-031]:**

1471 The measurementDescriptionListData read SHOULD be a "partial" read operation with the following
1472 Selectors:

1473 - scopeType = "acPowerApparentTotal"

1474 The measurementConstraintsListData read, measurementListData read and
1475 electricalConnectionParameterDescriptionListData read SHOULD be "partial" read operations with
1476 the following Selectors:

1477 - measurementId (derived from the measurementDescriptionListData reply)

1478 The electricalConnectionDescriptionListData read SHOULD be a "partial" read operation with the
1479 following Selectors:

1480 - electricalConnectionId (derived from the electricalConnectionParameterDescriptionListData
1481 reply)

1482 Note: If partial read is not supported a full read SHALL be performed.

1483

1484 **Partial data information regarding [MOI-032]:**

1485 The measurementDescriptionListData read SHOULD be a "partial" read operation with the following
1486 Selectors:

1487 - scopeType = "acPowerApparent"

1488 The measurementConstraintsListData read, measurementListData read and
1489 electricalConnectionParameterDescriptionListData read SHOULD be "partial" read operations with
1490 the following Selectors:

1491 - measurementId (derived from the measurementDescriptionListData reply)

1492 The electricalConnectionDescriptionListData read SHOULD be a "partial" read operation with the
1493 following Selectors:

1494 - electricalConnectionId (derived from the electricalConnectionParameterDescriptionListData
1495 reply)

1496 Note: If partial read is not supported a full read SHALL be performed.

1497

1498 **Partial data information regarding [MOI-033]:**

1499 The measurementDescriptionListData read SHOULD be a "partial" read operation with the following
1500 Selectors:

1501 - scopeType = "acPowerReactiveTotal"

1502 The measurementConstraintsListData read, measurementListData read and
1503 electricalConnectionParameterDescriptionListData read SHOULD be "partial" read operations with
1504 the following Selectors:

1505 - measurementId (derived from the measurementDescriptionListData reply)

1506 The electricalConnectionDescriptionListData read SHOULD be a "partial" read operation with the
1507 following Selectors:

1508 - electricalConnectionId (derived from the electricalConnectionParameterDescriptionListData
1509 reply)

1510 Note: If partial read is not supported a full read SHALL be performed.

1511

1512 **Partial data information regarding [MOI-034]:**

1513 The measurementDescriptionListData read SHOULD be a "partial" read operation with the following
1514 Selectors:

- 1515 - scopeType = "acPowerReactive"

1516 The measurementConstraintsListData read, measurementListData read and
1517 electricalConnectionParameterDescriptionListData read SHOULD be "partial" read operations with
1518 the following Selectors:

- 1519 - measurementId (derived from the measurementDescriptionListData reply)

1520 The electricalConnectionDescriptionListData read SHOULD be a "partial" read operation with the
1521 following Selectors:

- 1522 - electricalConnectionId (derived from the electricalConnectionParameterDescriptionListData
1523 reply)

1524 Note: If partial read is not supported a full read SHALL be performed.

1525

1526 **3.4.5 Scenario 4 - Monitor Inverter AC energy details**

1527 **3.4.5.1 Additional information**

1528 The data points Inverter Total Yield Day ([MOI-042a]) and Inverter Total Yield Day Historical ([MOI-
1529 042b]) share some of their data (the Functions measurementDescriptionListData,
1530 measurementConstraintsListData, electricalConnectionDescriptionListData and
1531 electricalConnectionParameterDescriptionListData). [MOI-042a] then uses the Function
1532 measurementListData for the current measurement value, whereas [MOI-042b] uses the Function
1533 measurementSeriesListData for the corresponding historical measurement data.

1534 The data points Inverter Total Yield Month ([MOI-043a]) and Inverter Total Yield Month Historical
1535 ([MOI-043b]) share some of their data (the Functions measurementDescriptionListData,
1536 measurementConstraintsListData, electricalConnectionDescriptionListData and
1537 electricalConnectionParameterDescriptionListData). [MOI-043a] then uses the Function
1538 measurementListData for the current measurement value, whereas [MOI-043b] uses the Function
1539 measurementSeriesListData for the corresponding historical measurement data.

1540 The data points Inverter Total Yield Year ([MOI-044a]) and Inverter Total Yield Year Historical ([MOI-
1541 044b]) share some of their data (the Functions measurementDescriptionListData,
1542 measurementConstraintsListData, electricalConnectionDescriptionListData and
1543 electricalConnectionParameterDescriptionListData). [MOI-044a] then uses the Function
1544 measurementListData for the current measurement value, whereas [MOI-044b] uses the Function
1545 measurementSeriesListData for the corresponding historical measurement data.

1546

1547 **3.4.5.2 Information access**

1548 **Partial data information regarding [MOI-041]:**

1549 The measurementDescriptionListData read SHOULD be a "partial" read operation with the following
1550 Selectors:

1551 - scopeType = "acYieldTotal"

1552 The measurementConstraintsListData read, measurementListData read and
1553 electricalConnectionParameterDescriptionListData read SHOULD be "partial" read operations with
1554 the following Selectors:

1555 - measurementId (derived from the measurementDescriptionListData reply)

1556 The electricalConnectionDescriptionListData read SHOULD be a "partial" read operation with the
1557 following Selectors:

1558 - electricalConnectionId (derived from the electricalConnectionParameterDescriptionListData
1559 reply)

1560 Note: If partial read is not supported a full read SHALL be performed.

1561

1562 **Partial data information regarding [MOI-042a]:**

1563 The measurementDescriptionListData read SHOULD be a "partial" read operation with the following
1564 Selectors:

1565 - scopeType = "acYieldDay"

1566 The measurementConstraintsListData read, measurementListData read and
1567 electricalConnectionParameterDescriptionListData read SHOULD be "partial" read operations with
1568 the following Selectors:

1569 - measurementId (derived from the measurementDescriptionListData reply)

1570 The electricalConnectionDescriptionListData read SHOULD be a "partial" read operation with the
1571 following Selectors:

1572 - electricalConnectionId (derived from the electricalConnectionParameterDescriptionListData
1573 reply)

1574 Note: If partial read is not supported a full read SHALL be performed.

1575

1576 **Partial data information regarding [MOI-042b]:**

1577 The measurementDescriptionListData read SHOULD be a "partial" read operation with the following
1578 Selectors:

1579 - scopeType = "acYieldDay"

1580 The measurementConstraintsListData read, measurementSeriesListData read and
1581 electricalConnectionParameterDescriptionListData read SHOULD be "partial" read operations with
1582 the following Selectors:

- 1583 - measurementId (derived from the measurementDescriptionListData reply)

1584 The electricalConnectionDescriptionListData read SHOULD be a "partial" read operation with the
1585 following Selectors:

- 1586 - electricalConnectionId (derived from the electricalConnectionParameterDescriptionListData
1587 reply)

1588 Note: If partial read is not supported a full read SHALL be performed.

1589

1590 **Partial data information regarding [MOI-043a]:**

1591 The measurementDescriptionListData read SHOULD be a "partial" read operation with the following
1592 Selectors:

- 1593 - scopeType = "acYieldMonth"

1594 The measurementConstraintsListData read, measurementListData read and
1595 electricalConnectionParameterDescriptionListData read SHOULD be "partial" read operations with
1596 the following Selectors:

- 1597 - measurementId (derived from the measurementDescriptionListData reply)

1598 The electricalConnectionDescriptionListData read SHOULD be a "partial" read operation with the
1599 following Selectors:

- 1600 - electricalConnectionId (derived from the electricalConnectionParameterDescriptionListData
1601 reply)

1602 Note: If partial read is not supported a full read SHALL be performed.

1603

1604 **Partial data information regarding [MOI-043b]:**

1605 The measurementDescriptionListData read SHOULD be a "partial" read operation with the following
1606 Selectors:

- 1607 - scopeType = "acYieldMonth"

1608 The measurementConstraintsListData read, measurementSeriesListData read and
1609 electricalConnectionParameterDescriptionListData read SHOULD be "partial" read operations with
1610 the following Selectors:

- 1611 - measurementId (derived from the measurementDescriptionListData reply)

1612 The electricalConnectionDescriptionListData read SHOULD be a "partial" read operation with the
1613 following Selectors:

- 1614 - electricalConnectionId (derived from the electricalConnectionParameterDescriptionListData
1615 reply)

1616 Note: If partial read is not supported a full read SHALL be performed.

1617

1618 **Partial data information regarding [MOI-044a]:**

1619 The measurementDescriptionListData read SHOULD be a "partial" read operation with the following
1620 Selectors:

- 1621 - scopeType = "acYieldYear"

1622 The measurementConstraintsListData read, measurementListData read and
1623 electricalConnectionParameterDescriptionListData read SHOULD be "partial" read operations with
1624 the following Selectors:

- 1625 - measurementId (derived from the measurementDescriptionListData reply)

1626 The electricalConnectionDescriptionListData read SHOULD be a "partial" read operation with the
1627 following Selectors:

- 1628 - electricalConnectionId (derived from the electricalConnectionParameterDescriptionListData
1629 reply)

1630 Note: If partial read is not supported a full read SHALL be performed.

1631

1632 **Partial data information regarding [MOI-044b]:**

1633 The measurementDescriptionListData read SHOULD be a "partial" read operation with the following
1634 Selectors:

- 1635 - scopeType = "acYieldYear"

1636 The measurementConstraintsListData read, measurementSeriesListData read and
1637 electricalConnectionParameterDescriptionListData read SHOULD be "partial" read operations with
1638 the following Selectors:

- 1639 - measurementId (derived from the measurementDescriptionListData reply)

1640 The electricalConnectionDescriptionListData read SHOULD be a "partial" read operation with the
1641 following Selectors:

- 1642 - electricalConnectionId (derived from the electricalConnectionParameterDescriptionListData
1643 reply)

1644 Note: If partial read is not supported a full read SHALL be performed.

1645

1646 **3.4.6 Scenario 5 - Monitor Inverter AC additional details**1647 **3.4.6.1 Additional information**

1648 None.

1649

1650 **3.4.6.2 Information access**1651 **Partial data information regarding [MOI-051]:**

1652 The measurementDescriptionListData read SHOULD be a "partial" read operation with the following
1653 Selectors:

- 1654 - scopeType = "acCosPhi"

1655 The measurementConstraintsListData read, measurementListData read and
1656 electricalConnectionParameterDescriptionListData read SHOULD be "partial" read operations with
1657 the following Selectors:

- 1658 - measurementId (derived from the measurementDescriptionListData reply)

1659 The electricalConnectionDescriptionListData read SHOULD be a "partial" read operation with the
1660 following Selectors:

- 1661 - electricalConnectionId (derived from the electricalConnectionParameterDescriptionListData
1662 reply)

1663 Note: If partial read is not supported a full read SHALL be performed.

1664

1665 **3.4.7 Scenario 6 - Monitor Inverter capabilities**1666 **3.4.7.1 Additional information**

1667 None.

1668

1669 **3.4.7.2 Information access**1670 **Partial data information regarding [MOI-061a]:**

1671 The electricalConnectionCharacteristicListData read SHOULD be a "partial" read operation with the
1672 following Selectors:

- 1673 - electricalConnectionId = <same value as used within [MPC-011]>
1674 - parameterId = <same value as used within [MPC-011]>
1675 - characteristicContext = "entity"
1676 - characteristicType = "powerProductionNominalMax"

1677 The electricalConnectionId and parameterId SHALL link to the electricalConnectionId and
1678 parameterId used for [MPC-011].

1679 Note: If partial read is not supported a full read SHALL be performed.

1680

1681 **Partial data information regarding [MOI-061b]:**

1682 The electricalConnectionCharacteristicListData read SHOULD be a "partial" read operation with the
1683 following Selectors:

- 1684 - electricalConnectionId = <same value as used within [MPC-011]>
- 1685 - parameterId = <same value as used within [MPC-011]>
- 1686 - characteristicContext = "entity"
- 1687 - characteristicType = "powerConsumptionNominalMax"

1688 The electricalConnectionId and parameterId SHALL link to the electricalConnectionId and
1689 parameterId used for [MPC-011].

1690 Note: If partial read is not supported a full read SHALL be performed.

1691

1692 **Partial data information regarding [MOI-062a]:**

1693 The electricalConnectionCharacteristicListData read SHOULD be a "partial" read operation with the
1694 following Selectors:

- 1695 - electricalConnectionId = <same value as used within [MOI-031]>
- 1696 - parameterId = <same value as used within [MOI-031]>
- 1697 - characteristicContext = "entity"
- 1698 - characteristicType = "apparentPowerProductionNominalMax"

1699 The electricalConnectionId and parameterId SHALL link to the electricalConnectionId and
1700 parameterId used for [MOI-031].

1701 Note: If partial read is not supported a full read SHALL be performed.

1702

1703 **Partial data information regarding [MOI-062b]:**

1704 The electricalConnectionCharacteristicListData read SHOULD be a "partial" read operation with the
1705 following Selectors:

- 1706 - electricalConnectionId = <same value as used within [MOI-031]>
- 1707 - parameterId = <same value as used within [MOI-031]>
- 1708 - characteristicContext = "entity"
- 1709 - characteristicType = "apparentPowerConsumptionNominalMax"

1710 The electricalConnectionId and parameterId SHALL link to the electricalConnectionId and
1711 parameterId used for [MOI-031].

1712 Note: If partial read is not supported a full read SHALL be performed.

1713

1714 Partial data information regarding [MOI-063a]:

1715 The electricalConnectionCharacteristicListData read SHOULD be a "partial" read operation with the
1716 following Selectors:

- 1717 - electricalConnectionId = <same value as used within [MPC-011]>
- 1718 - parameterId = <same value as used within [MPC-011]>
- 1719 - characteristicContext = "battery"
- 1720 - characteristicType = "powerConsumptionNominalMin"

1721 The electricalConnectionId and parameterId SHALL link to the electricalConnectionId and
1722 parameterId used for [MPC-011].

1723 Note: If partial read is not supported a full read SHALL be performed.

1724

1725 Partial data information regarding [MOI-063b]:

1726 The electricalConnectionCharacteristicListData read SHOULD be a "partial" read operation with the
1727 following Selectors:

- 1728 - electricalConnectionId = <same value as used within [MPC-011]>
- 1729 - parameterId = <same value as used within [MPC-011]>
- 1730 - characteristicContext = "battery"
- 1731 - characteristicType = "powerConsumptionNominalMax"

1732 The electricalConnectionId and parameterId SHALL link to the electricalConnectionId and
1733 parameterId used for [MPC-011].

1734 Note: If partial read is not supported a full read SHALL be performed.

1735

1736 Partial data information regarding [MOI-064]:

1737 The electricalConnectionCharacteristicListData read SHOULD be a "partial" read operation with the
1738 following Selectors:

- 1739 - electricalConnectionId = <same value as used within [MPC-011]>
- 1740 - parameterId = <same value as used within [MPC-011]>
- 1741 - characteristicContext = "battery"
- 1742 - characteristicType = "powerConsumptionMax"

1743 The electricalConnectionId and parameterId SHALL link to the electricalConnectionId and
1744 parameterId used for [MPC-011].

1745 Note: If partial read is not supported a full read SHALL be performed.

1746

1747 Partial data information regarding [MOI-065a]:

1748 The electricalConnectionCharacteristicListData read SHOULD be a "partial" read operation with the
1749 following Selectors:

- 1750 - electricalConnectionId = <same value as used within [MPC-011]>
- 1751 - parameterId = <same value as used within [MPC-011]>
- 1752 - characteristicContext = "battery"
- 1753 - characteristicType = "powerProductionNominalMin"

1754 The electricalConnectionId and parameterId SHALL link to the electricalConnectionId and
1755 parameterId used for [MPC-011].

1756 Note: If partial read is not supported a full read SHALL be performed.

1757

1758 Partial data information regarding [MOI-065b]:

1759 The electricalConnectionCharacteristicListData read SHOULD be a "partial" read operation with the
1760 following Selectors:

- 1761 - electricalConnectionId = <same value as used within [MPC-011]>
- 1762 - parameterId = <same value as used within [MPC-011]>
- 1763 - characteristicContext = "battery"
- 1764 - characteristicType = "powerProductionNominalMax"

1765 The electricalConnectionId and parameterId SHALL link to the electricalConnectionId and
1766 parameterId used for [MPC-011].

1767 Note: If partial read is not supported a full read SHALL be performed.

1768

1769 Partial data information regarding [MOI-066]:

1770 The electricalConnectionCharacteristicListData read SHOULD be a "partial" read operation with the
1771 following Selectors:

- 1772 - electricalConnectionId = <same value as used within [MPC-011]>
- 1773 - parameterId = <same value as used within [MPC-011]>
- 1774 - characteristicContext = "battery"
- 1775 - characteristicType = "powerProductionMax"

1776 The electricalConnectionId and parameterId SHALL link to the electricalConnectionId and
1777 parameterId used for [MPC-011].

1778 Note: If partial read is not supported a full read SHALL be performed.

1779

1780 Partial data information regarding [MOI-067a]:

1781 The deviceConfigurationKeyValueDescriptionListData read SHOULD be a "partial" read operation
1782 with the following Selectors:

1783 - keyName = "timeToAcChargePowerMax"

1784 The deviceConfigurationKeyValueListData read SHOULD be a "partial" read operation with the
1785 following Selectors:

1786 - keyId (derived from the deviceConfigurationKeyValueDescriptionListData reply)

1787 Note: If partial read is not supported a full read SHALL be performed.

1788

1789 **Partial data information regarding [MOI-067b]:**

1790 The deviceConfigurationKeyValueDescriptionListData read SHOULD be a "partial" read operation
1791 with the following Selectors:

1792 - keyName = "timeToAcDischargePowerMax"

1793 The deviceConfigurationKeyValueListData read SHOULD be a "partial" read operation with the
1794 following Selectors:

1795 - keyId (derived from the deviceConfigurationKeyValueDescriptionListData reply)

1796 Note: If partial read is not supported a full read SHALL be performed.

1797

1798 **3.4.8 Scenario 7 - Monitor Inverter internal data**

1799 **3.4.8.1 Additional information**

1800 None.

1801

1802 **3.4.8.2 Information access**

1803 **Partial data information regarding [MOI-071]:**

1804 The measurementDescriptionListData read SHOULD be a "partial" read operation with the following
1805 Selectors:

1806 - scopeType = "componentTemperature"

1807 The measurementConstraintsListData read and measurementListData read SHOULD be "partial" read
1808 operations with the following Selectors:

1809 - measurementId (derived from the measurementDescriptionListData reply)

1810 Note: If partial read is not supported a full read SHALL be performed.

1811